S8 Technology Family Topological Design Rules

Section A. Introduction

1.0 Purpose/Scope:

1.1 Purpose
Topological Design Rules (TDR) describe guidelines for circuit design based on the manufacturing capability and principle of operation of integrated circuits for the given technology.

1.2 Scope
This specification documents the topological design rules for the S8 technology family.
The products manufactured in this technology will be designed in S8 space
(drawn = final) without later shrink at the mask shop.

2.0 Responsibilities:

2.1 This specification is to be defined and revised by Technology engineers only. Revisions proposed by any other parties have to be reviewed and agreed upon by Technology.
2.2 Record keeping: All changes need to be documented in detail using History page.

3.0 Referenced Documents:

3.1 Referenced Document (DMS)
Click on the above link to access the Document Management System (DMS) details page for the spec. Then click on the “Reference Documents” button to bring up the list of referenced documents.

4.0 Materials:

The topological design rules are contained in this Excel spreadsheet with graphics.

5.0 Equipment: NA

6.0 Safety: NA

7.0 Critical Requirements Summary:

7.1 Process Restrictions
Top of the line photolithography tool: DUV scanner ASML5500/800.
See Technology TDR Criteria for implementation details.

8.0 Operating Procedures and Responsibilities:

8.1. Description of generic S8 Technology Family
Key features of S8 technology:
* Design in S8 space without optical shrink at mask shop
* Power supply: 1.8 V, 3.3V or 5V
* Single poly, non-salicided source/drain, single local interconnect and triple metal
* 32A gate oxide non-regulated, 110A (electrical) for regulated FET
* 2T (1 SONOS & 1NPASS) for Flash and EEPROM, 6T for SRAM and 12T for NVSRAM
* Triple-well on p-type substrate for device optimization and noise immunity.

INFORMATION COLORED II GRAY NOT IMPLEMENTED BY CAD

8.2 CAD Flows
See Table A1.

8.3 Logic product compatibility
Any S8 Family technology can be used without modification for logic products.

9 Quality Requirements:

Each section should be reviewed by the responsible module/integration engineers.
See also Spec # 01-30002.

10 Records:

10.1 All records shall be maintained as specified in 00-00064 - Record Retention Policy and Procedures

11 Preventive Maintenance:

This specification must be updated along with the objective specification change.

12 Posting Sheets/Forms/Appendix: N/A

Each section should be reviewed by the responsible module/integration engineers.

Table A1. S8 Mfg Fab flows and mask layers used by CAD flows

CAD flow order		1	2	4	6	7	11	15	16	17	18	19	20	21	23	24	25	28	29	34	45
	Fab Flow =>	***S8-SUPERSET-5R***	S8DIN[1,3,4]-5[P]&S8DI-5[P]	S8P[3,1]-5[P]	S8P[3,1]-5[P]	S8PM[3,4]-10[P]	S8PF[1,4]-10[P]	S8PFN-20P	S8PIR-10, S8PIM-10[P]	S8PR2-10, S8PR1-10[P]	S8SPF-10[P], S8SPM-10[P]	S8TM-5R	S8TMA-5[P,R]	S8TMC-5[P,R,RP]	S8TNV[1,3]-5[P,R]	S8P[3,1]-5[P]	S8PF[4,1]-10[P]	S8PR2-10, S8PR1-10[P]	S8PIRS	S8SPR1[P]	S8PHRC-20R
	CAD flow =>	**s8ss-5r**	**s8di-5r**	**s8p-10r**	**s8p-5r**	**s8p12-10r**	**s8pf-10r**	**s8pfn-20r**	**s8pir-10r**	**s8pr2-10r**	**s8spf-10r**	**s8tm-5r**	**s8tma-5r**	**s8tmc-5r**	**s8tnv-5r**	**sp8p-5r**	**s8pfhd-10r**	**s8phr-10r**	**s8phirs-10r**	**s8sphr-10r**	**s8phrc-20r**
Layer Description	**Mask Acronym**		2 flows	Shared	Shared	Shared	Shared							2 flows		Shared					
Field Oxide	FOM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Deep N-Well	DNM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
P-Well Block Mask	PWBM	**X**						X													
P-Well Drain Extended	PWDEM	**X**						X													
N-Well*	NWM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
High Vt PCh*	HVTPM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Low Vt Nch*	LVTNM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
HLow VT PCh Radio*	HVTRM	**X**							X									X			
N-Core Implant	NCM	**X**													X						
Tunnel Mask	TUNM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
ONO Mask	ONOM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Low Voltage Oxide	LVOM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Resistor Protect	RPM	**X**						X	X	X								X			X
Rev Resistor Protect	RRPM	**X**																			X
Poly 1	P1M	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
N-tip Implant	NTM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
High Volt. N-tip	HVNTM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Lightly Doped N-tip	LDNTM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Nitride Poly Cut	NPCM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
P+ Implant	PSDM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
N+ Implant	NSDM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Local Intr Cont.1	LICM1	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Local Intrcnct 1	LI1M	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Contact	CTM1	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Metal 1	MM1	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Via	VIM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Capacitor MiM on M2	CAPM	**X**											X	X							
Metal 2	MM2	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Via 2-TNV	VIM2	**X**													X						
Via 2-S8TM	VIM2	**X**										X	X	X							
Via 2-PLM	VIM2	**X**		X	X	X	X	X	X	X	X					X	X	X	X	X	X
Metal 3-TLM	MM3	**X**													X						
Metal 3-S8TM	MM3	**X**										X	X	X							
Capacitor MiM	CAPMM3	**X**																			X
Metal 3-PLM	MM3	**X**		X	X	X	X	X	X	X	X					X	X	X	X	X	X
Pad Via	VIPDM	**X**	X																		
Via3-PLM	VIM3	**X**		X	X	X	X	X	X	X	X					X	X	X	X	X	X
Inductor-TLM	INDM	**X**	X																		
Capacitor MiM on M4	CAPMM4	**X**																			X
Metal 4	MM4	**X**		X	X	X	X	X	X	X	X					X	X	X	X	X	X
Via4	VIM4	**X**		X	X	X	X	X	X	X	X					X	X	X	X	X	X
Metal 5	MM5	**X**		X	X	X	X	X	X	X	X					X	X	X	X	X	X
Nitride Seal Mask	NSM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
ViaTop	VIMT																				X
Mtest	MTEST																				X
Pad (scribe protect)	PDM	**X**	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Pad (scribe unprotect)	PDM	**X**																			
Polyimide	PMM	**X**	X	X	X	X	X	X			X	X	X	X	X	X	X		X	X	X
Polyimide_ExtFab	PMM[E]	**X**																			
Pad&Polyimide_ExtFab	PDMM[E]	**X**																			
DECA PBO	*PBO*	***X***							*X*	*X*								*X*			
Cu Inductor/Redist.	*CU1M*	***X***							*X*	*X*								*X*			
Polyimide 2 (2)	*PMM2*	**X**							*X*	*X*								*X*			
Under Bump Metal	*UBM*	**X**																			
Bumps	*BUMP*	**X**																			
Number of reticles		**52**	**26**	**30**	**30**	**30**	**30**	**33**	**34**	**33**	**30**	**26**	**27**	**27**	**27**	**30**	**30**	**34**	**30**	**30**	**35**
Flow #		1	2	4	6	7	11	15	16	17	18	19	20	21	23	24	25	28	29	34	45

* LVTNM, NWM, HVTPM: Mask sequence in S8TNV flow is different from other S8 flows (SUW-422).
(2) DECA/ Other BE Fab layers in italics.

(2 flows) - This CAD flow corresponds to two Fab flows, with and without polyimide
(Shared) - These fab flows are shared with 2 or 3 CAD flows with polyimide
CMI Traveller Nomenclature

[3] TIN BEOL
[P] Polyimide

Table A1. Custom CAD flows

A1.1. DECA CAD flow

(1) To enable Deca WLCSP rules, create a link in the design's config named "pack", and point it as in config dir below

Contents of config directory (example):

cds.lib
cydir -> pcios/
design.env
pack --> pcios/../../deca/p/deca/wlcsp/s8pir-10r
pcios -> /tools/stabflow/001-59001/pcios/001-59001/
tech -> pcios/s8/s8pir-10r/
topddc -> ../s8blerf_ver2/
"/tools/stabflow/ (PCIOS NUMBER) /deca/p/deca/wlcsp/ (TECHNOLOGY) /calibre".

* Masks past top metal (typically, MM5) sent to sub-cons for Wafer Level Chip Scale
* No verification is supported for the S8SS flow; shown for total mask count only.
* OFM is an optional masks in S8-DI* and other *-foundry CAD flows

A.1.2. Copper Back End (CU BE) CAD flow

PV Flow:

1. Conversion
2. Streamin
3. Checking in and initial PVs
4. CD and VPP cell replacement
5. PVs:
- DRC1 - std
- DRC2 - LVL based (**IR Drop** check)
6. CLDRC - S8
7 CLDRC - MB65

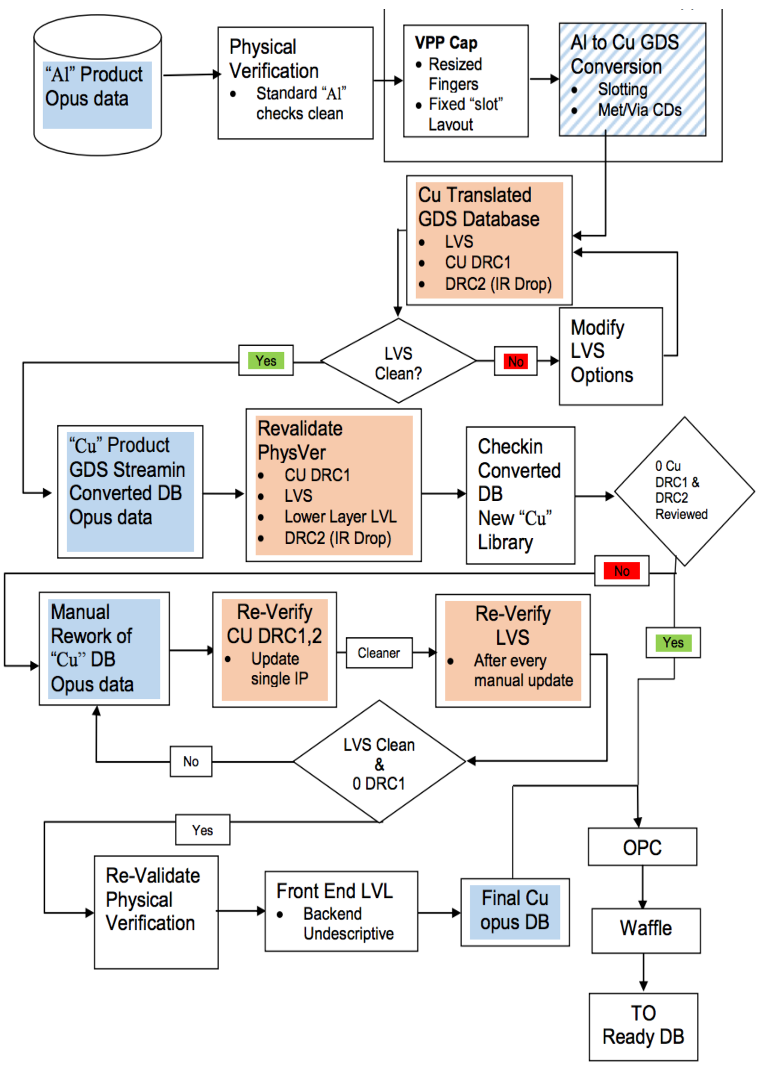

Table A2. Major CAD flows compared at a device level.

Process (CAD)	Process (fab)	PDK current?	# metals	SONOS	1.8V FET	5V FET	300ohm poly res	2kohm poly res	# MiM caps	20V FET	Top Metal Thickness (um)	inductor compatible	Comment
S130	S8PHRC20	yes	5	standard	X	X	X	X	2, M3/M4	X	1.26		PDK in new DRM
S8PHRC-20R	S8PHRC20	yes	5	standard	X	X	X	X	2, M3/M4	X	1.26		superset
S8PFN-20R	S8SPF-20P	no	5	standard	X	X	X			X	1.26		subset of S8PHRC-20R
S8PHIRS-10R	S8PIRS	no	5	"star" cell	X	X	X				1.26	post-fab	subset of S8PHRC-20R
S8PFHD-10R	S8PF1-10P	no	5	standard	X	X					1.26		subset of S8PHIRS
S8P-5R	S8P1-5P	no	5	standard	X	X					2.2		legacy only
S8DI-5R	S8DI-5R	no	3	standard	X	X					4.08	on-Si (M3)	legacy only
S8TMA-5R	S8TMA-5R	no	3	standard	X	X	X		1, M2		2.2		legacy only
S8TNV-5R	S8TNV1-5	no	3	NVSRAM	X	X					0.77		legacy only

Section B: Design Rule Criteria & Assumptions

Note: all numbers below are final (S8, on wafer) unless explicitly specified otherwise

1. General

Parameter	Units	Value	Variable name
Space to Draw		S8	
Grid Size - Drawn	um	0.005	GSF

2. Minimum CDs in Design or on Wafer, Required by Technology (Core or Periphery)

Layer Name		S8 feature	S8 space	Variable name
Field Oxide		0.140	0.270	FOMCD/FOMCDSP
Deep N-Well		3.000	6.300	DNMCD/DNMCDSP
P-Well Block Mask		0.840	1.270	PWBMCD/PWBMCDSP
P-Well Drain Extended		0.840	1.270	PWDEMCD/PWDEMCDSP
N-Well		0.840	1.270	NWMCD/NWMCDSP
High Vt PCh		0.380	0.380	HVTPMCD/HVTPMCDSP
Low Vt Nch		0.380	0.380	LVTNMCD/LVTNMCDSP
HLow VT PCh Radio		0.380	0.380	HVTRMCD/HVTRMCDSP
N-Core Implant		0.380	0.380	NCMCD/NCMCDSP
Tunnel Mask		0.410	0.500	TUNMCD/TUNMCDSP
ONO Mask		0.410	0.500	ONOMCD/ONOMCDSP
Low Voltage Oxide		0.600	0.700	LVOMCD/LVOMCDSPCSMC
Resistor Protect		1.270	0.840	RPMCD/RPMCDSP
Reverse Resistor Protect		0.600	0.700	RRPMCD/RRPMCDSP
Poly 1	Endcap/Gap	0.150	0.210	P1G
Poly 1		N/A	0.140	P1MCD/P1MCDSP
N-tip Implant		0.840	0.700	NTMCD/NTMCDSP
High Volt. N-tip		0.700	0.700	HVNTMCD/HVNTMCDSP
Lightly Doped N-tip		0.700	0.700	LDNTMCD/LDNTMCDSP
Nitride Poly Cut		0.270	0.270	NPCMCD/NPCMCDSP
P+ Implant		0.380	0.380	PSDMCD/PSDMCDSP
N+ Implant		0.380	0.380	NSDMCD/NSDMCDSP
Local Intr Cont.1	Slotted	0.170	0.170	LICM1SLCD/LICM1SLCDSP
Local Intr Cont.1	Core	0.190	0.350	LICM1CD/LICM1CDSP
Local Intrcnct 1	Core	0.140	0.140	LI1MCD/LI1MCDSP
Local Intrcnct 1		0.170	0.170	LI1MCD/LI1MCDSP
Contact		0.170	0.190	CTM1CD/CTM1CDSP
Metal 1		0.140	0.140	MM1CD/MM1CDSP
Metal 1 - Cu		0.140	0.140	MM1_CuCD/MM1_CuCDSP
Via		0.150	0.170	VIMCD/VIMCDSP
Via - Cu		0.180	0.130	VIM_CuCD/VIM_CuCDSP
Capacitor MiM on M2		2.000	0.840	CAPMCD/CAPMCDSP
Metal 2		0.140	0.140	MM2CD/MM2CDSP
Metal 2 - Cu		0.140	0.140	MM2_CuCD/MM2_CuCDSP
Via 2-TNV		0.280	0.280	VIM2CD/VIM2CDSP
Via 2-S8TM		0.800	0.800	VIM2CD/VIM2CDSP
Via 2-PLM		0.200	0.200	VIM2CD/VIM2CDSP
Via 2-Cu		0.210	0.180	VIM2_CuCD/VIM2_CuCDSP
Metal 3-TLM		0.360	0.360	MM3CD/MM3CDSP
Metal 3-S8TM		0.800	0.800	MM3CD/MM3CDSP
Capacitor MiM (5LM)		1.000	0.840	CAPMCD_P/CAPMSP_P
Metal 3-PLM		0.300	0.300	MM3CD/MM3CDSP
Metal 3-Cu		0.300	0.300	MM3_CuCD/MM3_CuCDSP
Pad Via		1.200	1.270	VIPDMCD/VIPDMCDSP
Via3-PLM		0.200	0.200	VIM3CD/VIM3CDSP
Via3-Cu		0.210	0.180	VIM3_CuCD/VIM3_CuCDSP
Inductor-TLM		2.500	2.500	INDMCD/INDMCDSP
Capacitor-2 MiM (5LM)		1.000	0.840	CAP2MCD_P/CAP2MSP_P
Metal 4		0.300	0.300	MM4CD/MM4CDSP
Metal 4-Cu		0.300	0.300	MM4_CuCD/MM4_CuCDSP
Via4		0.800	0.800	VIM4CD/VIM4CDSP
Metal 5	All flows except S8PF*/S8PIR*	0.800	0.800	MM5CD/MM5CDSP
Metal 5	S8PF*/S8PIR*	1.600	1.600	MM5CD/MM5CDSP
Nitride Seal Mask		3.000	4.000	NSMCD/NSMCDSP
ViaTop	Only flows with topside vias	1.000	1.200	VIMTCD/VIMTCDSP
Mtest	SKyWater use only	1.000	1.000	MTESTCD/MTESTCDSP
Pad (scribe protect)		2.000	1.270	PDMCD/PDMCDSP
Polyimide		5.000	15.000	PMMCD/PMMCDSP
Polyimide_ExtFab		5.000	15.000	PMM[E]CD/PMM[E]CDSP
DECA PBO		10.000	10.000	PBOCD/PBOCDSP
Cu Inductor/Redist.		20.000	20.000	CU1MCD/CU1MCDSP
Serifs		0.100	0.100	SERCD/SERCDSP

3. Semiconductor Criteria

Basic Parameters	Units	Value		Variable name
n-well peak concentration	cm-3	6.00E+17		NWPCONC
background concentration	cm-3	8.00E+14		NWBCONC
y.char	um	0.43		NWYCHAR
desired Nmin/Ns ratio		0.90		NMINNSRATIO
min n-well width to guarantee 90 % peak concentr.	um	0.55		MINNWWID
p-well peak concentration	cm-3	4.00E+17		PWPCONC
p-well peak coordinate	um	0.42		PWPCOORD
y.char	um	0.13		PWYCHAR
min. p-well width to guarantee 90 % peak concentr.	um	0.33		MINPWWID
Junction Depths		**Vertical**		**Variable name**
Baseline: N-Well	um	1.10		NWVDIM
P-Well	um	0.75		PWVDIM
N-w/P-w junction (from drawn edge)	um	*	0.034	WELLJCT
N+ or P+ S/D (XJ)	um	0.10	0.060	JCTD / LD
Max (N+ or P+ S/D outdiff.) next to isol. edge	um		0.007	LDST
Max (N+ or P+ S/D outdiff.) next to isol. edge for 6 V			0.050	LDST5
N Tip (As)	um		0.010	LDNTIP
Other Width Criteria		**Value (um)**		**Variable name**
Min. diff/tap width for reproducible resistivity		0.120		MINFWR
Min. width to open a strip of tap between two diffs		0.340		SDM3
Max s/d diff width without contact, consistent w/Ram4,5,6		5.700		XMAXCON
Minimum Spacing for 3.3 V Punchthrough (1.8V devices)		**Value (um)**		**Variable name**
n-well - n-well		0.835		NWPTS
n+ - n+ or p+-p+		0.230		DPTS
p+ in nwell to pwell		0.050		PPTS
n+ in pwell to nwell		0.150		PNPTS
Latch-up/ESD Criteria		**Value (um)**		**Variable name**
Minimum n+ or p+ - nwell spacing to prevent latch-up		0.230		NPNWLU
Min n-well enclos. of tap to ensure bkdwn N-w/P-w before N+/P-w (ESD)		0.040		XNWESD
Max. overlap of n-well by p+ tap		0.060		XNWPTS
Implant angles		**Angle (deg)**		**Variable name**
High current		0	0.00	HCIMPA
Angle for tip implant		7		TipAng
Angle for HV tip implant		40		HvTipAngle
Twist angle for HV Tip		23		HvTipTwist

4. Physical Criteria

Material Thicknesses	Value (um)	Variable name
field oxide (above silicon surface) ... underneath poly	0.070	FOXSTEP
min. etch and fill capability for isolation, licon, and met1	0.150	DEFC
min. etch and fill capability for mcon	0.140	CEFC
min. etch and fill capability for via	0.180	VEFC
poly cap after SPE	0.200	OVGTTH
poly thickness	0.180	POLYTH
oxide spacer	0.050	SpThickn
Pre-LI ILD thickness	0.500	ILDTHICKN
Licon1 etch angle (deg)	10	LICETANG
Standard Licon bottom CD	0.080	LBCD
Mcon enclosure by Li	0.000	mconLiEnclosure
Via1 slope	0.020	Via1Slope
Oxide Bias for MM1	0.600	BiasMM1
Oxide Bias for MM2	0.600	BiasMM2
Oxide Bias for MM3	1.150	BiasMM3
Oxide Bias for MM4	1.150	BiasMM4
LI1 thickness for antenna ratio calculations	0.100	LiThick
Metal 1 thickness for antenna ratio calculations (S8D*)	0.350	Met1Thick
Metal 2 thickness for antenna ratio calculations (S8D*)	0.350	Met2Thick
Inductor thickness for antenna ratio calculation (S8D*)	4.000	IndmThick
Metal 3 thickness for antenna ratio calculation (S8Q/SP8Q)	0.800	Met3thick_q
Metal4 thickness for antenna ratio calculation (S8Q*/SP8Q)	2.000	Met4Thick_q
Metal 3 thickness for antenna ratio calculation (S8P*/SP8P*)	0.800	Met3thick_p
Metal4 thickness for antenna ratio calculation (S8P*/SP8P*)	0.800	Met4Thick_p
Metal5 thickness for antenna ratio calculation (S8P*/SP8P* with 2um thick metal)	2.000	Met5Thick_p
Metal5 thickness for antenna ratio calculation (S8P*/SP8P* with 1.2um thick metal)	1.200	Met5Thickp_12
Metal 2 thickness for antenna ratio calculations (SP8T/S8T*)	0.350	Met2_Qthick
Metal 3 thickness for antenna ratio calculations (S8T* other than S8TM*)	0.850	Met3_Qthick
Metal 3 thickness for antenna ratio calculations (S8TM* flow)	2.000	Met3_TMthick
Metal 3 thickness for antenna ratio calculations (SP8T flow)	0.800	Met3_SP8Tthick
Photoresist thickness	1.140	PRTHICKN
Photoresist thickness for HV Tip Implants	0.300	PrThickImplant
Min width of tip implant opening	0.100	minTip_impW
NTM shadowing	0.160	ntmShadowing
HVNTM shadowing	0.232	hvntmShadowing
HVPTM shadowing	0.089	hvptmShadowing
pseudo-shadowing	0.045	pseudoShadowing
Channel length for low Vt PMOS	0.350	lvtpmos_poly
Width of the Low Leakage gate on each side of LowVt Pmos connected to power rails (requirement based on exp data)	0.280	LvtEnc_forPowerRail
CD tolerance for PDM (3s)	1.000	PdmCD_tol
Min process bias 3s tolerance	0.032	PHTOL
Min process bias 3s tolerance for poly	0.020	PHP1TOL

Minimum Space and Overlap	Value (um)	Variable name
Minimum mcon overlap onto LI for reproducible contact resistance	0.120	TCONOVLP
Dogbone PR decay length (SRS 8/4/99)	0.200	DBPRDEC
Bowing of rectangular contact (per edge) -- seal ring sizing	0.015	TBOWINGSEAL
Waffling / Pattern Density	**Value**	**Variable name**
S8 average FOM PD (extractions from logic device)	45.0%	FOMPDAVG
Size of small PD extraction box for rough tolerance (um)	700	SMALLPDBOX
Size of large PD extraction box for rough tolerance (um)	2000	LARGEPDBOX
Min pattern density for oxide	0.750	OxideMinPD
Min MM* PD range	30.0%	MMPDrange
FOM 700um box PD tolerance for CMP (SOI8 PCR2) for all technologies	15.0%	FOM700TOL
Stepping box shift as a percent of box size	50.0%	BOXSHIFT
Maximum metal waffle drop pattern density in the frame	55.0%	PD_FrameWP
Window size for frame waffle drop PD check	100.00	WP_PDWINDOW
Step size for frame waffle drop PD check	10.00	WP_PDSTEP
Other	**Value**	**Variable name**
Poly resistor width and spacing to reduce CD variation (um)	0.330	POLYRCD
Poly resistor width and spacing to reduce CD variation (um)	0.480	POLYRSPC
Spacing between slotted_licons (Not applicable when the two edges L= 0.19um)	0.510	LICM1SLSP1
Precision resistor width to accommodate 6 contacts across	2.030	PRECRESW
Li resistor width (to drop one Licon w/o dogbones)	0.290	LIRESCD
Correction factor for spacing to a wide metal line	2.000	BIGMF
Min spacing for created dnwell to pnp.dg (more restrictive than dnwell.4 rule)	5.000	cdnwPnpSpc
Min spacing between nwell and deep nwell on separate nets (Taken from dnwell.3 from S4* TDR *N plus rounded up, IGK request.)	6.000	nwellDnwellSpc
Min space between deep nwells used as photo diode (um)	5.000	PDDnwSpc
Min space between dnwell (used for photo diode and other deep nwell (um)	5.300	PDDnwSpc1
Min/Max width of nwell inside deep nwell (for photo diodes)	0.840	PDNwmCD
Min/Max enclosure of nwell by deep nwell (for photo diode)	1.080	PDNwmDnwEnc
Min/Max width of tap inside deep nwell (for photo diode)	0.410	PDTapCD
Min/Max enclosure of tap by nwell inside deep nwell (for photo diode)	0.215	PDTapNwmEnc

5. Laser Fuse Criteria

Min. spac. of laser spot to diffused junction to ensure jct integrity	0.600	XLASJUN
Max. width of a metal fuse line that can be removed reliably	0.800	FSW
Min. L of met. fuse at which damage doesn't extend beyond ends	6.605	FSLE
Max. extension of met2 beyond fuse boundary	0.005	FEXT
Min. distance between laser spot and active junction	0.545	LASJCT
Standard contact bottom CD	0.090	
Positioning tolerance of laser spot (3 s)	0.300	LASMA
Nominal effective laser spot diameter	3.500	LASSPT
Max. increase in spot diameter at max. distance from focus (3 s)	0.900	LASCDTOL
Fuse melting radius	3.600	MELTRAD
Melting related crack size in ILD	0.360	FUSECRACK
Min space between fuse and any feature not connected to it	0.200	MinFuseSpace
Space between fuse and any unrelated layer	0.500	SP_fuse_to_unrelated
DC offset in some fuse rules	0.870	LASDC1

6. Seal Ring Criteria

Parameter	Name in SR spec	Value (um)	
FOM offset	D7+D10+D14	9.180	SRFOMOFFS
FOM width	D4	3.000	SRFOMWIDTH
Licon1 size for seal ring	D2	0.230	licon1srsize
FOM-licon1 enc	D3	1.390	SRFOMLICON1ENC
LI1-licon1 encl	D1	2.420	SRLILICON1ENC
LI1-licon1 encl far side	D16	1.250	SRLILICON1ENCFS
licon1/mcon/via spaces	D8	1.170	SRCTSPACES
li1-mconenc	D5	1.170	SRLI1MCONENC
met1 width	D11	5.990	SRMET1WIDTH
Via size for seal ring	D12	0.280	viasrsize
met2-via enc	D9	5.710	SRMET2VIAENC
met2-via far side enc	D13	2.000	SRMET2VIAF
met2-met1 offset	D14	1.000	M2M1OFF
Moat (passivation opening) width	D15	2.000	MOATW
Pad to Scribe (areaid:seal) min space	per spec 01-70001	20.000	PADSEALSP
Passivation enclosure	D7	5.000	SRPASSENC

7. Other criteria and parameters

Layer / Design rule	CD	space	Comment
MOSFET width	0.135		FOMSE
MOSFET width in standard cells	0.075		FOMSESC
Spacing of poly on field to diff		0.065	PFDSE
Spacing of poly on field to tap		0.005	PFTSE
Enclosure of tap by nwell for pwell res		0.220	PTAP_NWL_SP
Grid conversion rounding factor		0.005	GRCF
Licon enclosure rounding		0.020	LICENCLR
LI1CD add/drop	0.010	0.040	
Huge metal X min. W and L	3.000		HugeM
Min Nsdm area	0.265		MinNsdmArea
Min Psdm area	0.255		MinPsdmArea
Min N/Psdm hole area	0.265		MinNPsdmHole
Large waffle size must be divisible by 4	7.200		waffle_large
P1M additional CD control	0.011		P1MCDcontrol
Li1 proximity correction		0.250	LI1PROXSpace
Serif added to nwell convex corner (SXX-572, 573)	0.220		NwellCvxSerif
Serif added to nwell concave corner (SXX-572, 573)	0.120		NwellCveSerif
NWM extension beyond nwell edge straddling *de_nFet_source* (for GSMC; QZM-133)	0.075		NvhvNwellExt
Min enclosure of pad by pmm for Cu inductor (JNET-80)	0.000		padPMMEncInd
Min enclosure of pmm by cu1m for Cu inductor (JNET-80)	10.750		pmmCu1mEncInd
Min enclosure of pbo by cu1m per DECA 000348 Rev S	10.000		pboCu1mEnc
Min enclosure of pmm by pmm2 for radio flow in the die (JNET-80)	13.000		pmmPmm2EncInd
Min enclosure of pmm by pmm2 inside frame	7.500		pmmPmm2EncIndFrame
Min space between pmm2 and Inductor.dg		7.500	pmm2IndSpc
Min cu1m PD across full chip	35.0%		MinCU1Mpd
Max cu1m PD across full chip	45.0%		MaxCU1Mpd
Spacing between RDL and outer edge of seal ring	15.000		RdlSealSpc
Spacing between RDL and pmm2	6.160		RdlPmm2Spc
Enclosure of etest module in die by cpmm2	0.000		EtestCpmm2Enc
Keepout of active, poly, li and metal to NSM (TCS-2253)		1.000	NSMKeepout
3 um keepout of active, poly, li and metal to areaid.dt/areaid.ft (TCS-2253)		3.000	NSMKeepout_3um
pnp_emitter sizing (S8P GSMC flow)		0.050	PnpEmitterSzGSMC
pnp_emitter sizing (other flows)		0.030	PnpEmitterSz
MiM Capacitor aspect ration	20.000		MiM_AR
Min NCM space to be used to preserve NCM CL algorithm (avoid LVL error)		1.270	NCM_0LVL
Min space of NCM between core and periphery due to existing layout restriction		0.960	NcmCorePeriSP
Multiplication factor		0.010	S8LVconv
Minimum scribe width	50.000		scribew
spacing of p-well outside deep n-well to deep n-well mask edge		0.120	NWDNWENCL
p-well in deep n-well to p-sub		1.200	NWDNWOL
Field oxide etchback after P1ME before implants		0.040	WFDEL

8. Criteria for High Voltage FET

Layer / Design rule	CD	space	Comment
Min HVNwell to any nwell space		2.000	HVNwell_Nwell_SP
Min HVDiff width	0.290		HVDiff_CD
Min HVDiff space		0.300	HVDiff_SP
Min HV Pmos gate width	0.500		HVP_gate_CD
Min space between HV poly		0.280	HVPoly_SP
Min HV Nmos gate width	0.370		HVPoly_CD
HV P+ Diff enclosure by Nwell	0.330		HVPdiff_nwell_enc
HV N+ diff space to Nwell		0.430	HVNdiff_nwell_SP
HV N+ tap enclosure by Nwell	0.330		HVNtap_nwell_enc
HV P+tap space to Nwell		0.430	HVPtap_nwell_SP
Photoresist tilted implant penetration	0.020		HVPrPenetration
Photoresist tilted implant blocking distance	0.013		HVPrBlocking
Min size of HVTip	0.100		HVTipMinSize
Extra CD tol for HVNTM to match Ram7 process	0.015		HVNTMExtraCdTol
Min HVDiff resistor width	0.290		HVDiff_Res_CD
High voltage n+-n+ or p+-p+		0.300	HVDPTS15
HV MOSFET channel length	0.500		HVPCD

9. Criteria for polyimide manufacturability

Layer / Design rule	CD	space	Comment
Enclosure of fuses by polyimide	12.000		PimFuseEnc
Enclosure of bondpad by polyimide (YUY-165)	0.500		PimPadEnc
Enclosure of pad:dg by PBO inside inductor capture pad, with DECA online monitoring	4.500		PBOPadEnc
Enclosure of pad:dg by PBO per standard DECA rules	7.500		PBOPadEncDECA
DECA PBO drawn-to-final process bias per side	0.500		PBOProcBiasPerSide
Polyimide CD tolerance	1.000		PimCD_tol
Min Pim width over pad openings	87.000		PimOverPad_CD
Polyimide slope (001-87400)	5.300		I_polyimide_slope
Enclosure of polyimide by polymer tolerance	7.700		Po_po_tol
Min/Max enclosure of pad.dg inside M5RDL by pmm	0.000		pmmM5RDLpadEnc
Min spacing of pmm to (rdl NOT (pad.dg sized by 0.5))		19.160	pmmRDLspc
Enclosure of laser targets in the die by polyimide	30.000		PimLaserEnc

10. Criteria for VPP capacitor

Layer / Design rule	CD	space	Comment
Min width of capacitor:dg	4.380		VppWidth
Max width of unit capacitor:dg	8.580		VppMaxWidth
Min spacing between two capacitor:dg	1.500		VppSpc
Min spacing of capacitor:dg to li1 or met1 or met2 or nwell	1.500		VppOtherSPc
Min enclosure of capacitor by nwell	1.500		VppNwmEnc
Min spacing of pmm to (rdl NOT (pad.dg sized by 0.5))		19.160	pmmRDLspc

Table C3: Device, LVS and other CAD definitions

Name	Defining algorithm	Used in …
AR_met2_A	Net Area Ratio of met2 not connected to via and of via2 >=0.05 [Equation: (AREA(via2))/(2 * AREA(met2NotConnVia) + PERIMETER(met2NotConnVia) * 0.35)]	Rules
AR_met2_B	Net Area Ratio of *met2GroundOrFloat, v* ia, and via2 <=0.032 [Equation: (AREA(via2))/(2 * AREA(met2GroundOrFloatVia) + PERIMETER(met2GroundOrFloatVia) * 0.35)]	Rules
bondPad	pad:dg OUTSIDE areaid:ft	Rules
bottom_plate	(capm:dg AND met2:dg) sized by capm.3; Exclude all capm sharing same metal2 plate	Rules
Capacitor	Capm enclosing at least one via2	Rules
Chip_extent	Holes (areaid:sl) OR areaid.sl	Rules
Diecut_pmm	areaid.dt NOT (cfom.wp OR cp1m.wp OR cmm1.wp OR cmm2.wp)	Rules
drain_diffusion	(diff NOT poly in nwell or pwell) not abutting tap in the same well or abutting tap in the opposite well	Rules
dummy_capacitor	Capm not overlapping via2	Rules
dummy_poly	poly overlapping text "dummy_poly" (written using text.dg)	Rules
ESD_nwell_tap	n+ tap coincident with nwell such that n+ tap and nwell are completely surrounded by and abutting n+ diff on all edges, within areaid:ed	Rules
fomDmy_keepout_1	(diff.dg OR tap.dg OR poly.dg OR pwell resistor OR pad OR cfom.dg OR cfom.mk OR *PhotoArray* OR cp1m.mk)	Rules
*floating_met**	met*.dg not connected to diffusion or tap through met(*+1) or met(*-1) and their respective vias and contacts	Rules
fom_waffles	fom.mk with dimensions (um x um): 0.5 x 0.5, 1.5 x 1.5, 2.5 x 2.5 and 4.08 x 4.08	Rules
gated_npn	cell name: s8rf_npn_1x1_2p0_HV	Rules
huge_metX	Metal X geometry wider and longer than 3.000um	Rules
hugePad	pad.mk with width > 100um	Rules
iso_pwell	(NOT nwell) AND dnwell	Rules
isolated_tap	tap that does not abut diff	Rules
laser_target	cell *lazX_* and *lazY_* OUTSIDE areaid:ft	Rules
LVnwell	nwell NOT hvi	Rules
LVTN_Gate	Gate overlapping lvtn	Rules
met2GroundOrFloat	met2 connected to ptap or met2 not connected to difftap	Rules
met2GroundOrFloatVia	*met2GroundOrFloat* interacting with via2 >2	Rules
N+_diff	Diff NOT Nwell	Rules
N+_tap	Tap AND Nwell	Rules
nsdmHoles	Hole(nsdm)	Rules
NSM_keepout	nsm.dg OR nsm.mk	Rules
nwell_all	nwell OR extension of cnwm beyond nwell edge straddling *de_nFet_source* by cnwm.3f (45 degree edges are retained for the NVHV device nwell); Rule cnwm.3f applies only to GSMC flows	Rules
P+_diff	Diff AND Nwell	Rules
P+_tap	Tap NOT Nwell	Rules
Pattern_density	(diff_tap area) / PD window (as specified in the rule section)	Rules
photoDiode	deep nwell overlapping areaid.po. Die+frame utility will use the mask data of dnwell in the implementation of this definition	Rules
poly_licon1	Any licon1 that does not overlap ((diff or tap) NOT poly)	Rules
poly_waffles	p1m.mk with dimensions (um x um): 0.48 x 0.48, 0.54 x 0.54 and 0.72 x 0.72	Rules
prec_resistor	rpm AND (poly overlapping poly.rs) AND psdm	Rules
prec_resistor_terminal	*prec_resistor* AND li	Rules
psdmHoles	Hole(psdm)	Rules
pwell	NOT nwell (default substrate area)	Rules
pwres_terminal	P+tap abutting pwell.rs	Rules
pnp_emitter	diff AND pnp.dg AND psdm	Rules
routing_terminal	metX.pin sized inside of metX.drawing by 1/2 * metalX min width; Similar definition applies to Li1 layer	Rules
scribe_line	areaid:ft NOT areaid:dt	Rules
slotted_licon	licon1.dg of size 0.19um x 2.0um	Rules
slotted_licon_edge1	2.0um edge of the *slotted_licon*	Rules
source_diffusion	(diff NOT poly in nwell or pwell) abutting tap in same well	Rules
tap_licon	Tap AND Licon1	Rules
tap_notPoly	tap NOT poly	Rules
top_indmMetal	met3 for S8D*	Rules
top_metal	met3.dg OR mm3.mk (for S8T*/SP8TEE-5R); met3.dg OR indm.mk (for S8D*); met4.dg OR mm4.mk (for SP8Q/S8Q*); met5.dg OR mm5.mk (for SP8P*/S8P*)	Rules
top_padVia	Via2 for S8D*	Rules
top_plate	capm:dg	Rules
Var_channel	poly AND tap AND (nwell NOT hvi) NOT areaid.ce	Rules
VaracTap	Tap overlapping *Var_channel*	Rules
vpp_with_noLi	vpp with cell names: s8rf2_xcmvpp3, s8rf2_xcmvpp4, s8rf2_xcmvpp5, s8rf2_xcmvpp4p4x4p6_m1m2, s8rf2_xcmvpp11p5x11p7_m1m2, s8rf2_xcmvpp11p5x11p7_m1m4,	Rules
vpp_with_Met3Shield	vpp with cell names: s8rf_xcmvpp1p8x1p8_m3shield, s8rf_xcmvpp4p4x4p6_m3shield, s8rf_xcmvpp8p6x7p9_m3shield, s8rf_xcmvpp11p5x11p7_m3shield	Rules
vpp_with_LiShield	vpp with cell names: s8rf*_li*shield	Rules
vpp_over_MOSCAP	vpp with cell names: s8rf_xcmvpp2 when over nhvnative W/L=10x4, s8rf_xcmvpp2_nwell when over phv/pshort/phighvt/plowvt W/L=5x4	Rules
vpp_with_Met5PolyShield	vpp with cell names: s8rf2_xcmvpp11p5x11p7_polym5shield, s8rf2_xcmvpp11p5x11p7_polym4shield, s8rf2_xcmvpp2x4_2xnhvnative10x4, s8rf2_xcmvpp6p8x6p1_polym4shield, s8rf2_xcmvpp11p5x11p7_polym50p4shield, s8rf2_xcmvpp_hd5_*	Rules

vpp_with_Met5	vpp with cell names: s8rf2_xcmvpp11p5x11p7_m5shield, s8rf2_xcmvpp11p5x11p7_polym5shield, s8rf2_xcmvpp11p5x11p7_lim5shield, s8rf2_xcmvpp8p6x7p9_m3_lim5shield, s8rf2_xcmvpp2x4_2xnhvnative10x4, s8rf2_xcmvpp11p5x11p7_m4shield, s8rf2_xcmvpp11p5x11p7_polym4shield, s8rf2_xcmvpp6p8x6p1_polym4shield, s8rf2_xcmvpp6p8x6p1_lim4shield, s8rf2_xcmvpp11p5x11p7_m3_lim5shield, s8rf2_xcmvpp4p4x4p6_m3_lim5shield, s8rf2_xcmvpp11p5x11p7_m1m4m5shield, s8rf2_xcmvpp11p5x11p7_m1m4 s8rf2_xcmvpp11p5x11p7_polym50p4shield, s8rf2_xcmvpp_hd5_*	Rules
cp1m_HV	cp1m AND Hvi	Rules (HV)
de_nFet_drain	((isolated tap) AND areaid.en) overlapping nwell	Rules (HV)
de_nFET_gate	*deFET_gate* overlapping (diff NOT dnwell)	Rules (HV)
de_nFet_source	(diff AND areaid.en) overlapping *de_nFET_gate*	Rules (HV)
de_pFet_drain	((isolated tap) AND areaid.en) not overlapping nwell	Rules (HV)
de_pFET_gate	*deFET_gate* overlapping (diff AND dnwell)	Rules (HV)
de_pFet_source	(diff AND areaid.en) overlapping *de_pFET_gate*	Rules (HV)
deFET_gate	(poly AND areaid.en) not overlapping pwm ; For CAD flows that do not have pwm layer, it is (poly AND areaid.en)	Rules (HV)
Hdiff	Diffusion AND Hvi	Rules (HV)
Hgate	Hpoly AND diff	Rules (HV)
Hnwell	Nwell AND Hvi	Rules (HV)
Hpoly	Poly AND Hvi	Rules (HV)
Htap	Tap AND Hvi	Rules (HV)
hv_source/drain	= (diff andNot poly) that overlaps diff.hv	Rules (HV)
hvFET_gate	= FET_gate butting *hv_source/drain*	Rules (HV)
hvPoly	= poly electrically connected to *hv_source/drain*	Rules (HV)
HV_nwell	(nwell AND hvi) OR (nwell overlapping areaid.hl)	Rules (HV)
stack_hv_lv_diff	(diff And Hvi NOT nwell) abutting (diff NOT nwell)	Rules (HV)
SHVdiff	Diff And shvi	Rules (SHV)
SHVGate	SHVPoly AND diff	Rules (SHV)
SHVPoly	Poly OVERLAP shvi:dg	Rules (SHV)
SHVSourceDrain	Diff And shvi NOT poly NOT diff:rs	Rules (SHV)
VHVdiff	Diff And vhvi	Rules (VHV)
VHVGate	VHVPoly AND diff	Rules (VHV)
VHVPoly	Poly OVERLAP vhvi:dg	Rules (VHV)
VHVSourceDrain	(Diff AND tap) And vhvi NOT poly NOT diff:rs	Rules (VHV)
background	Area where waffling grid is defined, sized to avoid waffle shift between runs	Waffles
die	Holes (areaid:sl)	Waffles
frame	(areaid.ft SIZE by -(max of s.2e/h)) NOT (OR areaid.dt SEALIDandHole)	Waffles
inductor_metal	(inductor:dg AND (met1 OR met2 OR met3)) size by 10 um [For all flows except S8PIR-10R]	Waffles
*mm*_slot*	mm* slots are defined as empty holes in metal that are located in (areaid.cr OR areaid.cd)	Waffles
nwellDnwellHoles	(inner HOLES of nwellAndDnwell). Die+frame utility will use the mask data of nwell and dnwell in the implementation of this definition	Waffles
photoArray	(OR nwellAndDnwell nwellDnwellHoles) enclosing *photoDiode* . Die+frame utility will use the mask data of nwell and dnwell in the implementation of this definition	Waffles
gate	poly AND diff	pfet, nfet (LVS)
nfet	Gate NOT nwell	pfet, nfet (LVS)
pfet	Gate AND nwell	pfet, nfet (LVS)
nDiode	Ndiff AND DiodeID	Diodes (LVS)
Pdiff	diff AND nwell	Diodes (LVS)
pDiode	Pdiff AND DiodeID	Diodes (LVS)
diff_hole	Hole(diff)	ESD (LVS)
diff_tap_nwell	tap_nwell INSIDE diff_hole	ESD (LVS)
esd_diff_tap_nwell	ESDID AND diff_tap_nwell	ESD (LVS)
Ndiff	diff NOT nwell	ESD (LVS)
tap_nwell	tap INSIDE nwell	ESD (LVS)
ESD_diffusion	A+B31ny diffusion or ESD_nwell_tap connected directly or through a resistor to a Pad or to Vss/Vcc that is covered by areaid.ed and located within a double tap guardrings.	Latch up rules
ESD_cascode_diffusion	Diffusion covered by areaid.ed between two minimum spaced poly gates and located within a pair of double tap guardrings. (There should be no licons on the diffusion.)	Latch up rules
ESD_diode	Any nwell (other than ESD_nwell_tap) covered by areaid.ed and areaid.de that does not contain poly	Latch up rules
ESD_FET	(any Pdiff covered by areaid:ed within a double tap guardrings) Or *ESD_NFET*	Latch up rules
ESD_NFET	(any Ndiff covered by areaid:ed abutting ESD_nwell_tap) Or (any Ndiff covered by areaid:ed abutting gate within 3.5um of ESD_nwell_tap) Or (any Ndiff abutting ESD_nwell_tap within areaid.ed) a double tap	Latch up rules
I/O_or_Output_Pmos	ESD P+ diffusion overlapping poly and overlapping ESD source/drain diffusion connected to I/O or output net	Latch up rules
I/O_Pmos_w/series_R	ESD PMOS connected to I/O or output net through series resistors	Latch up rules
met_ESD_resistor	Metal resistor inside areaid:ed	Latch up rules
Non_Vcc_nwell	Any nwell connected to any bias other than power supply	Latch up rules
Nwell_area	Is determined using the following steps: (a) Grow pdiff by 1.5 mm (b) Merge (c) And Nwell:dg	Latch up rules

Pwell_area	Is determined using the following steps: (a) Grow ndiff by 1.5 mm (b) Merge (c) NOT Nwell:dg	Latch up rules
Series_transistors	Merged diffusion determined by Nwell_area and Pwell_area	Latch up rules
fuse:dg	met2:fe for S8D*/S8TM*, met3.fe for S8TEE*/S8TNV/S8Q*/SP8TEE-5R/SP8Q*, met4.fe for S8P*/SP8P*	Fuse rules
fuse_contact	(*fuse_metal* overlapping *fuse:dg*) NOT *fuse:dg*	Fuse rules
fuse_metal	met3 for S8TEE*/S8TNV/S8Q*/SP8TEE-5R/SP8Q*; met2 for S8D*/S8TM*, met4 for S8P*/SP8P*	Fuse rules
fuse_shield	Metal line (same metal level as fuse) between fuse and periphery, not overlapping contacts or vias, with specified dimensions	Fuse rules
non-isolated fuse edge	Long edge of the fuse spaced to Met2/Met3/Met4 less than a specified amount	Fuse rules
single_fuses	Fuses without neighboring fuses within specified distance	Fuse rules

Table C4: Purpose layer description in LSW window and Auxiliary Layers

Layer purpose	icfb ver 5.0	icfb ver 5.1
drawing	dg	drw
pin	pn	pin
boundary	by	bnd
net	nt	net
res	rs	res
label	ll	lbl
cut	ct	cut
short	st	sho
pin	pn	pin
gate	ge	gat
probe	pe	pro
blockage	be	blo
model	ml	mod
optionX (X = 1…n)	oX (X = 1..n)	opt*(X=1..n)
fuse	fe	fus
mask	mk	mas*
maskAdd	md	mas*
maskDrop	mp	mas*
waffleAdd1	w1	waffleAdd1
waffleAdd2	w2	waffleAdd2
waffleDrop	wp	waf
error	er	err
warning	wg	wng
dummy	dy	dmy

Layout Data Name & GDSII No.	Brief description	icfb ver 5.1	Identifies (See WOLF-41, SPR 95111 for more details)	Who	Use
areaid.sl{81:1}	areaid sealring	areaid.sea	The area of the Seal ring	Tech	
areaid.ww{81:13}	areaid Waffle Window	areaid.waf	Used to prevent waffle shifting. When larger than areaid:sl re-defines the placement of waffles.	Frame	CLDRC
areaid.dn{81:50}	areaid dead Zon	areaid.dea	“deadzone” area in the DieSealR pcell (Seal Ring) for metal stress relief rule checks	Tech	
areaid.cr{81:51}	areaid critCorner	areaid.cri*	For portions of layout that are not to be put in the critical side due to stress constraints. Should be used sparingly and only over the portion of the layout to remove DRC violations. Avoid using a blanket polygon over the entire layout. This layer is to be used instead of using the noCritSideReg verification option in Stress. “critical corner” area in the DieSealR pcell (Seal Ring) for metal stress relief rule checks	Tech	Stress
areaid.cd{81:52}	areaid critSid	areaid.cri*	“criticalsid” area in the DieSealR pcell (Seal Ring) for metal stress relief rule checks	Tech	Stress
areaid.ce{81:2}	areaid core	areaid.cor	Memory core (memory cells and approved on-pitch only)	Tech	DRC
areaid.fe{81:3}	areaid frame	areaid.fra*	Pads in the frame	Frame	DRC
areaid.ed{81:19}	areaid ESD	areaid.esd	ESD devices- Surrounds any diffusion or ESD nwell tap connected to a signal pad. (only over ESD devices with special poly/tap exemption rules per LFL)	ESD, Des	DRC
areaid.dt{81:11}	areaid die cut	areaid.die	Location of the die within the frame used in frame builder generation to create blanking for die and other drop-ins. Also used in cldrc/drc for rules in frame to die edge (waffles, nsm, metals etc)	Frame	Tech
areaid.mt{81:10}	areaid module cut	areaid.mod	Location of e-test modules within the frame used in frame builder generation to create data in scribe lane(example: opaque/clear masks) and to mark location of cells (etest and fab)for frame reports. Also used in drc/cldrc for rules to cell edge.	Frame	Tech

areaid.ft{81:12}	areaid frameRect	areaid.fra*	Boundary of the frame used in frame builder generation to mark boundary of frame. Also used in cldrc/drc for rules to frame edge	Frame	DRC/CLDRC
areaid.de{81:23}	areaid Diode	areaid.dio	The area occupied by diodes; Used to identify diodes during LVS	All	LVS
areaid.sc{81:4}	areaid standardc	areaid.sta	Cells in the standard cell library (over standard cell IP blocks only) .	Standard cell	DRC
areaid.st{81:53}	areaid SubstrateCut	areaid.sub	Regions to be considered as isolated substrates (only to designate 2 different resistively connected substrate regions, >100um apart)	Tech, Des, ESD	Latch up, LVS, soft
areaid.en{81:57}	areaid extended drain	areaid.ext	Used to identify the extended drain devices	Tech, Des, ESD	LVS
areaid.le{81:60}	areaid LV Native	areaid.lvn	Used to identify the 3V Native NMOS versus 5V Native NMOS	Tech, Des	LVS
areaid.po{81:81}	areaid photo	areaid.pho	The areaid id is to identify the dnwell photo diode	Tech, Des	DRC
areaid.et{81:101}	areaid etest	areaid.ete	Used in etest modules	Frame	DRC
areaid.ld{81:14}	areaid low tap density	areaid.low	6um tap to diff rule will not be checked in this region Diffusion >6u from related tap, requiring >50u from sigPadDiff && sigPadMetNtr) Should be used sparingly and only over the portion of the layout to remove DRC violations. This layer is not to be used if a tapping solution can be found. This layer can only be used if there is low risk for latchup. This layer will be reviewed during PDQC.	All	DRC
areaid.ns{81:15)	areaid not-crtical side	areaid .not	critSideReg stress rules will not be checked in this region Cannot be placed in the critical side – uncommon, or where stress errors can't be fixed)	All	DRC
areaid.ij{81:17}	areaid injection	areaid.inj	Identify all circuits that are susceptible to injection and ensure no signal-pad connected diffusion is within 100u. “areaid.inj” encloses any circuitry deemed sensitive (by design team) to injected substrate areaid.inj encloses any PVT compliant circuitry	All	DRC
areaid.hl{81:63}	areaid.hvnwell	areaid.hvn	Identify nwell hooked to HV but containing FETs with thin oxide; Potential difference across the FET terminals is LV Used over lv devices, operating in lv mode, placed in hv nwells, and should NOT have hvi	All	DRC
areaid.re{81:125}	areaid rf diode	areaid.rfd	Defines rf diodes that need to be extracted with series resistance (memo GCZ-124/125)	All	LVS
areaid.rd{81:24}	areaid.rdlprobepad	areaid.rdl	Ignore RDL keepouts when opening up PMM2	All	CLDRC
areaid.sf{81:6}	areaid sigPadDiff		Identify all srdrn diffusions and tap which are intended to be connected to signal pad (io Nets) Goes over diffusions connected to a signal pad - including through a poly resistor	All	LATCHUP
areaid.sl{81:7}	areaid.sigPadWell		Identify all nwells and pwells which are intended to be connected to signal pad (io Nets) Goes over wells with tap connected to a signal pad, including through a poly resistor	All	LATCHUP
areaid.sr{81:8}	areaid sigPadMetNtr		Identify all srcdrn, tap, and wells which are intended to be metallically connected to signal pad (io Nets) not through a resistor. Must be used in unison with areaid.sigPadDifff or areaid.sigPadWell. Used with one of the above 2 areaids, nodes metallically connection to a sigPad (not through res)	All	LATCHUP
areaid.low_vt	ID layer for low_vt		Identify "low_vt" regions of >20V devices for proper diode extractions	Tech, Des	LVS
inductor:dg{82:24}	ID layer for inductor		Inductors	Tech, Des	DRC
t1,2,3 {82:26, 27, 28}	terminal labels for inductor		Labels required for inductor terminals recognition as device	Tech, Des	LVS
poly:ml {66:83}	poly device model		Model name extraction	Tech, Des, ESD	LVS
ncm {92:44}	N-Core Implant		Ncm.dg is available as a drawn layer	All	DRC/CLDRC
protect)	VPP capacitor		Interdigitated, vertical Li1, M1 and M2 capacitor	All	LVS
capm_2t.dg	MIM caps (2 terminal model)		ID layer for MIMCAP that will be treated as 2T device	All	DRC/LVS
cpmm:dg{91}	Drawn compatible polyimide layer		Drawn compatible layer and used only inside S8 RF pad	Frame	
li1.be{67:10}	li1 blockage layer		Li1 blockage layer used for IP integration (per CWR 137)	All	DRC
met1.be{68:10}	Metal1 blockage layer		Metal 1 blockage layer used for IP integration and P&R	All	DRC

met2.be{69:10}	Metal2 blockage layer		Metal 2 blockage layer used for IP integration and P&R	All	DRC
met3.be{70:10}	Metal3 blockage layer		Metal 3 blockage layer used for IP integration and P&R	All	DRC
met4.be{71:10}	Metal4 blockage layer		Metal 4 blockage layer used for IP integration and P&R	All	DRC
met5.be{72:10}	Metal5 blockage layer		Metal 5 blockage layer used for IP integration and P&R	All	DRC
vhvi {74:21}	Very High voltage id layer		Used to identify nodes that operate at 12V nominal (16V max)	Des	VHV Rules
uhvi {74:22}	Ultra High voltage id layer		Used to identify nodes that operate at 20V nominal	Des	UHV Rules
areaid.e0{81:58}	Area extended drain	areaid.ext	Used to identify 20V drain extended devices	Des	LVS
areaid.zr{81:18}	Area zener diode	areaid.zen	Used to identify Zener diodes	Des	LVS
fom.dy{}	FOM dummy		FOM waffle drawn in this layer	All	Waffles
prune:dg{84:44}	prune		Areas ignored by LVS	Frame	LVS
areaid:cr {81:55}	copper pillar (.cuPillar)	areaid.cup	Placement of Cu pillar over the pad area, streamed out to Amkor, s8pfhd-10r flow only	Die	CLDRC s8pfhd-10r
cyprotect.dg {56.44}	External F25 layer	cyprotect.dg	Switch to direct streaming to drawn (no protect) or mask layer (with protect)	Frame	CLDRC
cytextmc.dg {50:44}	Locations for mask compose	cytextmc.dg	Text to extract placement for Fab25 tool	Frame	CLDRC
cypsbr.dg {51:44}	No phaseshift allowed	cypsbr.dg	Phaseshift layer common to all F25 phaseshift masks	Frame	
areaid:ag{81:79}	analog	areaid.ana	Used to identify analog circuits	All	Analog
natfet.dg {124:21}	DEFETs	natfet.dg	Add TUNM for SONOS channel implants. See SPR 117559, SGL-529	All	DRC/CLDRC
areaid:lw	Ultra High voltage id layer		Areaid low voltage: UHV box to put all HV/LV circuits in	All	Analog

* To distinguish the layers, the full name of the layer needs to be turned on in the LSW window
As the layers are displayed in LSW window in icfb version 5.0; For purpose layer displayed in version 5.1, pls refer table C3

Table F2a: Devices and Layout vs. Schematic (LVS):

Refer to Opus Library Elements Documentation Manual in the CAD online documentation

Category	Name	diff.dg	diff.rs	diff.ct	cfom.wp	tap.dg	dnwell:dg	pwbm.dg	pwdem:dg	hvtr.dg	nwell.dg	hvtp:dg	lvtn:dg	pwell.rs	pwell.ct	ncm.dg (2)	tunm:dg	hvi:dg	rpm:dg	urpm:dg	poly.dg	poly.rs	poly.ct	poly:ml	ldntm:dg	npc:dg	nsdm.dg	psdm.dg	licon.dg	li1.dg	li.rs	li.ct	capm:dg	capm_2t.dg	cap2m:dg	metX.dg	metX.fe	met1:dg	met2:dg	met3:dg	met4.dg	met5.dg	rdl.dg	inductor:dg	capacitor.dg	areaid.le	areaid.en	pnp.dg	npn.dg	areaid.st	areaid.de	areaid.re	areaid.po	areaid:ce	areaid.ed	areaid.ext	UHVI	areaid.low_vt
RESISTOR	n diff resistor	D	D	D	+	-	+	-	+	-	-	-	-	+	+	-	+	-	+	+	-	-	-	+	+	+	D	-	+	+	+	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	+		-	-
	HV n diff resistor	D	D	D	+	-	+	-	+	-	-	-	-	+	+	-	+	D	+	+	-	-	-	+	+	+	D	-	+	+	+	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	+		-	-
	p diff resistor	D	D	D	+	-	+	-	+	-	D	-	-	+	+	C	+	-	+	+	-	-	-	+	+	+	-	D	+	+	+	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	+		-	-
	p diff resistor NV	D	D	D	+	-	+	-	+	-	D	-	-	+	+	C	+	-	+	+	-	-	-	+	+	+	-	D	+	+	+	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	+		-	-
	HV p diff resistor	D	D	D	+	-	+	-	+	-	D	-	-	+	+	-	+	D	+	+	-	-	-	+	+	+	-	D	+	+	+	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	+		-	-
	Isolated Pwell resistor	+	+	+	+	D	D	-	-	-	-	+	+	D	D	+	+	+	+	+	-	-	-	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	-	-	-	+	+	+	+	+	-	-	+		-	-
	n+ poly resistor	-	-	-	+	-	+	-	+	-	+	-	-	+	+	-	+	+	-	-	D	D	D	+	+	-	+	+	+	+	+	-	+	+	+	+	-	+	+	+	+	+	+	+	-[20]	-	-	-	-	+	-	+	-	-	+		-	-
	p- poly resistor	-	-	-	+	-	+	-	+	-	+	-	-	+	+	+	+	+		D	D	D	D	+	+	D	-	D	+	+	+	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	+		-	-
	p+ poly resistor	-	-	-	+	-	+	-	+	-	+	-	-	+	+	+	+	+	D		D	D	D	+	+	D	-	D	+	+	+	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	+		-	-
	li resistor	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	D	D	D	+	+	+	+	+	+	+	+	+	+	+	+	-	-	-	+	+	+	+	+	-	-	+		-	-
	metal fuse	-	-	-	+	-	+	-	+	-	-	-	-	+	+	-		-			-	-	-	+	+	+	+	+	-	-	-	-	+	+	+	D	D	+	+	+	+	+	+	-	-	-	-	-	-	+	-	+	-	-	-		-	-
32 A CMOS	nmos 1.8V	D	-	-	+	-	+	-	-	-	-	-	-	+	+	-	-	-	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	+	+		-	-
	pmos 1.8V	D	-	-	+	-	+	-	-	-	D	C	-	+	+	C	-	-	-		D	-	-	+	-	-	-	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	+	-	-	-	-	+	-	+	-	-	+		-	-
	Low Vt pmos 1.8V 32A	D	-	-	+	-	+	-	-	-	D	-	D	+	+	-	-	-	-		D	-	-	+	-	-	-	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	+	-	-	-	-	+	-	+	-	-	-		-	-
	High Vt pmos 1.8V 32A (10)	D	-	-	+	-	+	-	-	-	D	D	C	+	+	+	-	-	-		D	-	-	+	-	-	-	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	+	-	-	-	-	+	-	+	-	+	-		-	-
	Low Vt nmos 1.8V 32A	D	-	-	+	-	+	-	-	-	-	-	D	+	+	-	-	-	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	-		-	-
	nmos_core (5)	D	-	-	+	-	+	-	-	-	-	-	-	+	+	-	-	-	-		D	-	-	D	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	D	-		-	-
	nmos_core NV (5)	D	-	-	+	-	+	-	-	-	-	-	-	+	+	D	-	-	-		D	-	-	D	D	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	D	-		-	-
	pmos_core [5]	D	-	-	+	-	+	-	-	-	D	D	C	+	+	D	-	-	-		D	-	-	D	-	-	-	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	D	-		-	-
	pmos_core NV [5]	D	-	-	+	-	+	-	-	-	D	D	C	+	+	-	-	-	-		D	-	-	D	-	-	-	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	D	-		-	-
	Low Vt nmos_core [5]	D	-	-	+	-	+	-	-	-	-	-	D	+	+	-	-	-	-		D	-	-	D	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	D	-		-	-
	Low Vt Varactor	-	-	-	+	D	+	-	-	-	D	-	C	+	+	-	-	-	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	-		-	-
	High Vt Varactor	-	-	-	+	D	+	-	-	-	D	D	C	+	+	C	-	-	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	-		-	-
	HV varactor (floating gate)	-	-	-	+	D	+	-	-	-	D	-	D	+	+	-	-	D	-		D	-	-	+	-	-	D	-	-	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	-		-	-
SONOS (& SONOS Latch)	SONOS fet [11]	D	-	-	+	-	D	-	-	-	-	-	D	+	+	-	D	-	-		D	-	-	+	D	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	D	-		-	-
	SONOS fet [11]	D	-	-	+	-	D	-	-	-	-	-	D	+	+	-	D	-	-		D	-	-	+	D	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	D	-		-	-
	NV SONOS fet [11]	D	-	-	+	-	+	-	-	-	-	-	D	+	+	-	D	-	-		D	-	-	D	D	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	D	-		-	-
110A CMOS	HV nmos 5/10.5V	D	-	-	+	-	+	-	-	-	-	-	-	+	+	-	-	D	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	+		-	-
	HV nmos 5/10.5V	D	-	-	+	-	+	-	-	-	-	-	-	+	+	-	-	D	-		D	-	-	D	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	+		-	-
	HV pmos 5/10.5 V	D	-	-	+	-	+	-	-	-	D	-	-	+	+	-	-	D	-		D	-	-	+	-	-	-	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	+	-	-	-	-	+	-	+	-	-	+		-	-
	HV pmos 5/10.5 V	D	-	-	+	-	+	-	-	-	D	-	-	+	+	-	-	D	-		D	-	-	D	-	-	-	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	+		-	-
	Native nmos 5V	D	-	-	+	-	+	-	-	-	-	-	D	+	+	-	-	D	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	+	-	-	-	-	+	-	+	-	-	-		-	-
	Native nmos 3V	D	-	-	+	-	+	-	-	-	-	-	D	+	+	-	-	D	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	D	-	-	-	+	-	+	-	-	-		-	-
	Flash npass [5]	D	-	-	+	-	+	-	-	-	-	-	-	+	+	-	-	D	-		D	-	-	D	D	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	D	-		-	-
	Flash npass NV	D	-	-	+	-	+	-	-	-	-	-	-	+	+	-	-	D	-		D	-	-	D	D	-	D(3	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	D	-		-	-
	VHV nmos 5/16V DE	D (7)	-	-	+	D (6)	-	-	-	-	D	-	-	+	+	-	-	D	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	D	-	-	+	-	+	-	-	-		-	-
	VHV pmos 5/16V DE	D (7)	-	-	+	D (6)	D	-	-	-	D (3	-	-	+	+	-	-	D	-		D	-	-	+	-	-	-	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	D	-	-	+	-	+	-	-	-		-	-
	UHV nmos 5/20V DE	D (7)	-	-	+	D (6)	D(3	D(3	-	-	D(3	-	D(3	+	+	-	-	D	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+				+	-	-	-	-	-		-	+	-	-	-	D	D	D
	UHV iso nmos 5/20V DE	D (7)	-	-	+	D (6)	D(3	D(3	-	-	D(3	-	D(3	+	+	-	-	D	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+				+	-	-	-	-	-		-	+	-	-	-	D	D	D
	UHV Native nmos 5/20V DE	D (7)	-	-	+	D (6)	D(3	D(3	-	-	D(3	-	D	+	+	-	-	D	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+				+	-	-	-	-	-		-	+	-	-	-	D	D	D
	UHV Zvt nmos 5/20V DE	D (7)	-	-	+	D (6)	D(3	D	-	-	D(3	-	D	+	+	-	-	D	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+				+	-	-	-	-	-		-	+	-	-	-	D	D	D
	UHV pmos 5/20V DE	D (7)	-	-	+	D (6)	D	D(3	D(3	-	D(3	-	D(3	+	+	-	-	D	-		D	-	-	+	-	-	-	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-		-	-		-	+	-	-	-	D	D	D

Table F2a: Devices and Layout vs. Schematic (LVS):

Refer to Opus Library Elements Documentation Manual in the CAD online documentation

Category	Name	diff.dg	diff.rs	diff.ct	cfom.wp	tap.dg	dnwell:dg	pwbm.dg	pwdem:dg	hvtr.dg	nwell.dg	hvtp:dg	lvtn:dg	pwell.rs	pwell.ct	ncm.dg (2)	tunm:dg	hvi:dg	rpm:dg	urpm:dg	poly.dg	poly.rs	poly.ct	poly:ml	ldntm:dg	npc:dg	nsdm.dg	psdm.dg	licon.dg	li1.dg	li.rs	li.ct	capm:dg	capm_2t.dg	cap2m:dg	metX.dg	metX.fe	met1:dg	met2:dg	met3:dg	met4.dg	met5.dg	rdl.dg	inductor:dg	capacitor.dg	areaid.le	areaid.en	pnp.dg	npn.dg	areaid.st	areaid.de	areaid.re	areaid.po	areaid:ce	areaid.ed	areaid.ext	UHVI	areaid.low_vt
CAPACITOR	MiM (3 Terminal) (s8t*)	-	+	+	+	-	+	-	+	-	+	+	+	+	+	+	+	+	+		-	+	+	+	+	+	+	+	+	-	+	+	D	-	D	+	-	-	D	+	+	+	+	-	-	-	-	+	+	+	+	+	-	-	+		-	-
	MiM (2 Terminal) (s8t*)	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	D	D	D	+	-	+	D	+	+	+	+	-	-	-	-	+	+	+	+	+	-	-	+		-	-
	MiM (2 Terminal) (s8p*)	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	D	D	+	+	-	+	+	D	+	+	+	-	-	-	-	+	+	+	+	+	-	-	+		-	-
	MiM2 (2 Terminal) (s8p*)	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	D	D	+	-	+	+	+	D	+	+	-	-	-	-	+	+	+	+	+	-	-	+		-	-
	VPP (9)	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	-	D	+	+	-	+	-	+	-	D	D	+	+	+	+	+	D	-	-	+	+	+	+	+	-	-	+		-	-
	VPP over NHVNATIVE	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	D	+	+	-	+	-	+	-	D	D	+	+	+	+	+	D	-	-	+	+	+	+	+	-	-	+		-	-
	VPP over PHV	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	D	+	+	-	+	-	+	-	D	D	+	+	+	+	+	D	-	-	+	+	+	+	+	-	-	+		-	-
	VPP and NHVNATIVE	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	D	+	+	-	+	-	+	-	D	D	D	+	+	+	+	D	-	-	+	+	+	+	+	-	-	+		-	-
	4-Terminal VPP (with Met3 shield) [12]	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	D	+	+	-	+	-	+	-	D	D	D	+	+	+	+	D	-	-	+	+	+	+	+	-	-	+		-	-
	4-terminal VPP (with M5 shield) [12]	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	D	+	+	-	+	-	+	-	D	D	D	D	D	+	+	D	-	-	+	+	+	+	+	-	-	+		-	-
	4-terminal VPP (with M4 shield) [12]	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	D	+	+	-	+	-	+	-	D	D	D	D	+	+	+	D	-	-	+	+	+	+	+	-	-	+		-	-
	3-Terminal VPP [12]	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	D	+	+	-	+	-	+	-	D	D	+	+	+	+	+	D	-	-	+	+	+	+	+	-	-	+		-	-
	3-Terminal VPP [12] (for S8Q/S8P ONLY)	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	D	+	+	-	+	-	+	-	D	D	D	+	+	+	+	D	-	-	+	+	+	+	+	-	-	+		-	-
	3-Terminal VPP [12] (for S8P ONLY)	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	D	+	+	-	+	-	+	-	D	D	D	D	+	+	+	D	-	-	+	+	+	+	+	-	-	+		-	-
INDUCTOR	Inductor	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	D	+	+	+	D	-	-	-	+	+	+	+	+	-	-	+		-	-
	Cu Inductor	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	D	+	D [16]	D [16]	D	D [16]	D	-	-	-	-	-	-	-	-	-	-	-		-	-
	Balun Inductor [15]	+	+	+	+	+	+	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	D	D	D	D	-	-	-	-	-	-	-	-	-	-	-		-	-

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DIODE	nDiode	D	-	-	+	-	+	-	+	-	-	-	-	+	+	-	+	-	+		-	-	-	+	+	+	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	D	+	-	+	+		-	-
	HV nDiode	D	-	-	+	-	+	-	+	-	-	-	-	+	+	-	+	D	+		-	-	-	+	+	+	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	D	+	-	-	+		-	-
	RF ESD HV nDiode [13]	D	-	-	+	-	-	-	+	-	-	-	-	+	+	-	+	D	+		-	-	-	+	+	+	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	D	+	-	-	D		-	-
	RF ESD HV Deep Nwell nDiode [13]	D	-	-	+	-	D	-	+	-	-	-	-	+	+	-	+	D	+		-	-	-	+	+	+	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	D	+	-	-	D		-	-
	pDiode	D	-	-	+	-	+	-	+	-	D	-	-	+	+	C	+	-	+		-	-	-	+	+	+	-	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	D	+	-		+		-	-
	pDiode NV	D	-	-	+	-	+	-	+	-	D	-	-	+	+	C	+	-	+		-	-	-	+	+	+	-	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	D	+	-		+		-	-
	HV pDiode	D	-	-	+	-	+	-	+	-	D	-	-	+	+	-	+	D	+		-	-	-	+	+	+	-	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	D	+	-	-	+		-	-
	RF ESD HV pDiode [13]	D	-	-	+	-	+	-	+	-	D	-	-	+	+	-	+	D	+		-	-	-	+	+	+	-	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	D	+	-	-	D		-	-
	Photo Diode	-	-	-	+	D	D	-	-	-	D	-	-	-	-	-	-	-	-		-	-	-	-	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	D	+	D	-	-		-	-
	Low Vt pdiode [8]	D	+	+	+	+	+	-	+	-	D	-	D	+	+	+	+	+	+		+	+	+	+	+	+	+	D	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	+	+		-	-
	High Vt pDiode [8]	D	+	+	+	+	+	-	+	-	D	D	C	+	+	+	+	-	+		+	+	+	+	+	+	+	D	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	+	+		-	-
	High Vt pDiode NV [8]	D	+	+	+	+	+	-	+	-	D	D	C	+	+	C	+	-	+		+	+	+	+	+	+	+	D	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	+	+		-	-
	Low Vt nDiode [8]	D	+	+	+	+	+	-	+	-	+	+	D	+	+	+	+	+	+		+	+	+	+	+	+	D	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	+	+		-	-
	NV SONOS Diode [8]	D	+	+	+	+	+	-	+	-	+	+	D	+	+	+	D	+	+		+	+	+	+	D	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	D	+		-	-
	Native nDiode [8]	D	+	+	+	+	+	-	+	-	+	+	D	+	+	+	+	D	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	+	+		-	-
	Nwell Diode [8]	+	+	+	+	+	-	-	+	-	D	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	-	+	+		-	-
	Nwdiode_victim [17]	+	+	+	+	+	-	-	+	-	D	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	-	+	+		-	-
	Nwdiode_aggressor [17]	+	+	+	+	+	-	-	+	-	D	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	-	+	+		-	-
	RF Nwell Diode [8]	+	+	+	+	+	-	-	+	-	D	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	D	-	+	+		-	-
	Pwell-Deep Nwell Diode [8]	+	+	+	+	+	D	-	+	-	-	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	-	+	+		-	-
	RF Pwell-Deep Nwell Diode [8]	+	+	+	+	+	D	-	+	-	-	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	D	-	+	+		-	-
	Psub-Deep Nwell Diode [8]	+	+	+	+	+	D	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	+	+		-	-
	Psub-Deep Nwell Diode [17]	+	+	+	+	+	D	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	+	+		-	-
	Psub-Deep Nwell Diode [17]	+	+	+	+	+	D	-	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	+	+		-	-
	Psub-Deep Nwell Diode [8,19]	+	+	+	+	+	D	D	+	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	+	+		D (18)	D (18)
	HV Pwell-Deep Nwell Diode [8, 19]	+	+	+	+	+	D	-	D	-	+	+	+	+	+	+	+	+	+		+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	+	+		-	-
PNP	Parasitic PNP	D	-	-	+	D	-	-	-	-	D	+	-	+	+	-	+	-	+		-	-	-	+	+	+	D	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	D	-	+	-	+	-	-	-		-	-
	Parasitic HV Gated NPN	D	-	-	+	+	D	-	-	-	D	+	-	-	-	-	-	+	D	+		D	-	-	+	+	+	D	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	D	+	-	+	-	-	-		-	-
	Parasitic NPN	D	-	-	+	+	D	-	-	-	D	+	-	-	-	-	+	-	+		-	-	-	+	+	+	D	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	D	+	-	+	-	-	-		-	-
ESD transistor	LV nESD transistor	D	-	-	+	-	+	-	-	-	-	-	-	+	+	-	-	-	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	D		-	-
	HV nESD transistor	D	-	-	+	-	+	-	-	-	-	-	-	+	+	-	-	D	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	D		-	-
	HV Native nESD transistor	D	-	-	+	-	+	-	-	-	-	-	D	+	+	-	-	D	-		D	-	-	+	-	-	D	-	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	D		-	-
	HV pESD transistor	D	-	-	+	-	+	-	-	-	D	-	-	+	+	-	-	D	-		D	-	-	+	-	-	-	D	+	+	-	-	+	+	+	+	-	+	+	+	+	+	+	+	-	-	-	-	-	+	-	+	-	-	D		-	-

Explanation of symbols:
- = Layer illegal for the device
+ = Layer allowed to overlap
D = DRAWN indicates that a layer is drawn by Design.
C = CREATED indicates that the layer is only created by CAD.
(1) Low vt needs to be set on the schematic element
(2) Ncm is drawn inside core. Otherwise it is created in periphery. See rules ncm.X.* for details
(3) Drawn over half of device
(4) ASSUMPTION: FET models will be same regardless of backend flow
(5) The 2 core FETs and flash npass must have a poly.ml label with their model name.
(6) over the drain
(7) over the source
(8) Information for RCX
(9) Uses a black box for LVS. This is a fixed layout; Use symbol provided by modeling group

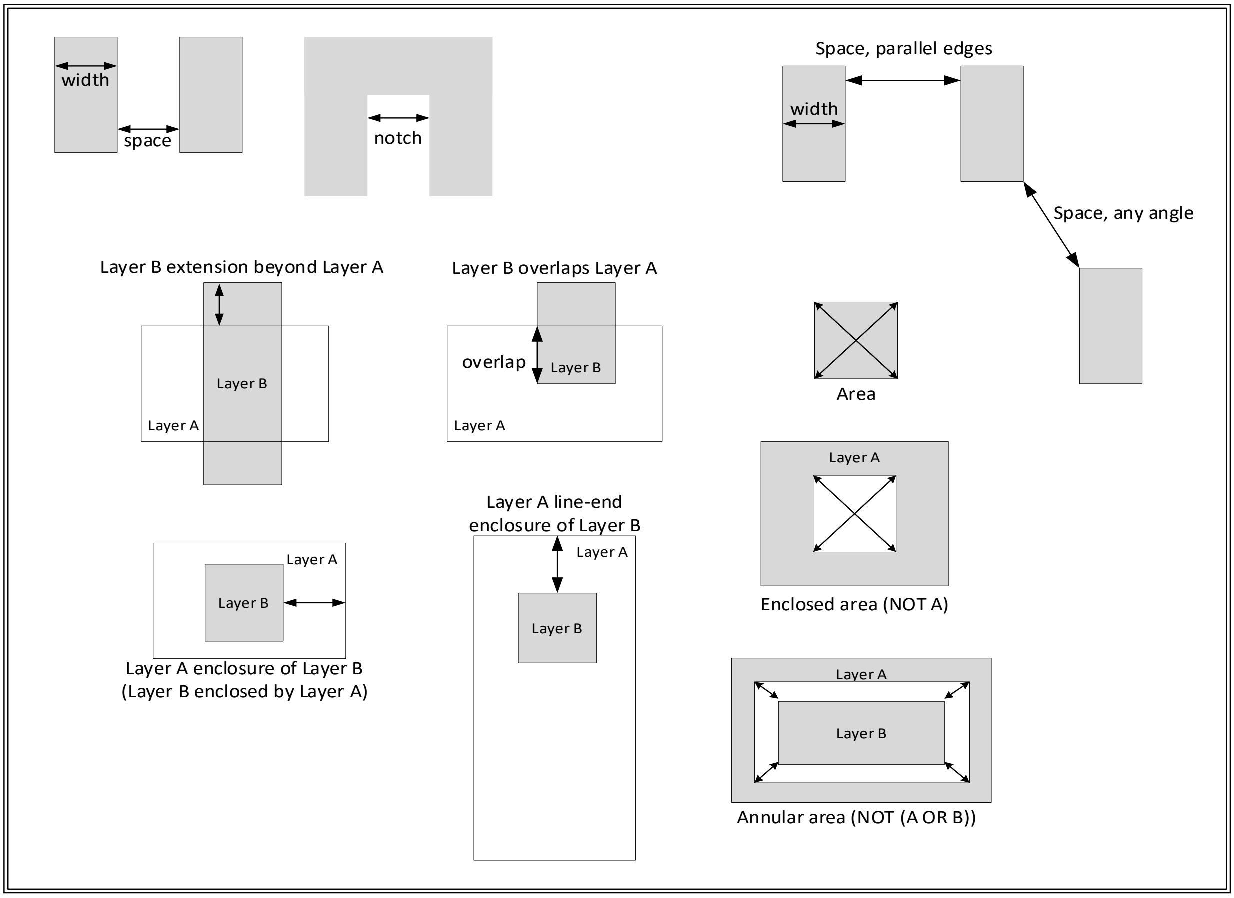
Category	Name	Drawn Route / Comments	Layout Model and required schematic element		s8di-5r*	s8tnv-5r	s8p-5r	sp8p-5r*	s8tm-5r*	s8tma-5r*	s8tmc-5r*	s8p-10r*	s8pir-10r	s8phirs-10r	s8pr2-10r	s8pf-10r*	s8pfn-20r	s8p12-10r*	s8spf-10r*	s8*phr-10r	s8phrc*
RESISTOR	n diff resistor		mrdn	res, resn	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	HV n diff resistor		mrdn_hv	res, resnhv	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	p diff resistor		mrdp	res, resp	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	p diff resistor NV		mrdp	res, resp		X															
	HV p diff resistor		mrdp_hv	res, resphv	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Isolated Pwell resistor		xpwres	respw	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	n+ poly resistor		mrp1	res	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	p- poly resistor		xuhrpoly_*	res3																	
	p+ poly resistor		xhrpoly_*	res3						X			X	X	X		X			X	X
	li resistor		mrl1	res	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	metal fuse	metX AND metX.fe	mrmX	mrmX	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
32 A CMOS	nmos 1.8V		nshort	nfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	pmos 1.8V		pshort	pfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Low Vt pmos 1.8V 32A		plowvt	pfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	High Vt pmos 1.8V 32A (10)		phighvt	pfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Low Vt nmos 1.8V 32A		nlowvt	nfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	nmos_core (5)		npass, npd	nfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	nmos_core NV (5)		npassll, npdll	nfet		X															
	pmos_core [5]		ppu	pfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	pmos_core NV [5]		ppull	pfet		X															
	Low Vt nmos_core [5]		nlvtpass	nfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Low Vt Varactor		xcnwvc	capbn_b	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	High Vt Varactor		xcnwvc2	capbn_b	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	HV varactor (floating gate)		xchvnwc	capbn_b	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
SONOS (& SONOS Latch)	SONOS fet [11]		sonos_p/e	nfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	SONOS fet [11]		sonos_de/dp	nfet				X													
	NV SONOS fet [11]		nvssonos_p/e	nfet		X															
110A CMOS	HV nmos 5/10.5V		nhv	nfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	HV nmos 5/10.5V		nhvcore	nfet				X													
	HV pmos 5/10.5 V		phv	pfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	HV pmos 5/10.5 V		phvcore	pfet				X													
	Native nmos 5V		nhvnative	nfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Native nmos 3V		ntvnative	nfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Flash npass [5]		fnpass	nfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Flash npass NV		nvsrnpass	nfet		X															
	VHV nmos 5/16V DE		nvhv	nfetextd	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	VHV pmos 5/16V DE		pvhv	pfetextd	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	UHV nmos 5/20V DE		n20vhv1	nfetextd													X				
	UHV iso nmos 5/20V DE		n20vhviso1	nfetextdiso													X				
	UHV Native nmos 5/20V DE		n20nativevhv1	nfetextd													X				
	UHV Zvt nmos 5/20V DE		n20zvtvhv1	nfetextd													X				
	UHV pmos 5/20V DE		p20vhv1	pfetextd													X				

Category	Name	Drawn Route / Comments	Layout Model and required schematic element		s8di-5r*	s8tnv-5r	s8p-5r	sp8p-5r*	s8tm-5r*	s8tma-5r*	s8tmc-5r*	s8p-10r*	s8pir-10r	s8phirs-10r	s8pr2-10r	s8pf-10r*	s8pfn-20r	s8p12-10r*	s8spf-10r*	s8*phr-10r	s8phrc*
CAPACITOR	MiM (3 Terminal) (s8t*)		xcmimc	cmim3c						X	X										
	MiM (2 Terminal) (s8t*)		xcmimc2	cmimc						X	X										
	MiM (2 Terminal) (s8p*)		xcmimc	cmim3c						X	X										X
	MiM2 (2 Terminal) (s8p*)		xcmimc2	cmimc						X	X										X
	VPP (9)		xcmvpp, xcmvpp_2		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	VPP over NHVNATIVE		xcmvpp2_nhvnative10x4	cap_int3	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	VPP over PHV		xcmvpp2_phv5x4	cap_int3	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	VPP and NHVNATIVE		xcmvppx4_2xnhvnative10x4	vppcap			X	X				X	X	X	X	X	X	X	X	X	X
	4-Terminal VPP (with Met3 shield) [12]		xcmvpp1p8x1p8_m3shield, xcmvpp8p6x7p9_m3shield, xcmvpp4p4x4p6_m3shield, xcmvpp11p5x11p7_m3shield	vppcap	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	4-terminal VPP (with M5 shield) [12]		xcmvpp11p5x11p7_m5shield, xcmvpp11p5x11p7_polym5shield, xcmvpp11p5x11p7_lim5shield, xcmvpp8p6x7p9_m3_lim5shield, xcmvpp11p5x11p7_m3_lim5shield, xcmvpp4p4x4p6_m3_lim5shield, xcmvpp11p5x11p7_m1m4m5shield xcmvpp11p5x11p7_polym50p4shield	vppcap			X	X				X	X	X	X	X	X	X	X	X	X
	4-terminal VPP (with M4 shield) [12]		xcmvpp11p5x11p7_m4shield, xcmvpp11p5x11p7_polym4shield, xcmvpp6p8x6p1_polym4shield, xcmvpp6p8x6p1_lim4shield	vppcap			X	X				X	X	X	X	X	X	X	X	X	X
	3-Terminal VPP [12]		xcmvpp1p8x1p8, xcmvpp3, xcmvpp4, xcmvpp5, xcmvpp4p4x4p6_m1m2, xcmvpp11p5x11p7_m1m2	cap_int3	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	3-Terminal VPP [12] (for S8Q/S8P ONLY)		xcmvpp8p6x7p9_m3_lishield, xcmvpp4p4x4p6_m3_lishield, xcmvpp11p5x11p7_m3_lishield	cap_int3			X	X				X	X	X	X	X	X	X	X	X	X
	3-Terminal VPP [12] (for S8P ONLY)		xcmvpp11p5xx11p7_m1m4, xcmvpp_hd5_*	cap_int3			X	X				X	X	X	X	X	X	X	X	X	X
INDUCTOR	Inductor		xind	inductor	X				X	X	X										
	Cu Inductor		*xind4*	ind4									X	X	X						
	Balun Inductor [15]		*balun*										X	X	X						

Category	Name	Drawn Route / Comments	Layout Model and required schematic element		s8di-5r*	s8tnv-5r	s8p-5r	sp8p-5r*	s8tm-5r*	s8tma-5r*	s8tmc-5r*	s8p-10r*	s8pir-10r	s8phirs-10r	s8pr2-10r	s8pf-10r*	s8pfn-20r	s8p12-10r*	s8spf-10r*	s8*phr-10r	s8phrc*
DIODE	nDiode		ndiode	lvsdiode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	HV nDiode		ndiode_h	lvsdiode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	RF ESD HV nDiode [13]		xesd_ndiode_h_X (where X=100, 200, 300)	lvsdiode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	RF ESD HV Deep Nwell nDiode [13		xesd_ndiode_h_dnwl_X (where X=100, 200, 300)	lvsdiode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	pDiode		pdiode	lvsdiode	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	pDiode NV		pdiode	lvsdiode		X															
	HV pDiode		pdiode_h	lvsdiode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	RF ESD HV pDiode [13]		xesd_pdiode_h_X (where X=100, 200, 300)	lvsdiode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Photo Diode		dnwdiode_psub	lvsdiode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Low Vt pdiode [8]		pdiode_lvt	diode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	High Vt pDiode [8]		pdiode_hvt	diode	X		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	High Vt pDiode NV [8]		pdiode_hvt	diode		X															
	Low Vt nDiode [8]		ndiode_lvt	diode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	NV SONOS Diode [8]		ndiode_nvs	diode		X															
	Native nDiode [8]		ndiode_native	diode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Nwell Diode [8]		nwdiode	diode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Nwdiode_victim [17]		nwdiode	lvsdiode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Nwdiode_aggressor [17]		nwdiode	lvsdiode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	RF Nwell Diode [8]		xnwdiode_rf	diode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Pwell-Deep Nwell Diode [8]		dnwdiode_pw	diode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	RF Pwell-Deep Nwell Diode [8]		xdnwdiode_pwell_rf	diode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Psub-Deep Nwell Diode [8]		dnwdiode_psub	diode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Psub-Deep Nwell Diode [17]		dnwdiode_psub	lvsdiode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Psub-Deep Nwell Diode [17]		dnwdiode_psub	lvsdiode	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Psub-Deep Nwell Diode [8,19]		dnwhvdiode_psub	diode													X				
	HV Pwell-Deep Nwell Diode [8, 19]		dnwdiode_hvpw	diode																	
PNP	Parasitic PNP	Layout provided by technology	pnppar, pnppar5x	pnp4	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Parasitic HV Gated NPN	Layout provided by technology	npn_1x1_2p0_hv	npn4	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	Parasitic NPN	Layout provided by technology	npnpar1x1, npnpar1x2	npn4	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
ESD transistor	LV nESD transistor	NMOS with ESD_nwell_tap	nshortesd	nfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	HV nESD transistor	NMOS with *ESD_nwell_tap*	nhvesd	nfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	HV Native nESD transistor	NMOS with *ESD_nwell_tap*	nhvnativeesd	nfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	HV pESD transistor		phvesd	pfet	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

(10) LVS will check that phighvt inside areaid.ce overlaps ncm
(11) The default model is sonos_e, sonos_de and nvssonos_e. If sonos_p, sonos_dp and nvssonos_p model are required, poly.ml must be used
(12) The capacitor.dg is drawn 0.17um from the edge of the cell to be LVS clean
(13) Devices are LVS'ed by cell name, m=1 per cell, fixed area and perimeter (see QHC-18)
(14) (dnwell not (pwres or pnp or npn or areaid.en or areaid.de or areaid.po)) not nwell must have condiode text; Refer to VUN-104, 192 for condiode usage
(15) Tech element is created by the user, no CAD supplied tech element
(16) There are multiple configurations of the Cu inductor. Layers present in one configuration may not be drawn in other configurations. RDL is not routed over met5 cu inductor, not checkable in PDK
(17) Used for substrate noise isolation regions only
(18) Either UHVI or areaid.low_vt should be drawn over the structures
(19) Psub-Deep Nwell Diode must have condiode text "condiodeHvPsub"; CVA-596
(20) mrp1 can't overlay capacitor.dg: exempted s8rf2_xcmvpp11p5x11p7_lim5shield from the rule

Diagrams illustrating terminology used in design rule definitions:



Tables F3. Summary of key periphery rules for design reference

Table F3a. Front end layers (Low Voltage Devices)

Layer	CD	nwell		diff		tap		n/psdm		poly		npc		licon	Manual
Parameter	width	spc	enc	spc	enc	spc	enc	spc	enc	spc	enc	spc	enc	spc	**merge ?**
nwell	**0.840**	**1.270**	X	X	X	X	X	X	X	X	X	X	X	X	Yes
diff	**0.150**	0.340	0.180	**0.270**	X	X	X	X	X	X	X	X	X	X	-
tap	**0.150**	0.130	0.180	0.270	-	**0.270**	X	X	X	X	X	X	X	X	-
n/psdm	**0.380**	-	-	0.130	*0.130*	0.130	*0.130*	**0.380**	X	X	X	X	X	X	Yes
poly on diff	**0.150**	-	-	-	-	0.300	-	-	-	**0.210**	X	X	X	X	-
poly on field	**0.150**	-	-	0.075	-	0.055	-	-	-	**0.210**	X	X	X	X	-
npc	**0.270**	-	-	-	-	-	-	-	-	0.090	X	**0.270**	X	X	Yes
licon	**0.170**	-	-	-	0.04/ 0.06	-	0.000	-	-	0.055	-	0.090	-	**0.170**	**-**
poly_licon	**0.170**	-	-	0.190	illegal	0.190	illegal	-	-	-	0.080	-	0.100	**0.170**	**-**

Table F3b. Front end layers (High Voltage Devices)

Layer	CD	nwell		diff		tap		poly		lvom		Manual
Parameter	width	spc	enc	spc	enc	spc	enc	spc	enc	spc	enc	**merge ?**
hnwell	**0.840**	**2.000**	X	X	X	X	X	X	X	X	X	Yes
hvi	**0.600**	**0.700**	-	0.180	*0.180*	0.180	*0.180*	-	-	**0.700**	X	Yes
hdiff	**0.290**	0.430	0.330	**0.300**	X	X	X	X	X	X	X	-
htap	**0.150**	0.430	0.330		-	**0.270**	X	X	X	X	X	-
HV poly	**0.500**	-	-	0.075	-	0.055	-	**0.210**	X	-	-	-

Italics: enclosure of layer from vertical menu by layer from horizontal menu.
Manual merge means that features below min. space should be manually merged by drawing.

Table F3c. Back end layers for S8D* flow

Layer	CD	licon		li1		mcon		metal1		via		metal2		via2		metal3	
Parameter	width	spc	enc	spc	enc	spc	enc	spc	enc	spc	enc	spc	enc	spc	enc	spc	enc
li1	**0.170**	undefined	0.000	**0.170**	X	X	X	X	X	X	X	X	X	X	X	X	X
mcon	**0.170**	-	-	-	0.000	**0.190**	X	X	X	X	X	X	X	X	X	X	X
metal1	**0.140**	-	-	-	-	-	0.03/ 0.06	**0.140**	X	X	X	X	X	X	X	X	X
via	**0.150**	-	-	-	-	-	-	-	0.055 / 0.085	**0.170**	X	X	X	X	X	X	X
metal2	**0.140**	-	-	-	-	-	-	-	-	-	0.055 / 0.085	**0.140**	X	X	X	X	X
via2	**0.280**	-	-	-	-	-	-	-	-	-	-	-	0.040	**0.280**	X	X	X
metal3	**0.360**	-	-	-	-	-	-	-	-	-	-	-	-	-	0.045 / 0.07	**0.360**	X

All enclosures in tables are nominal and do not apply to butting edges or corners.

Table F3d. Back end layers for S8T* flow

Layer	CD	licon		li1		mcon		metal1		via		metal2		via2		metal3	
Parameter	width	spc	enc	spc	enc	spc	enc	spc	enc	spc	enc	spc	enc	spc	enc	spc	enc
li1	**0.170**	undefined	0.000	**0.170**	X	X	X	X	X	X	X	X	X	X	X	X	X
mcon	**0.170**	-	-	-	0.000	**0.190**	X	X	X	X	X	X	X	X	X	X	X
metal1	**0.140**	-	-	-	-	-	0.03/ 0.06	**0.140**	X	X	X	X	X	X	X	X	X
via	**0.150**	-	-	-	-	-	-	-	0.055 / 0.085	**0.170**	X	X	X	X	X	X	X
metal2	**0.140**	-	-	-	-	-	-	-	-	-	0.055	**0.140**	X	X	X	X	X
via2	**0.280**	-	-	-	-	-	-	-	-	-	-	-	0.190	**1.200**	X	X	X
metal3	**2.500**	-	-	-	-	-	-	-	-	-	-	-	-	-	0.310	**2.500**	X

All enclosures in tables are nominal and do not apply to butting edges or corners.

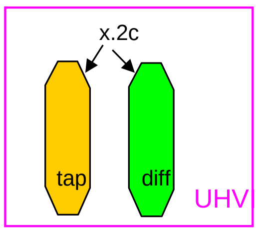
Table F4: Connectivity of Drawn and Mask Layers (1)

	Deep N We	**N Well**	**Diff**	**Tap**	**Poly**	**Li1**	**Capm**	**Met1**	**Met2**	**Met3**	**Met4**	**Met5**	**rdl**
Deep N Well	N/A												
N Well	Over	N/A											
Diff	X	X	N/A										
Tap	X	Over	X	N/A									
Poly	X	X	X	X	N/A								
Li1	X	X	Licon1	Licon1	Licon1 AND Npc	N/A							
Capm	X	X	X	X	X	X	N/A						
Met1	X	X	X	X	X	Mcon	X	N/A					
Met2	X	X	X	X	X	X	X	Via	N/A				
Met3	X	X	X	X	X	X	Via2	X	Via2	N/A			
Met4	X	X	X	X	X	X	X	X	X	Via3	N/A		
Met5	X	X	X	X	X	X	X	X	X	X	Via4	N/A	
rdl	X	X	X	X	X	X	X	X	X	X	X	(pad AND pmm) for s8pir/s8pr 2-10r flows (1)	N/A
bump	X	X	X	X	X	X	X	X	X	X	X	X	pi2 AND ubm

(1) All layers drawn except pmm which is created as cpmm:mask over bond pads or converted into cpbo:mask
(2) Entries in this table show the layer (or combination of layers) that act as connecting layers listed in the row/column headings. An X indicates that there is no direct connection between these layers. N/A is entered along the diagonal;
Over- Layers contacted by overlapping. A layer is always connected to itself.
(3) (Met5 AND pad AND rdl) should have one of the following sizes for LVS to work with WLCSP option: 60x60, 50x70, 60x80, and 80x80

Table F5: Device Connectivity Table

Devices	LVS	Latch up	Soft
Transistors	open	open	open
resistor	open	open	open
diode	open	open	open
pnp	open	open	open
Inductor	open	short	open
capacitors	open	open	open

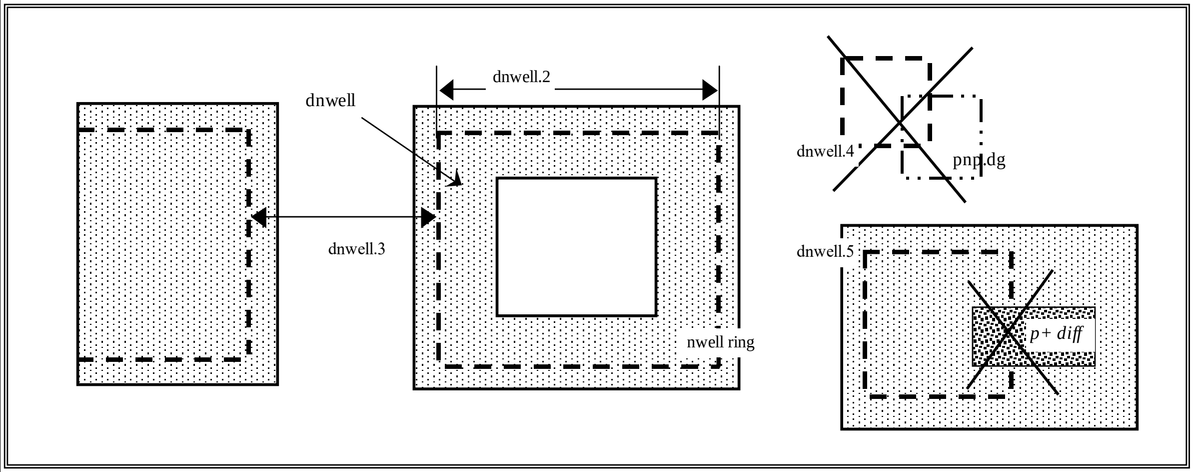
Rule	Section G2: Design Rules for S8*	Use								
	Section G2a. Periphery Rules									
(x.-)	**General**		**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
1a	p1m.md (OPC), DECA and AMKOR layers (pi1.dg, pmm.dg, rdl.dg, pi2.dg, ubm.dg, bump.dg) and mask data for p1m, met1, via, met2 must be on a grid of [mm]		0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
1b	Data for S8 layout and mask on all layers except those mentioned in 1a must be on a grid of [mm] (except inside Seal ring)		0.005	0.005	0.005	0.005	0.005	0.005	0.005	0.005
2	Angles permitted on: diff		n x 90	n x 90	n x 90	N/A	N/A	N/A	N/A	N/A
	Angles permitted on: diff except for: - diff inside "advSeal_6um* OR cuPillarAdvSeal_6um*" pcell, - diff rings around the die at min total L>1000 um and W=0.3 um		N/A	n x 90	N/A	n x 90	n x 90	n x 90	n x 90	n x 90
	Angles permitted on: tap (except inside areaid.en), poly (except for ESD flare gates or *gated_npn*), li1(periphery), licon1, capm, mcon, via, via2. Anchors are exempted.		n x 90	n x 90	n x 90	n x 90	n x 90	n x 90	n x 90	n x 90
	Angles permitted on: via3 and via4. Anchors are exempted.		N/A	N/A	N/A	n x 90	n x 90	n x 90	n x 90	n x 90
2a	Analog circuits identified by areaid.analog to use rectangular diff and tap geometries only; that are not to be merged into more complex shapes (T's or L's)									
2c	45 degree angles allowed on diff, tap inside UHVI									
										
3	Angles permitted on all other layers and in the seal ring for all the layers		n x 45	n x 45	n x 45	n x 45		n x 45	n x 45	n x 45
3a	Angles permitted on all other layers except WLCSP layers (pmm, rdl, pmm2, ubm and bump)						n x 45			
4	Electrical DR cover layout guidelines for electromigration	**NC**								
5	All "pin"polygons must be within the "drawing" polygons of the layer	**AI**								
6	All intra-layer separation checks will include a notch check									
7	Mask layer line and space checks must be done on all layers (checked with s.x rules)	**NC**								
8	Use of areaid "core" layer ("coreid") must be approved by technology	**NC**								
9	Shapes on maskAdd or maskDrop layers ("serifs") are allowed in core only. Exempted are: - cfom md/mp inside "advSeal_6um* OR cuPillarAdvSeal_6um*" pcell - diff rings around the die at min total L>1000 um and W=0.3 um, and PMM/PDMM inside areaid:sl		N/A		N/A					
	Shapes on maskAdd or maskDrop layers ("serifs") are allowed in core only. PMM/PDMM inside areaid:sl are excluded.					N/A	N/A	N/A	N/A	N/A
10	Res purpose layer for (diff, poly) cannot overlap licon1									
11	Metal fuses are drawn in met2	**LVS**		N/A		N/A	N/A	N/A	N/A	N/A
	Metal fuses are drawn in met3	**LVS**	N/A		N/A	N/A	N/A	N/A	N/A	N/A
	Metal fuses are drawn in met4	**LVS**	N/A	N/A	N/A					
 12a **12b** **12c**	To comply with the minimum spacing requirement for layer X in the frame: - Spacing of areaid.mt to any non-ID layer - Enclosure of any non-ID layer by areaid.mt - Rules exempted for cells with name "*_buildspace"	**F**								
12d	- Spacing of areaid.mt to *huge_metX* (Exempt met3.dg)	**F**				N/A	N/A	N/A	N/A	N/A

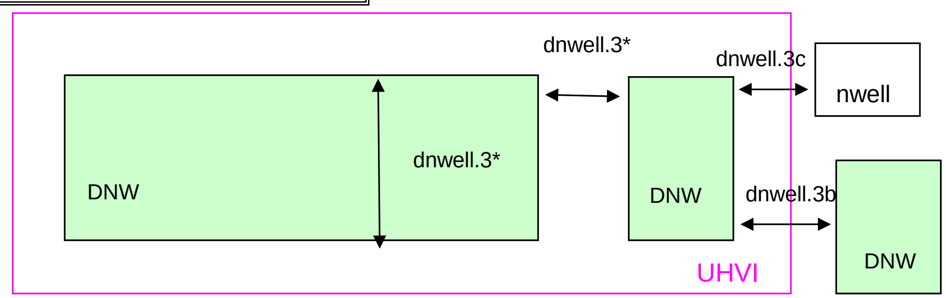
Rule	Section G2: Design Rules for S8*	Use								
12d	- Spacing of areaid.mt to *huge_metX* (Exempt met5.dg)	F	N/A	N/A	N/A					
12e	- Enclosure of *huge_metX* by areaid.mt (Exempt met3.dg)	F				N/A	N/A	N/A	N/A	N/A
12e	- Enclosure of *huge_metX* by areaid.mt (Exempt met5.dg)	F	N/A	N/A	N/A					
13	Spacing between features located across areaid:ce is checked by periphery rule									
14	Width of features straddling areaid:ce is checked by periphery rule									
15a	Drawn compatible, mask, and waffle-drop layers are allowed only inside areaid:mt (i.e., etest modules), or inside areaid:sl (i.e., between the outer and inner areaid:sl edges, but not in the die) or inside areaid:ft (i.e., frame, blankings). Exception: FOM/P1M/Metal waffle drop are allowed inside the *die*	P								
15b	Rule X.15a exempted for cpmm.dg inside cellnames "PadPLfp", "padPLhp", "padPLstg" and "padPLwlbi" (for the S8di-5r-gsmc flow)	Exempt								
16	*Die* must not overlap areaid.mt (rule waived for test chips and exempted for cellnames "*tech_CD_*", "*_techCD_*", "lazX_*" or "lazY_*")									
17	All labels must be within the "drawing" polygons of the layer; This check is enabled by using switch "floating_labels"; Identifies floating labels which appear as warnings in LVS. Using this check would enable cleaner LVS run; Not a gate for tapeout									
18	Use redundant mcon, via, via2, via3 and via4 (Locations where additional vias/contacts can be added to existing single vias/contacts will be identified by this rule). Single via under areaid.core and areaid.standarc are excluded from the single via check	RR	N/A	N/A	N/A					
19	Lower left corner of the seal ring should be at origin i.e (0,0)									
20	Min spacing between pins on the same layer (center to center); Check enabled by switch "IP_block"									
21	prunde.dg is allowed only inside areaid.mt or areaid.sc									
22	No floating interconnects (poly, li1, met1-met5) or capm allowed; Rule flags interconnects with no path to poly, difftap or metal pins. Exempt floating layers can be excluded using poly_float, li1_float, m1_float, m2_float, m3_float, m4_float and m5_float text labels. Also flags an error if these text labels are placed on connected layers (not floating) and if the labels are not over the appropriate metal layer. If floating interconnects need to be connected at a higher level (Parent IP or Full chip), such floating interconnects can be exempted using poly_tie, li1_tie, m1_tie, m2_tie, m3_tie, m4_tie and m5_tie text labels. It is the responsibility of the IP owner and chip/product owner to communicate and agree to the node each of these texted lines is connected to, if there is any risk to how a line is tied, and to what node. Only metals outside areaid.stdcell are checked.	RC								
	The following are exempt from x.22 violations: *_techCD_ , inductor.dg, modulecut, capacitors* and *s8blerf*									
	The 'notPublicCell' switch will deactivate this rule									
23a	areaid.sl must not overlap diff			N/A		N/A	N/A	N/A	N/A	N/A
23b	diff cannot straddle areaid.sl		N/A		N/A					
23c	areaid.sl must not overlap tap, poly, li1 and metX			N/A						
23d	areaid.sl must not overlap tap, poly		N/A		N/A	N/A	N/A	N/A	N/A	N/A
23e	areaid:sl must not overlap li1 and metX for pcell "advSeal_6um"		N/A		N/A	N/A	N/A	N/A	N/A	N/A
23f	areaid:SubstrateCut (areaid.st, local_sub) must not straddle p+ tap	RR								
24	condiode label must be in *iso_pwell*									
25	pnp.dg must be only within cell name "s8rf_pnp", "s8rf_pnp5x" or "s8tesd_iref_pnp", "stk14ecx_*"									
26	"advSeal_6um" pcell must overlap diff		N/A	N/A	N/A					
27	If the sealring is present, then partnum is required. To exempt the requirement, place text.dg saying "partnum_not_necessary". "partnum*block" pcell should be used instead of "partnum*" pcells	RR	N/A	N/A	N/A	N/A	N/A			N/A
28	Min width of areaid.sl		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
29	nfet must be enclosed by dnwell. Rule is checked when switch nfet_in_dnwell is turned on.									

Rule	Section G2: Design Rules for S8*	Use	
Use	**Explanation**		
MR	Manufacturing Rules, may not be waived by customer without written approval by SW fab mgt.		
P	Rule applies to periphery only (outside areaid.ce). A corresponding core rule may or may not exist.		
NE	Rule not checked for *esd_nwell_tap* . There are no corresponding rule for *esd_nwell_tap* .		
NC	Rule not checked by DRC. It should be used as a guideline only.		
TC	Rule not checked for cell name "*_tech_CD_top*"		
A	Rule documents a functionality implemented in CL algorithms and may not be checked by DRC.		
AD	Rule documents a functionality implemented in CL algorithms and checked by DRC.		
DE	Rule not checked for source of Drain Extended device		
LVS	Rule handled by LVS		
F	Rule intended for Frame only, not checked inside *Die*		
DNF	Drawn Not equal Final. The drawn rule does not reflect the final dimension on silicon. See table J for details.		
RC	Recommended rule at the chip level, required rule at the IP level.		
RR	Recommended rule at any IP level		
Al Cu	Rules applicable only to Al or Cu BE flows		
IR	IR drop check comparing Al database and slotted Cu database for the same product (2 gds files) must be clean	**Cu**	

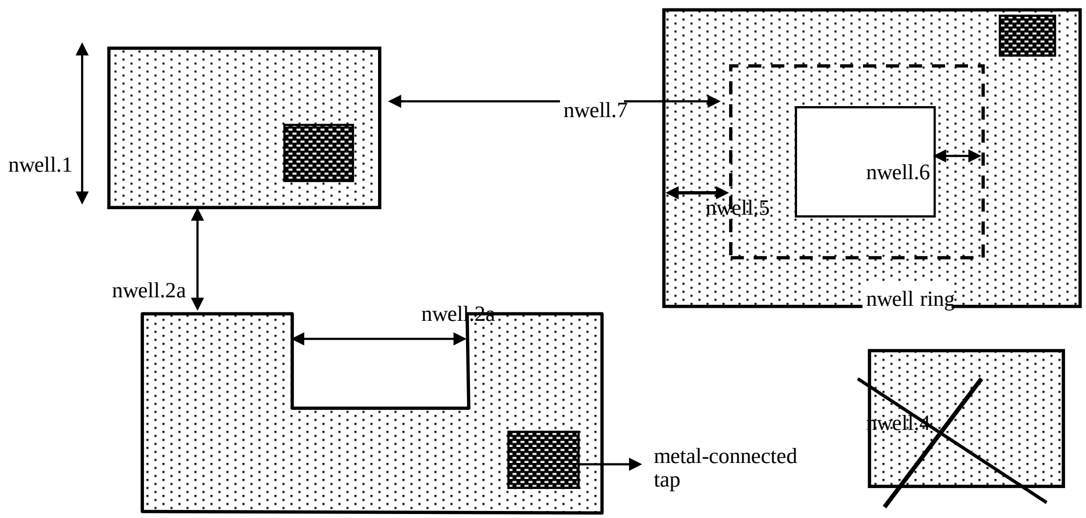
Note: some rules contain correction factors to compensate possible mask defect and unpredicted process biases

Rule	Section G2: Design Rules for S8*	Use								
(dnwell.-)	**Deep Nwell**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Define deep nwell for isolating pwell and noise immunity									
2	Min width of deep nwell	**MR**	3.000	3.000	3.000	3.000	3.000	3.000	3.000	3.000
3	Min spacing between deep nwells. Rule exempt inside UHVI.		6.300	6.300	6.300	6.300	6.300	6.300	6.300	6.300
3a	Min spacing between deep nwells on same net inside UHVI.		N/A	N/A	N/A	N/A	N/A	2.500	N/A	N/A
3b	Min spacing between deep-nwells inside UHVI and deep-nwell outside UHVI		N/A	N/A	N/A	N/A	N/A	12.000	N/A	N/A
3c	Min spacing between deep-nwells inside UHVI and nwell outside UHVI		N/A	N/A	N/A	N/A	N/A	9.500	N/A	N/A
3d	Min spacing between deep-nwells inside UHVI on different nets		N/A	N/A	N/A	N/A	N/A	12.000	N/A	N/A
4	Dnwell can not overlap pnp:dg									
5	*P+_diff* can not straddle Dnwell									
6	RF NMOS must be enclosed by deep nwell (RF FETs are listed in $DESIGN/config/tech/model_set/calibre/fixed_layout_model_map of corresponding techs)		N/A	N/A						
7	Dnwell can not straddle areaid:substratecut									

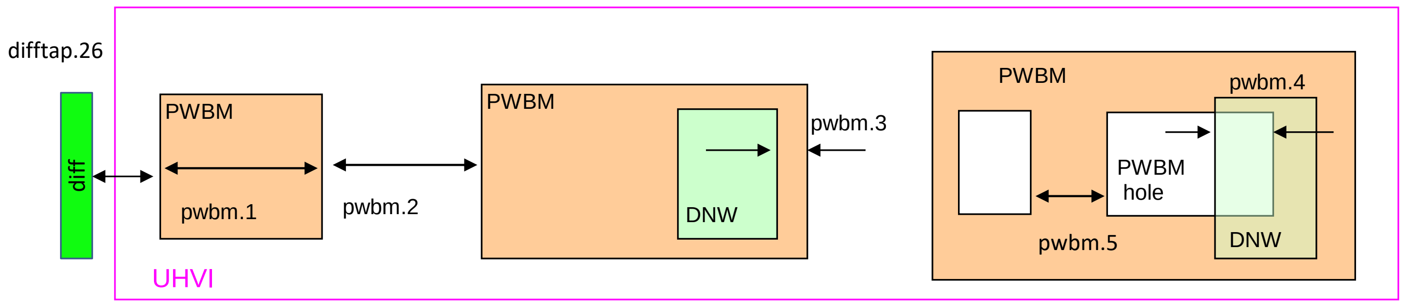




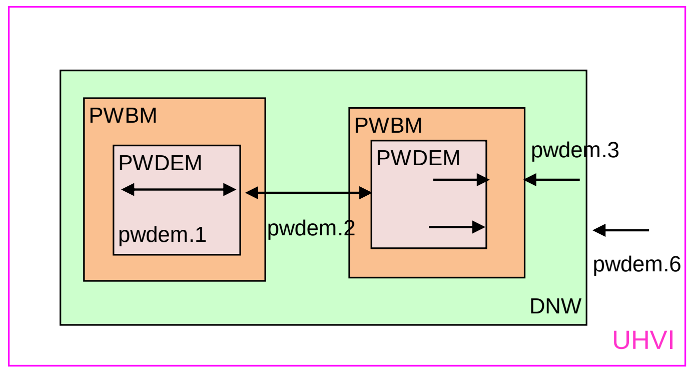
Rule	Section G2: Design Rules for S8*	Use								
(nwell.-)	**Nwell**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Define nwell implant regions									
1	Width of nwell	**MR**	0.840	0.840	0.840	0.840	0.840	0.840	0.840	0.840
2a	Spacing between two n-wells	**MR**	1.270	1.270	1.270	1.270	1.270	1.270	1.270	1.270
2b	Manual merge wells if less than minimum									
4	All n-wells will contain metal-contacted tap (rule checks only for licon on tap) . Rule exempted from high voltage cells inside UHVI									
5	Deep nwell must be enclosed by nwell by at least... Exempted inside UHVI or areaid.lw	**TC**	0.400	0.400	0.400	0.400	0.400	0.400	0.400	0.400
	Nwells can merge over deep nwell if spacing too small (as in rule nwell.2)									
5a	min enclosure of nwell by dnwell inside UHVI		N/A	N/A	N/A	N/A	N/A	0.000	N/A	N/A
5b	nwell inside UHVI must not be on the same net as nwell outside UHVI		N/A	N/A	N/A	N/A	N/A		N/A	N/A
6	Min enclosure of nwell hole by deep nwell outside UHVI	**TC**	1.030	1.030	1.030	1.030	1.030	1.030	1.030	1.030
7	Min spacing between nwell and deep nwell on separate nets	**TC**	4.500	4.500	4.500	4.500	4.500	4.500	4.500	4.500
	Spacing between nwell and deep nwell on the same net is set by the sum of the rules nwell.2 and nwell.5. By default, DRC run on a cell checks for the separate-net spacing, when nwell and deep nwell nets are separate within the cell hierarchy and are joined in the upper hierarchy. To allow net names to be joined and make the same-net rule applicable in this case, the "joinNets" switch should be turned on.									



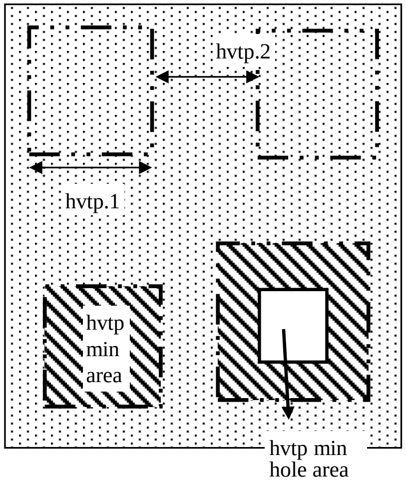
Rule	Section G2: Design Rules for S8*	Use								
(pwbm.-)	**Pwbm**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Define p-well block									
1	Min width of pwbm.dg		N/A	N/A	N/A	N/A	N/A	0.840	N/A	N/A
2	Min spacing between two pwbm.dg inside UHVI		N/A	N/A	N/A	N/A	N/A	1.270	N/A	N/A
3	Min enclosure of dnwell:dg by pwbm.dg inside UHVI (exempt pwbm hole inside dnwell)		N/A	N/A	N/A	N/A	N/A	0.000	N/A	N/A
4	dnwell inside UHVI must be enclosed by pwbm (exempt pwbm hole inside dnwell)		N/A	N/A	N/A	N/A	N/A		N/A	N/A
5	Min Space between two pwbm holes inside UHVI		N/A	N/A	N/A	N/A	N/A	0.840	N/A	N/A

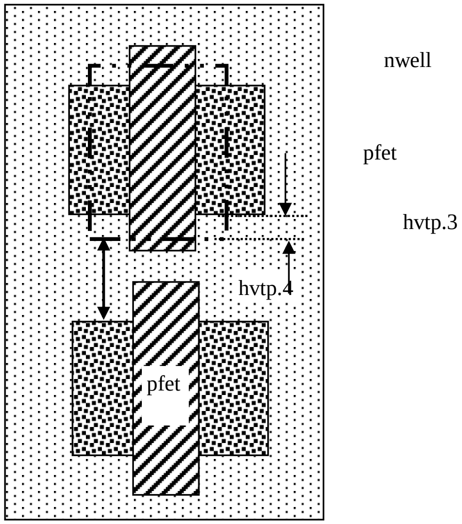


Rule	Section G2: Design Rules for S8*	Use								
(pwdem.-)	**Pwdem**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
1	Min width of pwdem.dg	**MR**	N/A	N/A	N/A	N/A	N/A	0.840	N/A	N/A
2	Min spacing between two pwdem.dg inside UHVI on same net	**MR**	N/A	N/A	N/A	N/A	N/A	1.270	N/A	N/A
3	Min enclosure of pwdem:dg by pwbm.dg inside UHVI		N/A	N/A	N/A	N/A	N/A	0.000	N/A	N/A
4	pwdem.dg must be enclosed by UHVI		N/A	N/A	N/A	N/A	N/A		N/A	N/A
5	pwdem.dg inside UHVI must be enclosed by deep nwell		N/A	N/A	N/A	N/A	N/A		N/A	N/A
6	Min enclosure of pwdem:dg by deep nwell inside UHVI		N/A	N/A	N/A	N/A	N/A	1.000	N/A	N/A

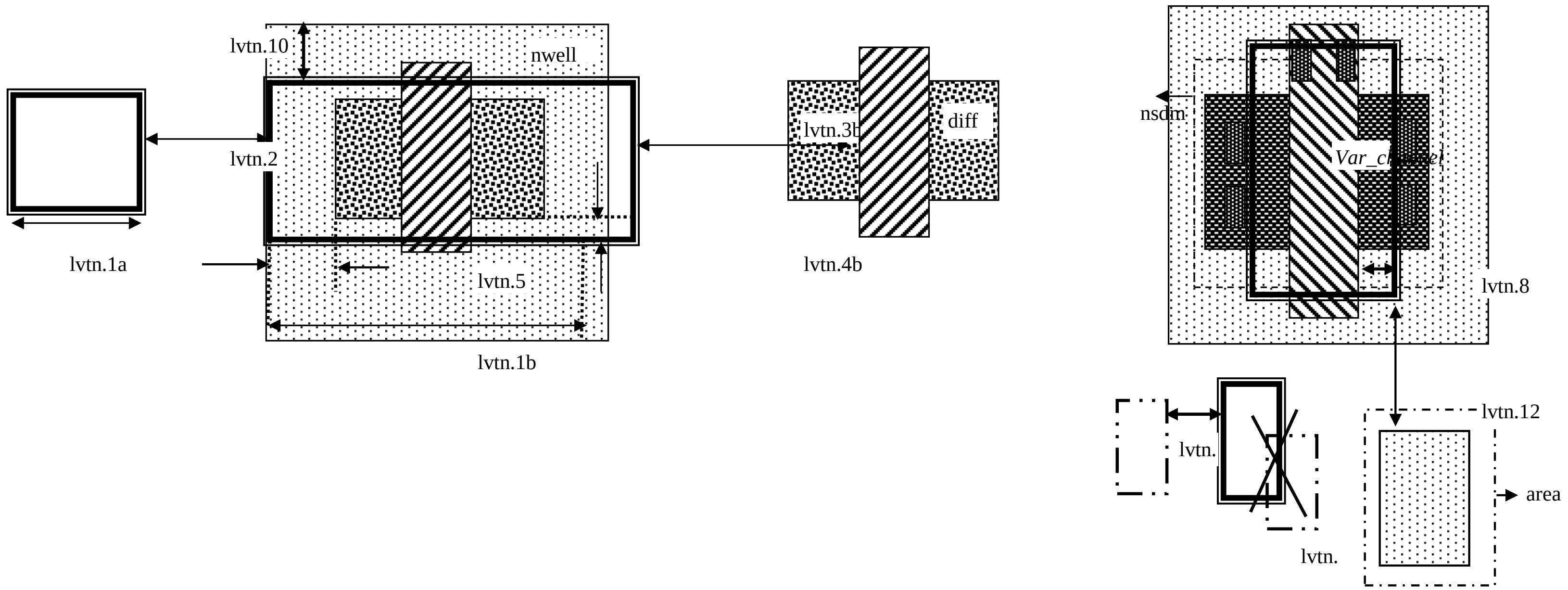


Rule	Section G2: Design Rules for S8*	Use								
(hvtp.-)	**Hvtp**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Define Vt adjust implant region for high Vt LV PMOS;									
1	Min width of hvtp	**MR**	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
2	Min spacing between hvtp to hvtp	**MR**	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
3	Min enclosure of *pfet* by hvtp	**P**	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180
4	Min spacing between *pfet* and hvtp	**P**	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180
5	Min area of hvtp (um^2)		0.265	0.265	0.265	0.265	0.265	0.265	0.265	0.265
6	Min area of hvtp Holes (um^2)		0.265	0.265	0.265	0.265	0.265	0.265	0.265	0.265



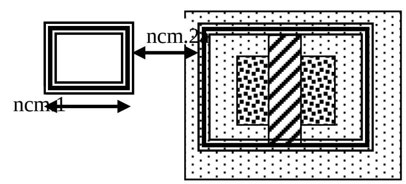


Rule	Section G2: Design Rules for S8*	Use								
(lvtn.-)	**Lvtnm**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Define regions to block Vt adjust implant for low Vt LV PMOS/NMOS, SONOS FETs and Native NMOS									
1a	Min width of lvtn	**MR**	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
2	Min space lvtn to lvtn	**MR**	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
3a	Min spacing of lvtn to *gate.* Rule exempted inside UHVI.	**P**	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180
3b	Min spacing of lvtn to *pfet* along the S/D direction	**P**	0.235	0.235	0.235	0.235	0.235	0.235	0.235	0.235
4b	Min enclosure of *gate* by lvtn. Rule exempted inside UHVI.	**P**	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180
9	Min spacing, no overlap, between lvtn and hvtp		0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
10	Min enclosure of lvtn by (nwell not overlapping *Var_channel)* (exclude coincident edges)		0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
12	Min spacing between lvtn and (nwell inside areaid.ce)		0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
13	Min area of lvtn (um^2)		0.265	0.265	0.265	0.265	0.265	0.265	0.265	0.265
14	Min area of lvtn Holes (um^2)		0.265	0.265	0.265	0.265	0.265	0.265	0.265	0.265

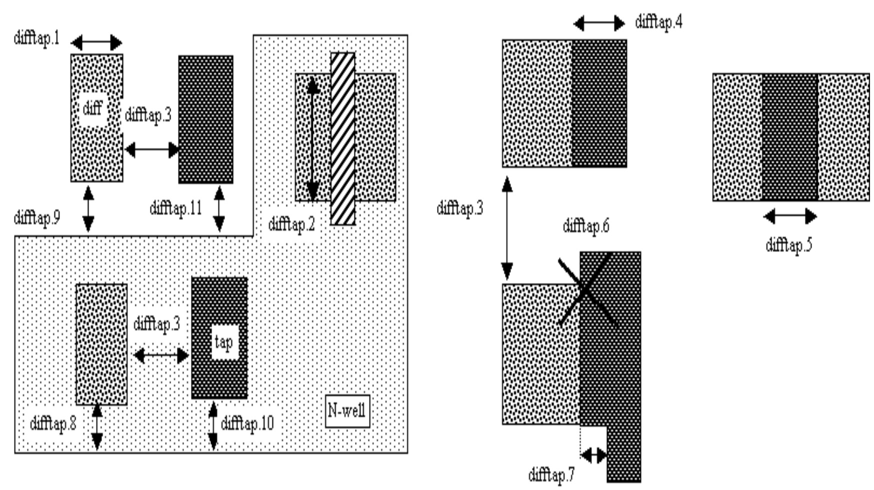


Rule	Section G2: Design Rules for S8*	Use								
(hvtr.-)	**Hvtr**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Define low VT adjust implant region for pmedlvtrf;									
1	Min width of hvtr	MR	N/A	N/A	N/A	N/A	0.380	N/A	N/A	N/A
2	Min spacing between hvtp to hvtr	MR	N/A	N/A	N/A	N/A	0.380	N/A	N/A	N/A
3	Min enclosure of *pfet* by hvtr	P	N/A	N/A	N/A	N/A	0.180	N/A	N/A	N/A

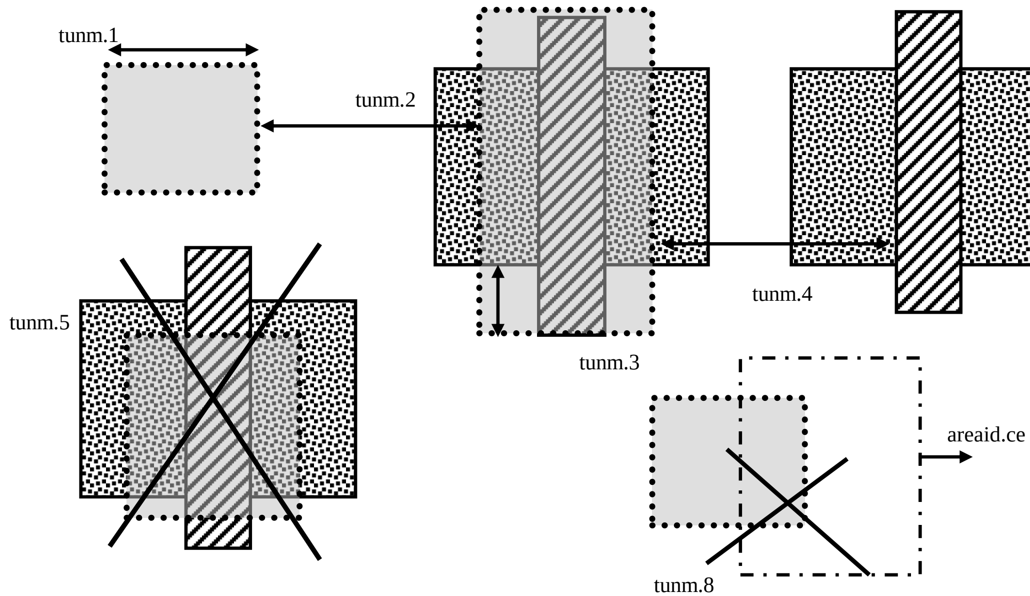
(ncm.-)	Ncm	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Define Vt adjust implant region for LV NMOS in the core of NVSRAM									
X.2	Ncm overlapping areaid:ce is checked for core rules only									
X.3	Ncm overlapping core cannot overlap N+diff in periphery	TC								
1	Width of ncm	MR	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
2a	Spacing of ncm to ncm	MR	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
2b	Manual merge ncm if space is below minimum									
7	Min area of ncm (um^2)		0.265	0.265	0.265	0.265	0.265	0.265	0.265	0.265
8	Min area of ncm Holes (um^2)		0.265	0.265	0.265	0.265	0.265	0.265	0.265	0.265



Rule	Section G2: Design Rules for S8*	Use								
(difftap.-)	**Diff/tap**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines active regions and contacts to substrate									
1	Width of diff or tap, outside of areaid:core	**MR, P**	0.150	0.150	0.150	0.150	0.150	0.150	0.150	0.150
2	Minimum channel width (Diff And Poly) except for FETs inside areaid.sc: Rule exempted in the SP8* flows only, for the cells listed in rule difftap.2a	**P**	0.420	0.420	0.420	0.420	0.420	0.420	0.420	0.420
2a	Minimum channel width (Diff And Poly) for cell names "s8cell_ee_plus_sseln_a", "s8cell_ee_plus_sseln_b", "s8cell_ee_plus_sselp_a", "s8cell_ee_plus_sselp_b" , "s8fpls_pl8", "s8fpls_rdrv4" , "s8fpls_rdrv4f" and "s8fpls_rdrv8"	**P**	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2b	Minimum channel width (Diff And Poly) for FETs inside areaid.sc	**P**	0.420	0.420	0.420	0.420	0.360	0.360	0.360	0.360
3	Spacing of diff to diff, tap to tap, or non-abutting diff to tap	**MR**	0.270	0.270	0.270	0.270	0.270	0.270	0.270	0.270
4	Min tap bound by one diffusion		0.290	0.290	0.290	0.290	0.290	0.290	0.290	0.290
5	Min tap bound by two diffusions	**P**	0.400	0.400	0.400	0.400	0.400	0.400	0.400	0.400
6	Diff and tap are not allowed to extend beyond their abutting edge									
7	Spacing of diff/tap abutting edge to a non-coinciding diff or tap edge	**NE**	0.130	0.130	0.130	0.130	0.130	0.130	0.130	0.130
8	Enclosure of (p+) diffusion by N-well. Rule exempted inside UHVI.	**DE, NE, P**	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180
9	Spacing of (n+) diffusion to N-well outside UHVI	**DE, NE, P**	0.340	0.340	0.340	0.340	0.340	0.340	0.340	0.340
10	Enclosure of (n+) tap by N-well. Rule exempted inside UHVI.	**NE, P**	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180
11	Spacing of (p+) tap to N-well. Rule exempted inside UHVI.		0.130	0.130	0.130	0.130	0.130	0.130	0.130	0.130
12	*ESD_nwell_tap* is considered shorted to the abutting diff	**NC**								
13	Diffusion or the RF FETS in Table H5 is defined by Ldiff and Wdiff.		N/A	N/A	N/A	N/A		N/A	N/A	N/A

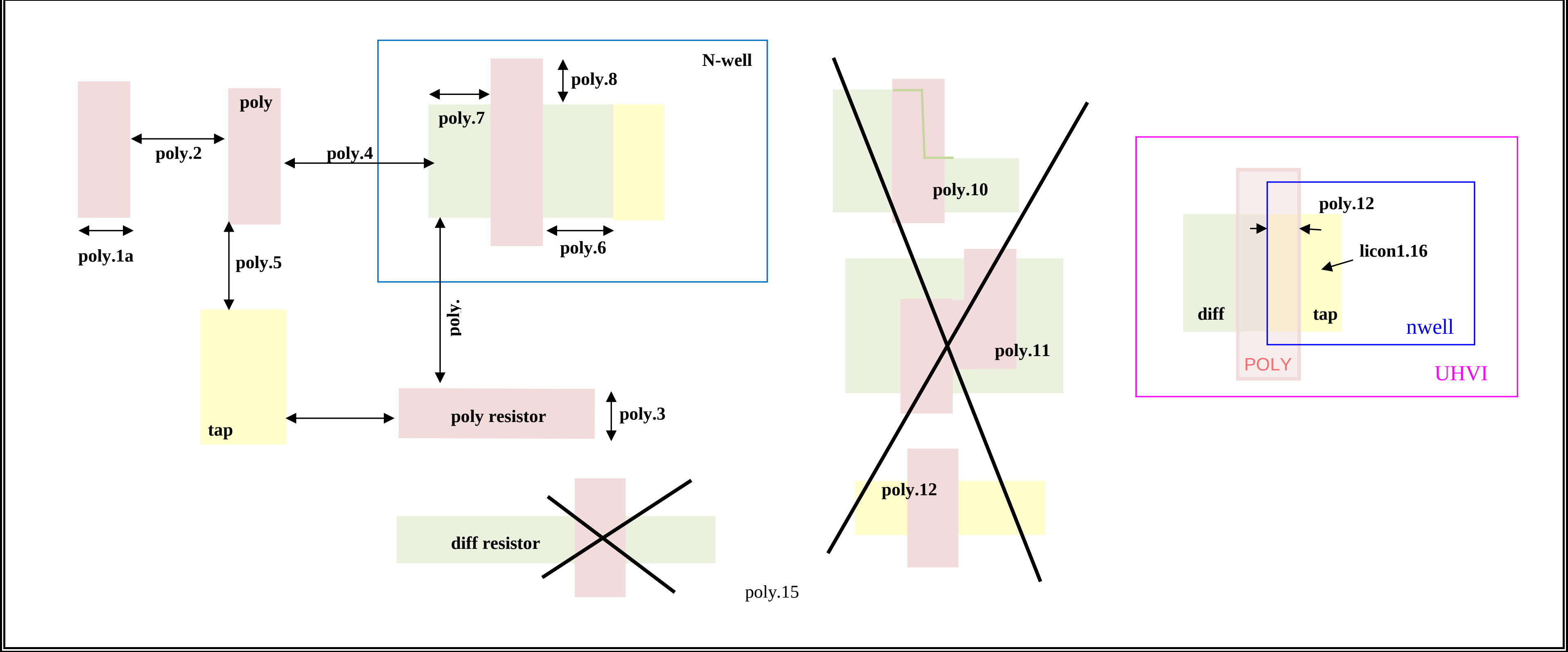


Rule	Section G2: Design Rules for S8*	Use								
(tunm.-)	Tunnel	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Defines SONOS FETs									
1	Min width of tunm	MR	0.410	0.410	0.410	0.410	0.410	0.410	0.410	0.410
2	Min spacing of tunm to tunm	MR	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500
3	Extension of tunm beyond (poly and diff)		0.100	0.100	0.100	0.100	0.095	0.095	0.100	0.095
4	Min spacing of tunm to (poly and diff) outside tunm		0.100	0.100	0.100	0.100	0.095	0.095	0.100	0.095
5	(poly and diff) may not straddle tunm									
6a	Tunm outside deep n-well is not allowed	TC		N/A						
7	Min tunm area		0.672	0.672	0.672	0.672	0.672	0.672	0.672	0.672
8	tunm must be enclosed by areaid.ce									



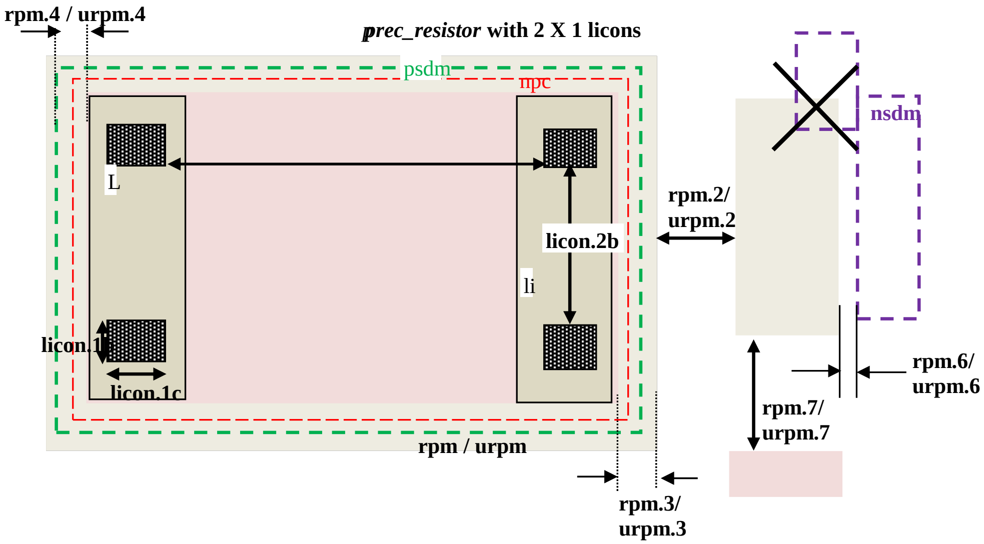
Rule	Section G2: Design Rules for S8*	Use								
(poly.-)	**Poly**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines FET gates, interconnects and resistors									
X.1	All FETs would be checked for W/Ls as documented in spec 001-02735 (Exempt FETs that are pruned; exempt for W/L's inside areaid.sc and inside cell name scs8*decap* and listed in the MRGA as a decap only W/L)									
X.1a	Min & max *dummy_poly* L is equal to min L allowed for corresponding device type (exempt rule for *dummy_poly* in cells listed on Table H3)									
1a	Width of poly	**MR**	0.150	0.150	0.150	0.150	0.150	0.150	0.150	0.150
1b	Min channel length (poly width) for *pfet* overlapping lvtn (exempt rule for *dummy_poly* in cells listed on Table H3)		0.350	0.350	0.350	0.350	0.350	0.350	0.350	0.350
2	Spacing of poly to poly outside areaid:core. Cases poly.c2 and poly.c3 are exempt.	**MR**	0.210	0.210	0.210	0.210	0.210	0.210	0.210	0.210
3	Min poly resistor width		0.330	0.330	0.330	0.330	0.330	0.330	0.330	0.330
4	Spacing of poly on field to diff (parallel edges only)	**P**	0.075	0.075	0.075	0.075	0.075	0.075	0.075	0.075
5	Spacing of poly on field to tap	**P**	0.055	0.055	0.055	0.055	0.055	0.055	0.055	0.055
6	Spacing of poly on diff to abutting tap (*min source*)	**P**	0.300	0.300	0.300	0.300	0.300	0.300	0.300	0.300
7	Extension of diff beyond poly (*min drain*)	**P**	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.250
8	Extension of poly beyond diffusion *(endcap)*	**P**	0.130	0.130	0.130	0.130	0.130	0.130	0.130	0.130
9a	Poly resistor spacing (no overlap) to diff/tap		0.480	0.480	0.480	0.480	0.480	0.480	0.480	0.480
10	Poly can't overlap inner corners of diff									
11	No 90 deg turns of poly on diff									
12	(Poly NOT (nwell NOT hvi)) may not overlap tap; Rule exempted for cell name "s8fgvr_n_fg2" and *gated_npn* and inside UHVI.	**P**								
15	Poly must not overlap diff:rs									
16	Inside RF FETs defined in Table H5, poly cannot overlap poly across multiple adjacent instances		N/A	N/A	N/A	N/A		N/A	N/A	N/A

Rule	Section G2: Design Rules for S8*	Use	



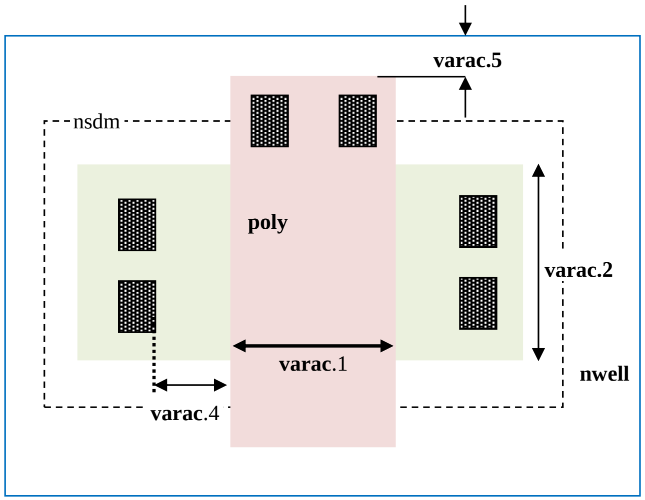
Rule	Section G2: Design Rules for S8*	Use								
(rpm.-)	**P+ Poly resistor**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines p+ poly resistors (330 ohm/sq)									
1a	Min width of rpm	**MR**	N/A	N/A	1.270	N/A	1.270	1.270	N/A	1.270
1b	Min/Max *prec_resistor* width xhrpoly_0p35		N/A	N/A	0.350	N/A	0.350	0.350	N/A	0.350
1c	Min/Max *prec_resistor* width xhrpoly_0p69		N/A	N/A	0.690	N/A	0.690	0.690	N/A	0.690
1d	Min/Max *prec_resistor* width xhrpoly_1p41		N/A	N/A	1.410	N/A	1.410	1.410	N/A	1.410
1e	Min/Max *prec_resistor* width xhrpoly_2p85		N/A	N/A	2.850	N/A	2.850	2.850	N/A	2.850
1f	Min/Max *prec_resistor* width xhrpoly_5p73		N/A	N/A	5.730	N/A	5.730	5.730	N/A	5.730
1g	Only 1 licon is allowed in xhrpoly_0p35 *prec_resistor_terminal*		N/A	N/A		N/A			N/A	
1h	Only 1 licon is allowed in xhrpoly_0p69 *prec_resistor_terminal*		N/A	N/A		N/A			N/A	
1i	Only 2 licons are allowed in xhrpoly_1p41 *prec_resistor_terminal*		N/A	N/A		N/A			N/A	
1j	Only 4 licons are allowed in xhrpoly_2p85 *prec_resistor_terminal*		N/A	N/A		N/A			N/A	
1k	Only 8 licons are allowed in xhrpoly_5p73 *prec_resistor_terminal*		N/A	N/A		N/A			N/A	
2	Min spacing of rpm to rpm	**MR**	N/A	N/A	0.840	N/A	0.840	0.840	N/A	0.840
3	rpm must enclose *prec_resistor* by at least		N/A	N/A	0.200	N/A	0.200	0.200	N/A	0.200
4	*prec_resistor* must be enclosed by psdm by at least		N/A	N/A	0.110	N/A	0.110	0.110	N/A	0.110
5	*prec_resistor* must be enclosed by npc by at least		N/A	N/A	0.095	N/A	0.095	0.095	N/A	0.095
6	Min spacing, no overlap, of rpm and nsdm		N/A	N/A	0.200	N/A	0.200	0.200	N/A	0.200
7	Min spacing between rpm and poly		N/A	N/A	0.200	N/A	0.200	0.200	N/A	0.200
8	poly must not straddle rpm		N/A	N/A		N/A			N/A	
9	Min space, no overlap, between *prec_resistor* and hvntm		N/A	N/A	0.185	N/A	0.185	0.185	N/A	0.185
10	Min spacing of rpm to pwbm		N/A	N/A	N/A	N/A	N/A	2.000	N/A	N/A
11	rpm should not overlap or straddle pwbm except cells s8usbpdv2_csa_top s8usbpdv2_20vconn_sw_300ma_ovp_ngate_unit s8usbpdv2_20vconn_sw_300ma_ovp s8usbpdv2_20sbu_sw_300ma_ovp		N/A	N/A	N/A	N/A	N/A		N/A	N/A
12	Poly resistor to poly resistor spacing (under same RPM)				0.210		0.210	0.210		0.210
13	Min spacing between prec_resistor and unrelated (poly NOT poly.rs)				0.480		0.480	0.480		0.480
14	Minimum resistor length (between licons) when RRPM or URPM is used		N/A	N/A	N/A	N/A	0.500	N/A	N/A	0.500
15	If RRPM or URPM are used, LI cannot cross the body of a P+ poly resistor (this is required to avoid double-implanting the resistor)		N/A	N/A	N/A	N/A		N/A	N/A	

Rule	Section G2: Design Rules for S8*	Use	

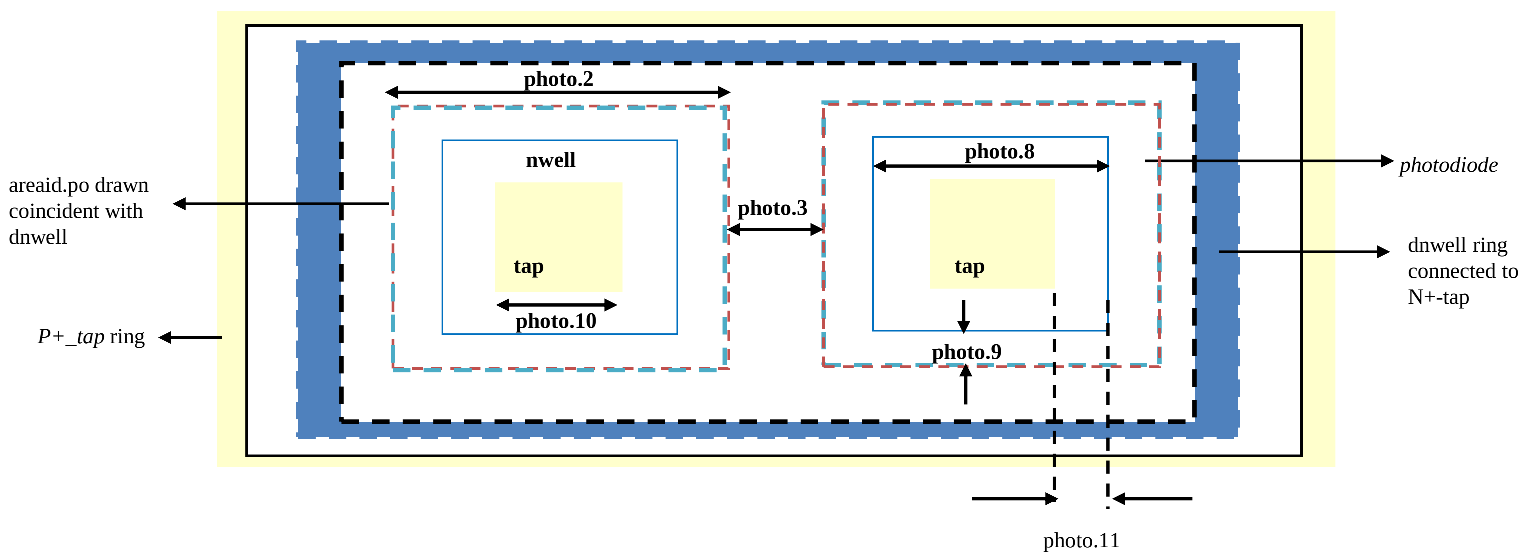


Rule	Section G2: Design Rules for S8*	Use								
(urpm.-)	**P- Poly resistor**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines p- poly resistors (2 kohm/sq)									
1a	Min width of urpm	**MR**	N/A	N/A	1.270	N/A	1.270	1.270	N/A	1.270
1b	Min/Max prec_resistor width xuhrpoly_0p35		N/A	N/A	0.350	N/A	0.350	0.350	N/A	0.350
1c	Min/Max prec_resistor width xuhrpoly_0p69		N/A	N/A	0.690	N/A	0.690	0.690	N/A	0.690
1d	Min/Max prec_resistor width xuhrpoly_1p41		N/A	N/A	1.410	N/A	1.410	1.410	N/A	1.410
1e	Min/Max prec_resistor width xuhrpoly_2p85		N/A	N/A	2.850	N/A	2.850	2.850	N/A	2.850
1f	Min/Max prec_resistor width xuhrpoly_5p73		N/A	N/A	5.730	N/A	5.730	5.730	N/A	5.730
1g	Only 1 licon is allowed in xuhrpoly_0p35 prec_resistor_terminal		N/A	N/A		N/A			N/A	
1h	Only 1 licon is allowed in xuhrpoly_0p69 prec_resistor_terminal		N/A	N/A		N/A			N/A	
1i	Only 2 licons are allowed in xuhrpoly_1p41 prec_resistor_terminal		N/A	N/A		N/A			N/A	
1j	Only 4 licons are allowed in xuhrpoly_2p85 prec_resistor_terminal		N/A	N/A		N/A			N/A	
1k	Only 8 licons are allowed in xuhrpoly_5p73 prec_resistor_terminal		N/A	N/A		N/A			N/A	
2	Min spacing of urpm to urpm	**MR**	N/A	N/A	0.840	N/A	0.840	0.840	N/A	0.840
3	urpm must enclose *prec_resistor* by at least		N/A	N/A	0.200	N/A	0.200	0.200	N/A	0.200
4	*prec_resistor* must be enclosed by psdm by at least		N/A	N/A	0.110	N/A	0.110	0.110	N/A	0.110
5	*prec_resistor* must be enclosed by npc by at least		N/A	N/A	0.095	N/A	0.095	0.095	N/A	0.095
6	Min spacing, no overlap, of urpm and nsdm		N/A	N/A	0.200	N/A	0.200	0.200	N/A	0.200
7	Min spacing between urpm and poly		N/A	N/A	0.200	N/A	0.200	0.200	N/A	0.200
8	poly must not straddle urpm		N/A	N/A		N/A			N/A	
9	Min space, no overlap, between *prec_resistor* and hvntm		N/A	N/A	0.185	N/A	0.185	0.185	N/A	0.185
10	Min spacing of urpm to pwbm		N/A	N/A	N/A	N/A	N/A	2.000	N/A	N/A
11	urpm should not overlap or straddle pwbm except cells s8usbpdv2_csa_top s8usbpdv2_20vconn_sw_300ma_ovp_ngate_unit s8usbpdv2_20vconn_sw_300ma_ovp s8usbpdv2_20sbu_sw_300ma_ovp		N/A	N/A	N/A	N/A	N/A		N/A	N/A
12	Poly resistor to poly resistor spacing (under same URPM)				0.210		0.210	0.210		0.210
13	Min spacing between prec_resistor and unrelated (poly NOT poly.rs)				0.480		0.480	0.480		0.480
14	Minimum resistor length (between licons) when RRPM or URPM is used		N/A	N/A	N/A	N/A	0.500	N/A	N/A	0.500
15	If RRPM or URPM are used, LI cannot cross the body of a P- poly resistor (this is required to avoid double-implanting the resistor)		N/A	N/A	N/A	N/A		N/A	N/A	

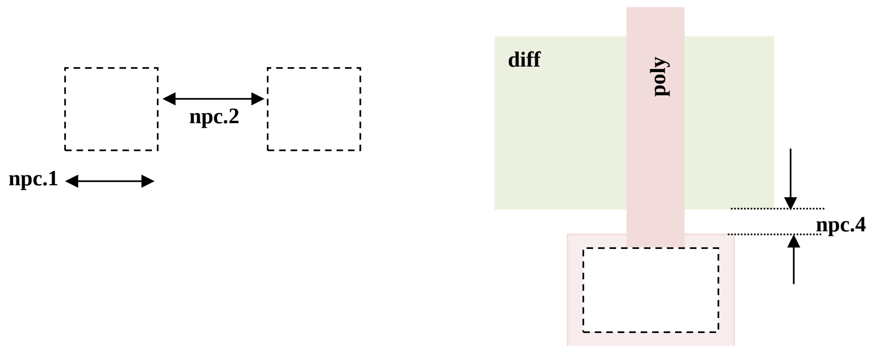
Rule	Section G2: Design Rules for S8*	Use								
(varac.-)	**Varactor**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines varactors									
1	Min channel length (poly width) of *Var_channel*		0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180
2	Min channel width (tap width) of *Var_channel*		1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3	Min spacing between hvtp to *Var_channel*		0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180
4	Min spacing of licon on tap to *Var_channel*		0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.250
5	Min enclosure of poly overlapping *Var_channel* by nwell		0.150	0.150	0.150	0.150	0.150	0.150	0.150	0.150
6	Min spacing between *VaracTap* and difftap		0.270	0.270	0.270	0.270	0.270	0.270	0.270	0.270
7	Nwell overlapping *Var_channel* must not overlap P+ diff									
8	Min enclosure of *Var_channel* by hvtp		0.255	0.255	0.255	0.255	0.255	0.255	0.255	0.255



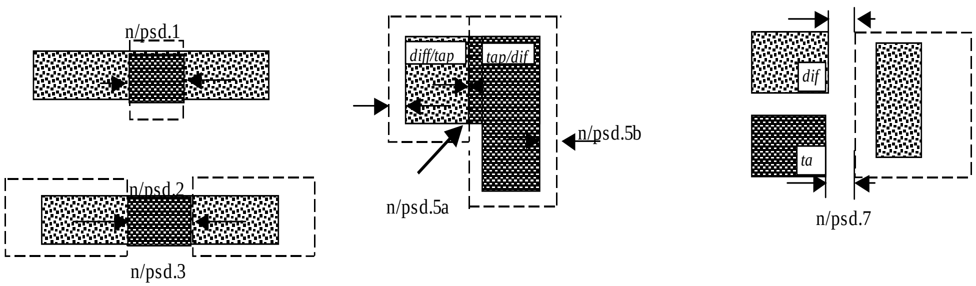
Rule	Section G2: Design Rules for S8*	Use								
(photo.-)	Photo diode	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Photo diode for sensing light									
1	Rules dnwell.3 and nwell.5 are exempted for *photoDiode*									
2	Min/Max width of *photoDiode*		3.000	3.000	3.000	3.000	3.000	3.000	3.000	3.000
3	Min spacing between *photoDiode*		5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000
4	Min spacing between *photoDiode* and deep nwell		5.300	5.300	5.300	5.300	5.300	5.300	5.300	5.300
5	*photoDiode* edges must be coincident with areaid.po									
6	*photoDiode* must be enclosed by dnwell ring									
7	*photoDiode* must be enclosed by p+ tap ring									
8	Min/Max width of nwell inside *photoDiode*		0.840	0.840	0.840	0.840	0.840	0.840	0.840	0.840
9	Min/Max enclosure of nwell by *photoDiode*		1.080	1.080	1.080	1.080	1.080	1.080	1.080	1.080
10	Min/Max width of tap inside *photoDiode*		0.410	0.410	0.410	0.410	0.410	0.410	0.410	0.410
11	Min/Max enclosure of tap by nwell inside *photoDiode*		0.215	0.215	0.215	0.215	0.215	0.215	0.215	0.215



Rule	Section G2: Design Rules for S8*	Use								
(npc.-)	**Nitride Poly Cut (NPC)**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines nitride openings to contact poly and Li1									
1	Min width of NPC	MR	0.270	0.270	0.270	0.270	0.270	0.270	0.270	0.270
2	Min spacing of NPC to NPC	MR	0.270	0.270	0.270	0.270	0.270	0.270	0.270	0.270
3	Manual merge if less than minimum									
4	Spacing (no overlap) of NPC to *Gate*		0.090	0.090	0.090	0.090	0.090	0.090	0.090	0.090
5	Max enclosure of poly overlapping *slotted_licon* by npcm (merge between adjacent short edges of the *slotted_licons* if space < min)		0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095

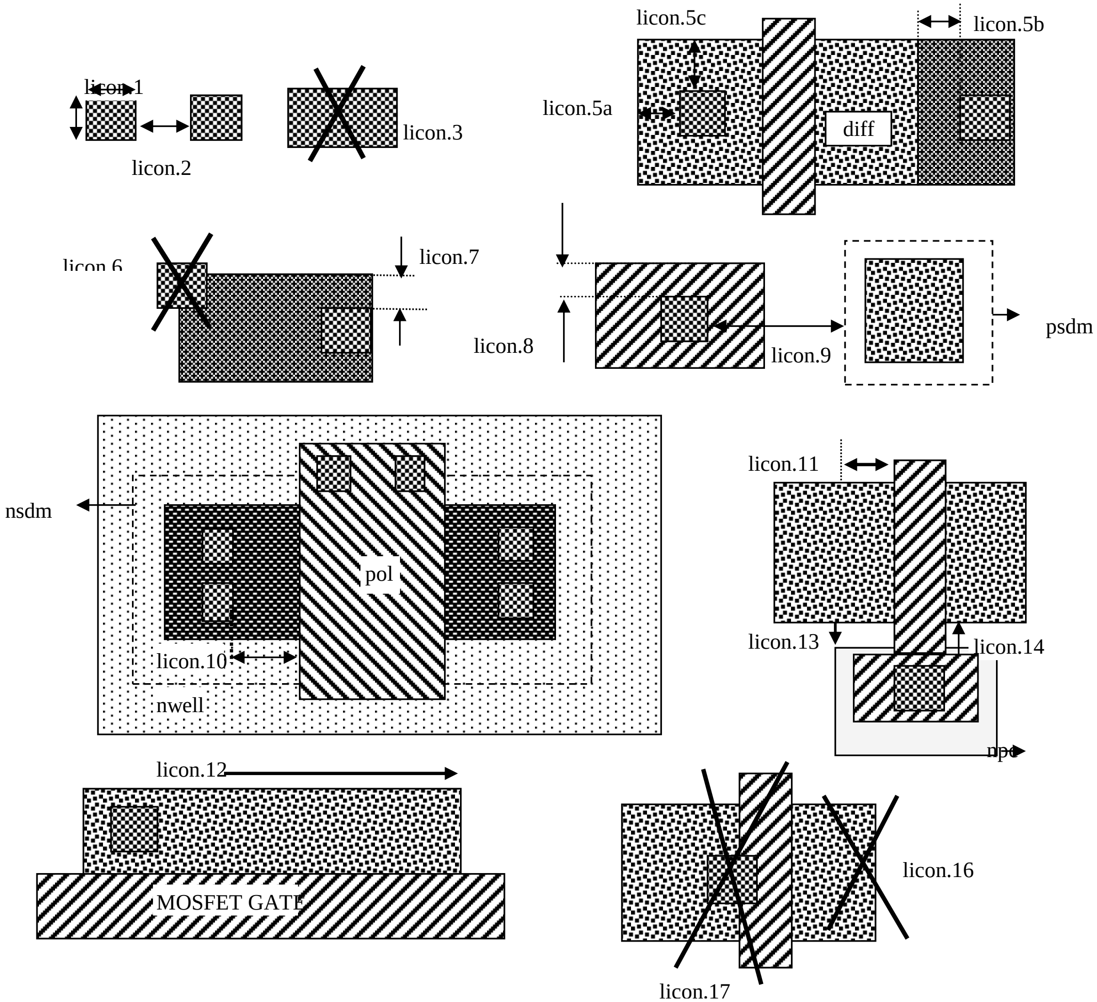


Rule	Section G2: Design Rules for S8*	Use								
(n/ psd.-)	N+/P+ Source/Drain Implants (Nsdm and Psdm)	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Defines opening for N+/P+ implants									
1	Width of nsdm(psdm), outside areaid:core	P, MR	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
2	Spacing of nsdm(psdm) to nsdm(psdm), outside areaid:core	P, MR	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
3	Manual merge if less than minimum									
5a	Enclosure of diff by nsdm(psdm), except for butting edge		0.130	0.130	0.130	0.130	0.125	0.125	0.125	0.125
5b	Enclosure of tap by nsdm(psdm), except for butting edge	P	0.130	0.130	0.130	0.130	0.125	0.125	0.125	0.125
6	Enclosure of diff/tap butting edge by nsdm (psdm)		0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
7	Spacing of NSDM/PSDM to opposite implant diff or tap (for non-abutting diff/tap edges)		0.130	0.130	0.130	0.130	0.130	0.130	0.130	0.130
8	Nsdm and psdm cannot overlap diff/tap regions of opposite doping	DE								
9	Diff and tap must be enclosed by their corresponding implant layers. Rule exempted for - diff inside "advSeal_6um* OR cuPillarAdvSeal_6um*" pcell for S8P*/SP8P*/S8DI-5R-CSMC flows - diff rings around the die at min total L>1000 um and W=0.3 um - *gated_npn* - *areaid.zer.*	DE								
10a	Min area of Nsdm (um^2)		0.265	0.265	0.265	0.265	0.265	0.265	0.265	0.265
10b	Min area of Psdm (um^2)		0.255	0.255	0.255	0.255	0.255	0.255	0.255	0.255
11	Min area of n/psdmHoles (um^2)		0.265	0.265	0.265	0.265	0.265	0.265	0.265	0.265

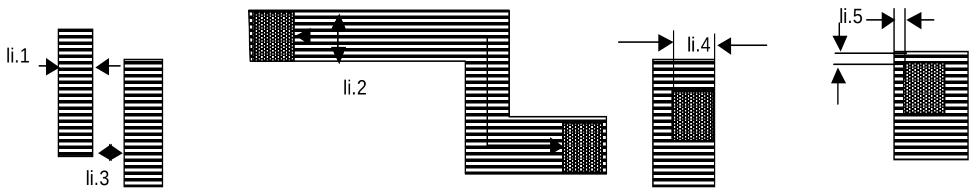


Rule	Section G2: Design Rules for S8*	Use								
(licon.-)	**Local Interconnect Contact (Licon)**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines contacts between poly/diff/tap and Li1									
1	Min and max L and W of licon (exempt licons inside *prec_resistor*)	**MR**	0.170	0.170	0.170	0.170	0.170	0.170	0.170	0.170
1b	Min and max width of licon inside *prec_resistor*		N/A	N/A	0.190	N/A	0.190	N/A	N/A	0.190
1c	Min and max length of licon inside *prec_resistor*		N/A	N/A	2.000	N/A	2.000	N/A	N/A	2.000
2	Spacing of licon to licon, outside of areaid:core	**P, MR**	0.170	0.170	0.170	0.170	0.170	0.170	0.170	0.170
2b	Min spacing between two *slotted_licon* (when the both the edges are 0.19um in length)		N/A	N/A	0.350	N/A	0.350	N/A	N/A	0.350
2c	Min spacing between two *slotted_licon* (except for rule licon.2b)		N/A	N/A	0.510	N/A	0.510	N/A	N/A	0.510
2d	Min spacing between a *slotted_licon* and 0.17um square licon		N/A	N/A	0.510	N/A	0.510	N/A	N/A	0.510
3	Only min. square licons are allowed except die seal ring where licons are (licon CD)*L		0.17*L	0.17*L	0.17*L	0.17*L	0.17*L	0.17*L	0.17*L	0.17*L
4	Licon1 must overlap li1 and (poly or diff or tap)									
5a	Enclosure of licon by diff	**P**	0.040	0.040	0.040	0.040	0.040	0.040	0.040	0.040
5b	Min space between *tap_licon* and diff-abutting tap edge	**P**	0.060	0.060	0.060	0.060	0.060	0.060	0.060	0.060
5c	Enclosure of licon by diff on one of two adjacent sides	**P**	0.060	0.060	0.060	0.060	0.060	0.060	0.060	0.060
6	Licon cannot straddle tap	**P**								
7	Enclosure of licon by one of two adjacent edges of isolated tap	**P**	0.120	0.120	0.120	0.120	0.120	0.120	0.120	0.120
8	Enclosure of *poly_licon* by poly	**P**	0.080	0.080	0.080	0.080	0.050	0.050	0.050	0.050
8a	Enclosure of *poly_licon* by poly on one of two adjacent sides	**P**	0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080
9	Spacing, no overlap, between *poly_licon* and psdm; In S8DIA/S8TMA/S8PIR-10 flows, the rule is checked only between (*poly_licon* outside rpm) and psdm	**P**	0.110	0.110	0.110	0.110	0.110	0.110	0.110	0.110
10	Spacing of licon on (tap AND (nwell NOT hvi)) to *Var_channel*	**P**	0.250	0.250	0.250	0.250	0.250	0.250	0.250	0.250
11	Spacing of licon on diff or tap to poly on diff (except for all FETs inside areaid.sc and except s8spf-10r flow for 0.5um phv inside cell names "s8fs_gwdlvx4", "s8fs_gwdlvx8", "s8fs_hvrsw_x4", "s8fs_hvrsw8", "s8fs_hvrsw264", and "s8fs_hvrsw520" and for 0.15um nshort inside cell names "s8fs_rdecdrv", "s8fs_rdec8", "s8fs_rdec32", "s8fs_rdec264", "s8fs_rdec520")	**P**	0.055	0.055	0.055	0.055	0.055	0.055	0.055	0.055
11a	Spacing of licon on diff or tap to poly on diff (for all FETs inside areaid.sc except 0.15um phighvt)	**P**	0.050	0.050	0.050	0.050	0.050	0.050	0.050	0.050
11b	Spacing of licon on diff or tap to poly on diff (for 0.15um phighvt inside areaid.sc)	**P**	0.055	0.055	0.055	0.055	0.050	0.050	0.050	0.050
11c	Spacing of licon on diff or tap to poly on diff (for 0.5um phv inside cell names "s8fs_gwdlvx4", "s8fs_gwdlvx8", "s8fs_hvrsw_x4", "s8fs_hvrsw8", "s8fs_hvrsw264", and "s8fs_hvrsw520")	**P**	N/A	N/A	N/A	N/A	0.040	0.040	N/A	0.040
11d	Spacing of licon on diff or tap to poly on diff (for 0.15um nshort inside cell names "s8fs_rdecdrv", "s8fs_rdec8", "s8fs_rdec32", "s8fs_rdec264", "s8fs_rdec520")	**P**	N/A	N/A	N/A	N/A	0.045	0.045	N/A	0.045
12	Max SD width without licon	**NC**	5.700	5.700	5.700	5.700	5.700	5.700	5.700	5.700
13	Spacing (no overlap) of NPC to licon on diff or tap	**P, MR**	0.090	0.090	0.090	0.090	0.090	0.090	0.090	0.090
14	Spacing of *poly_licon* to diff or tap	**P**	0.190	0.190	0.190	0.190	0.190	0.190	0.190	0.190
15	Enclosure of *poly_licon* by npc	**P**	0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100
16	Every *source_diff* and every tap must enclose at least one licon1, including the diff/tap straddling areaid:ce. Rule exempted inside UHVI.	**P**								
17	Licons may not overlap both poly and (diff or tap)	**MR**								
18	Npc must enclose *poly_licon*									
19	poly of the HV varactor must not interact with licon	**P**								

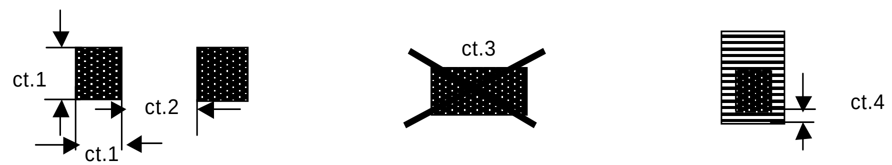
Rule	Section G2: Design Rules for S8*	Use	



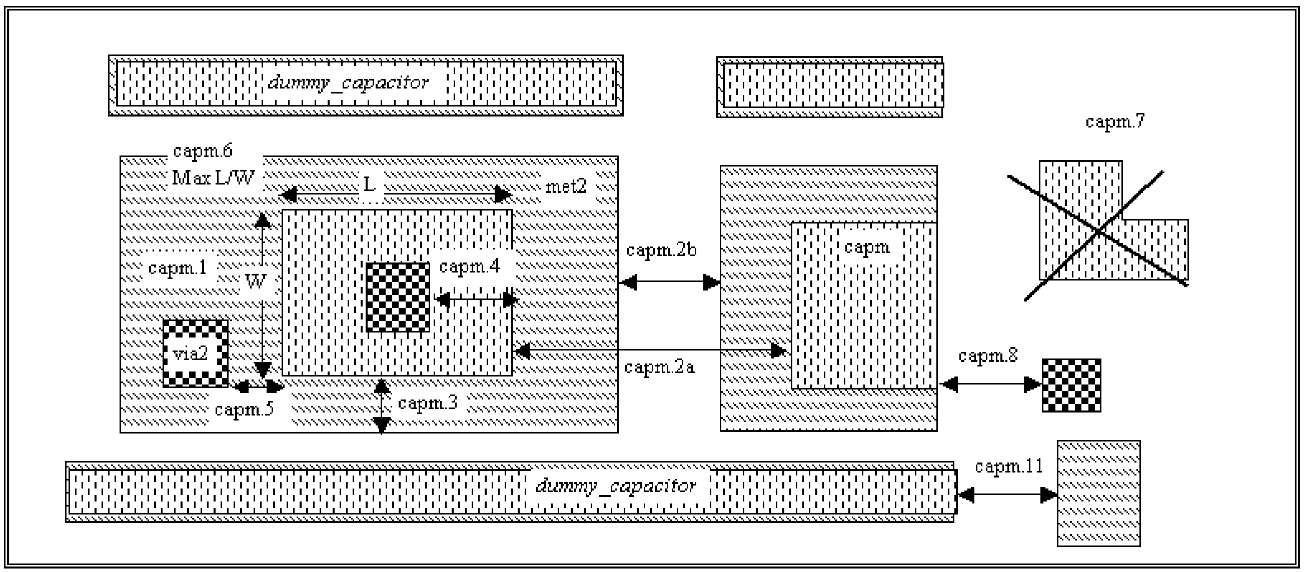
Rule	Section G2: Design Rules for S8*	Use								
(li.-.-)	**Local Interconnect (LI)**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines local interconnect to diff/tap and poly									
1	Width of LI outside of areaid:core (except for li.1a)	**P, MR**	0.170	0.170	0.170	0.170	0.170	0.170	0.170	0.170
1a	Width of LI inside of cells with name s8rf2_xcmvpp_hd5_*	**P**	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
2	Max ratio of length to width of LI without licon or mcon	**NC**	10.000	10.000	10.000	10.000	10.000	10.000	10.000	10.000
3	Spacing of LI to LI outside of areaid:core (except for li.3a)	**P, MR**	0.170	0.170	0.170	0.170	0.170	0.170	0.170	0.170
3a	Spacing of LI to LI inside cells with names s8rf2_xcmvpp_hd5_*	**P**	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
5	Enclosure of licon by one of two adjacent LI sides	**P, MR**	0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080
6	Min area of LI	**P, MR**	0.0561	0.0561	0.0561	0.0561	0.0561	0.0561	0.0561	0.0561
7	Min LI resistor width (rule exempted within areaid.ed; Inside areaid.ed, min width of the li resistor is determined by rule li.1)		0.290	0.290	0.290	0.290	0.290	0.290	0.290	0.290



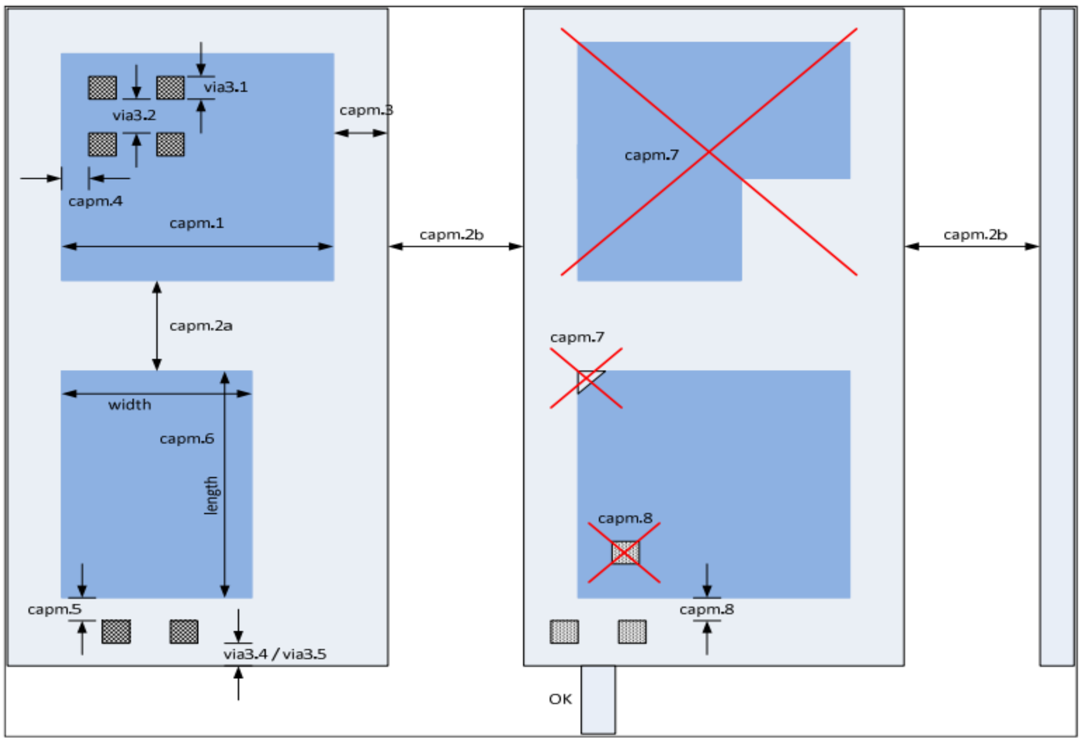
Rule	Section G2: Design Rules for S8*	Use								
(ct.-)	**Metal contact (Mcon)**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines contact between Li1 and met1									
1	Min and max L and W of mcon	**MR, DNF**	0.170	0.170	0.170	0.170	0.170	0.170	0.170	0.170
2	Spacing of mcon to mcon	**MR, DNF**	0.190	0.190	0.190	0.190	0.190	0.190	0.190	0.190
3	Only min. square mcons are allowed except die seal ring where mcons are…	**MR**	0.17*L	0.17*L	0.17*L	0.17*L	0.17*L	0.17*L	0.17*L	0.17*L
4	Minimum enclosure of mcon by LI	**P, MR**	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000



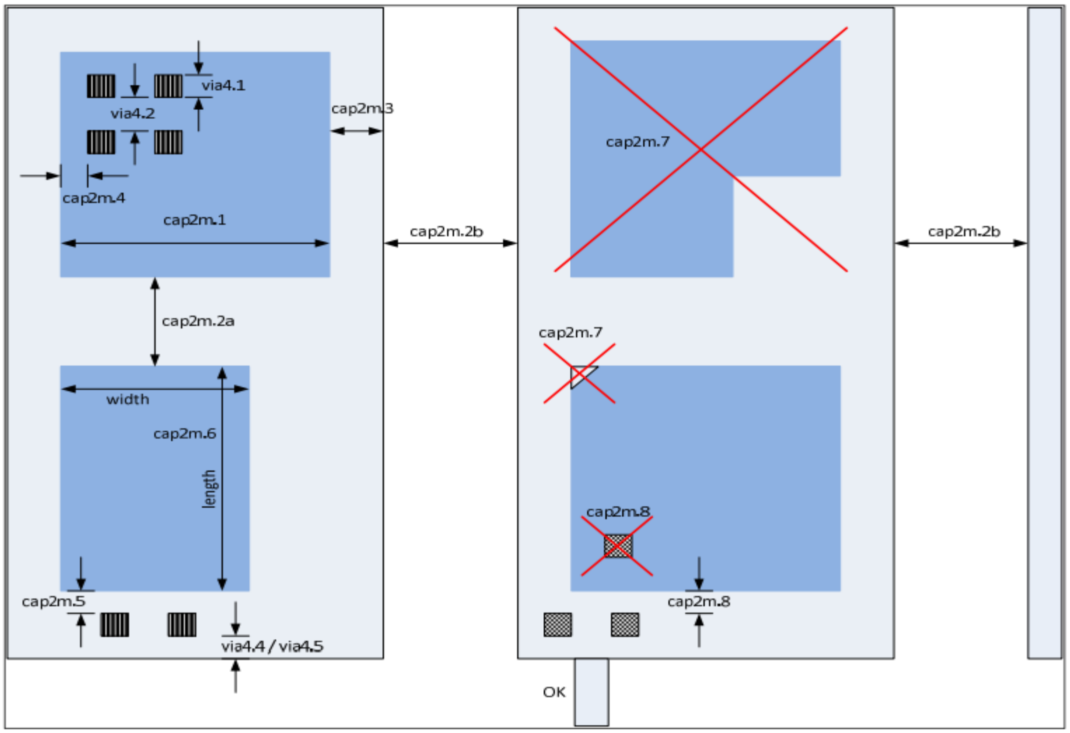
Rule	Section G2: Design Rules for S8*	Use								
(capm.-)	**MIM Capacitor (Capm)**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines MIM capacitor when placed over metal 2									
1	Min width of capm	**MR**	N/A	N/A	2.000	N/A	N/A	N/A	N/A	N/A
2a	Min spacing of capm to capm	**MR**	N/A	N/A	0.840	N/A	N/A	N/A	N/A	N/A
2b	Minimum spacing of capacitor *bottom_plate* to *bottom plate*		N/A	N/A	1.000	N/A	N/A	N/A	N/A	N/A
3	Minimum enclosure of capm (*top_plate*) by met2		N/A	N/A	0.140	N/A	N/A	N/A	N/A	N/A
4	Min enclosure of via2 by capm		N/A	N/A	0.200	N/A	N/A	N/A	N/A	N/A
5	Min spacing between capm and via2		N/A	N/A	1.270	N/A	N/A	N/A	N/A	N/A
6	Maximum Aspect Ratio (Length/Width)		N/A	N/A	20	N/A	N/A	N/A	N/A	N/A
7	Only rectangular capacitors are allowed		N/A	N/A		N/A	N/A	N/A	N/A	N/A
8	Min space, no overlap, between via and capm		N/A	N/A	0.140	N/A	N/A	N/A	N/A	N/A
10	capm must not straddle nwell, diff, tap, poly, li1 and met1 (Rule exempted for capm overlapping capm_2t.dg)	**TC**	N/A	N/A		N/A	N/A	N/A	N/A	N/A
11	Min spacing between capm to (met2 not overlapping capm)		N/A	N/A	0.500	N/A	N/A	N/A	N/A	N/A
12	Max area of capm (um^2)		N/A	N/A	1.E+07	N/A	N/A	N/A	N/A	N/A



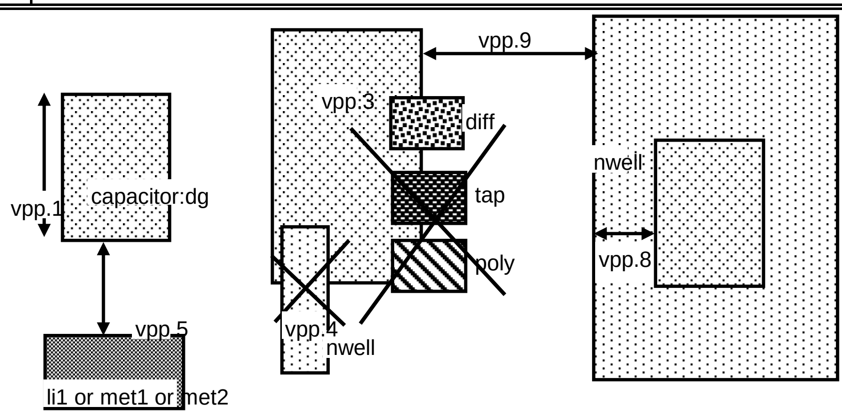
Rule	Section G2: Design Rules for S8*	Use								
(capm.-)	MIM Capacitor (Capm)	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Defines MIM capacitor when placed over met3									
1	Min width of capm	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.000
2a	Min spacing of capm to capm	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.840
2b	Minimum spacing of capacitor *bottom_plate* to any other met3	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.200
3	Minimum enclosure of capm (*top_plate*) by met3	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.140
4	Min enclosure of via3 by capm	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.140
5	Min spacing between capm and via3 (where via3 contacts met3)	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.140
6	Maximum Aspect Ratio (Length/Width)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	20.000
7	Only rectangular capacitors are allowed		N/A	N/A	N/A	N/A	N/A	N/A	N/A	
8	Min space, no overlap, between via2 and capm		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.140
11	Min spacing between capm to (met3 not overlapping capm)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.500
12	Max area of capm (um^2)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	1E+07



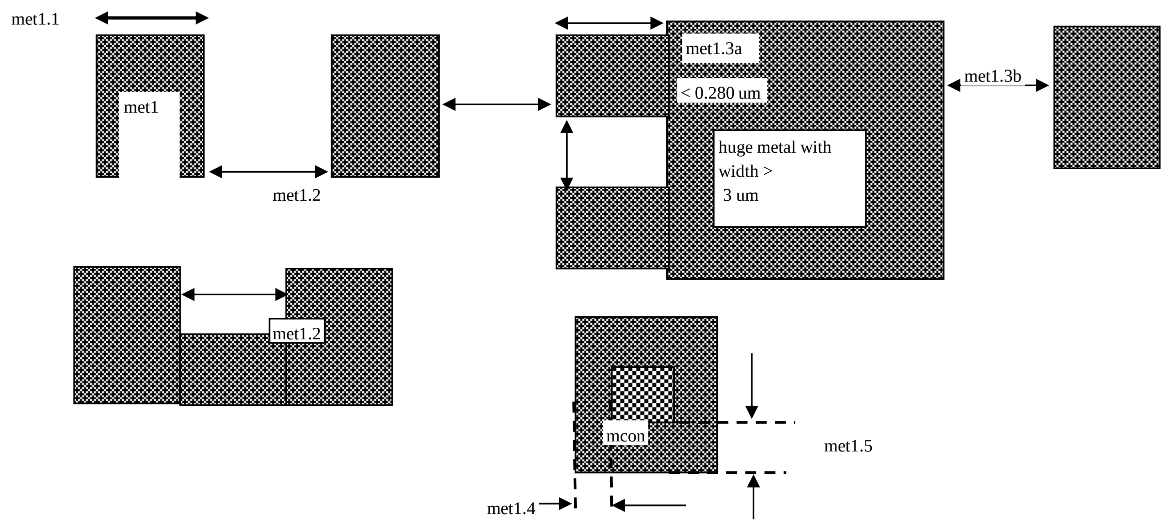
Rule	Section G2: Design Rules for S8*	Use								
(cap2m.-)	**MIM2 Capacitor (Cap2m)**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines MIM capacitor when placed over met4									
1	Min width of cap2m	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.000
2a	Min spacing of cap2m to cap2m	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.840
2b	Minimum spacing of capacitor *bottom_plate* to any other met4	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.200
3	Minimum enclosure of cap2m (*top_plate*) by met4	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.140
4	Min enclosure of via4 by cap2m	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.200
5	Min spacing between cap2m and via4 (where via4 contacts met4)	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.200
6	Maximum Aspect Ratio (Length/Width)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	20.000
7	Only rectangular capacitors are allowed		N/A	N/A	N/A	N/A	N/A	N/A	N/A	
8	Min space, no overlap, between via3 and cap2m		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.140
11	Min spacing between capm to (met3 not overlapping capm)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.500
12	Max area of capm (um^2)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	1E+07



Rule	Section G2: Design Rules for S8*	Use								
(vpp.-)	VPP Capacitor	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Defines VPP capacitor									
1	Min width of capacitor:dg		1.430	1.430	1.430	1.430	1.430	1.430	1.430	1.430
1b	Max width of capacitor:dg; Rule not applicable for *vpp_with_Met3Shield* and *vpp_with_LiShield* and *vpp_over_MOSCAP* and *vpp_with_Met5* and *vpp_with_noLi*		11.350	11.350	11.350	11.350	11.350	11.350	11.350	11.350
1c	Min/Max width of cell name "s8rf_xcmvpp1p8x1p8_m3shield "		3.880	3.880	3.880	3.880	3.880	3.880	3.880	3.880
3	capacitor:dg must not overlap (tap or diff or poly); (one exception: Poly is allowed to overlap *vpp_with_Met3Shield* and *vpp_with_Met5PolyShield*); (not applicable for *vpp_over_Moscap or "s8rf2_xcmvppx4_2xnhvnative10x4"* or *vpp_with_LiShield*)									
4	capacitor:dg must not straddle (nwell or dnwell)									
5	Min spacing between (capacitor:dg edge and (poly or li1 or met1 or met2)) to (poly or li1 or met1 or met2) on separate nets (Exempt area of the error shape less than 2.25 (um^2) and run length less than 2.0um); Rule not applicable for *vpp_with_Met3Shield* and *vpp_with_LiShield* and *vpp_over_MOSCAP* and *vpp_with_Met5* and *vpp_with_noLi*		1.500	1.500	1.500	1.500	1.500	1.500	1.500	1.500
5a	Max pattern density of met3.dg over capacitor.dg (not applicable for *vpp_with_Met3Shield* and *vpp_with_LiShield* and *vpp_over_MOSCAP* and *vpp_with_Met5*)		25.0%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%	25.0%
5b	Max pattern density of met4.dg over capacitor.dg (not applicable for *vpp_with_Met3Shield* and *vpp_with_Met5* and *vpp_over_MOSCAP*)		N/A	N/A	N/A	30.0%	30.0%	30.0%	30.0%	30.0%
5c	Max pattern density of met5.dg over capacitor.dg (not applicable for *vpp_with_Met3Shield* and *vpp_with_Met5* and *vpp_over_MOSCAP* and *vpp_with_noLi*); (one exception: rules does apply to cell "s8rf2_xcmvpp11p5x11p7_m1m4" and "s8rf2_xcmvpp_hd5_atlas*")		N/A	N/A	N/A	40.0%	40.0%	40.0%	40.0%	40.0%
8	Min enclosure of capacitor:dg by nwell		1.500	1.500	1.500	1.500	1.500	1.500	1.500	1.500
9	Min spacing of capacitor:dg to nwell (not applicable for *vpp_over_MOSCAP*)		1.500	1.500	1.500	1.500	1.500	1.500	1.500	1.500
10	vpp capacitors must not overlap; Rule checks for capacitor.dg overlapping more than one pwell pin									
11	Min pattern density of (poly and diff) over capacitor.dg; (*vpp_over_Moscap* only)		87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%	87.0%
12a	Number of met4 shapes inside capacitor.dg of cell "s8rf2_xcmvpp8p6x7p9_m3_lim5shield" must overlap with size 2.01 x 2.01 (no other met4 shapes allowed)		N/A	N/A	N/A	9.00	9.00	9.00	9.00	9.00
12b	Number of met4 shapes inside capacitor.dg of cell "s8rf2_xcmvpp11p5x11p7_m3_lim5shield" must overlap with size 2.01 x 2.01 (no other met4 shapes allowed)		N/A	N/A	N/A	16.00	16.00	16.00	16.00	16.00
12c	Number of met4 shapes inside capacitor.dg of cell "s8rf2_xcmvpp4p4x4p6_m3_lim5shield" must overlap with size 1.5 x 1.5 (no other met4 shapes allowed)		N/A	N/A	N/A	4.00	4.00	4.00	4.00	4.00

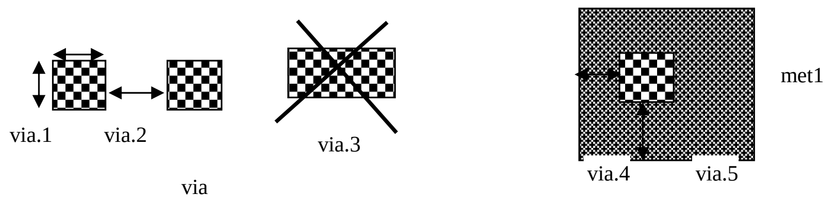


Rule	Section G2: Design Rules for S8*	Use								
(m1.-)	**Met1**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines first level of metal interconnects, buses etc;									
.X.1	Algorithm should flag errors, for met1, if ANY of the following is true: An entire 700x700 window is covered by cmm1 waffleDrop, and metX PD < 70% for same window. 80-100% of 700x700 window is covered by cmm1 waffleDrop, and metX PD < 65% for same window. 60-80% of 700x700 window is covered by cmm1 waffleDrop, and metX PD < 60% for same window. 50-60% of 700x700 window is covered by cmm1 waffleDrop, and metX PD < 50% for same window. 40-50% of 700x700 window is covered by cmm1 waffleDrop, and metX PD < 40% for same window. 30-40% of 700x700 window is covered by cmm1 waffleDrop, and metX PD < 30% for same window. Exclude cells whose area is below 40Kum2. NOTE: Required for IP, Recommended for Chip-level.	RC								
1	Width of metal1	MR	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
2	Spacing of metal1 to metal1	MR	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
3a	Min. spacing of features attached to or extending from huge_met1 for a distance of up to 0.280 um to metal1 (rule not checked over non-huge met1 features)	MR	0.280	0.280	0.280	0.280	0.280	0.280	0.280	0.280
3b	Min. spacing of huge_met1 to metal1 excluding features checked by m1.3a	MR	0.280	0.280	0.280	0.280	0.280	0.280	0.280	0.280
4	Minimum enclosure of mcon by Met1 (Rule exempted for cell names documented in rule m1.4a)	MR, P	0.030	0.030	0.030	0.030	0.030	0.030	0.030	0.030
4a	Mcon must be enclosed by Met1 by at least (for cell names "s8cell_ee_plus_sseln_a", "s8cell_ee_plus_sseln_b", "s8cell_ee_plus_sselp_a", "s8cell_ee_plus_sselp_b", "s8fpls_pl8", and "s8fs_cmux4_fm")	MR, P	NA	NA	NA	NA	0.005	0.005	NA	NA
5	Minimum enclosure of mcon by Met1 on one of two adjacent sides	MR, P, AI	0.060	0.060	0.060	0.060	0.060	0.060	0.060	0.060
6	Min metal 1 area [um^2]	MR	0.083	0.083	0.083	0.083	0.083	0.083	0.083	0.083
7	Min area of metal1 holes [um^2]	MR	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
pd.1	Min *MM1_oxide_Pattern_density*	RR, AI	70%	70%	70%	70%	70%	70%	70%	70%
pd.2a	Size of square regions used to check Rule m1.pd.1	A, AI	700	700	700	700	700	700	700	700
pd.2b	Stepping distance of sampling method used to check Rule m1.pd.1	A, AI	70	70	70	70	70	70	70	70

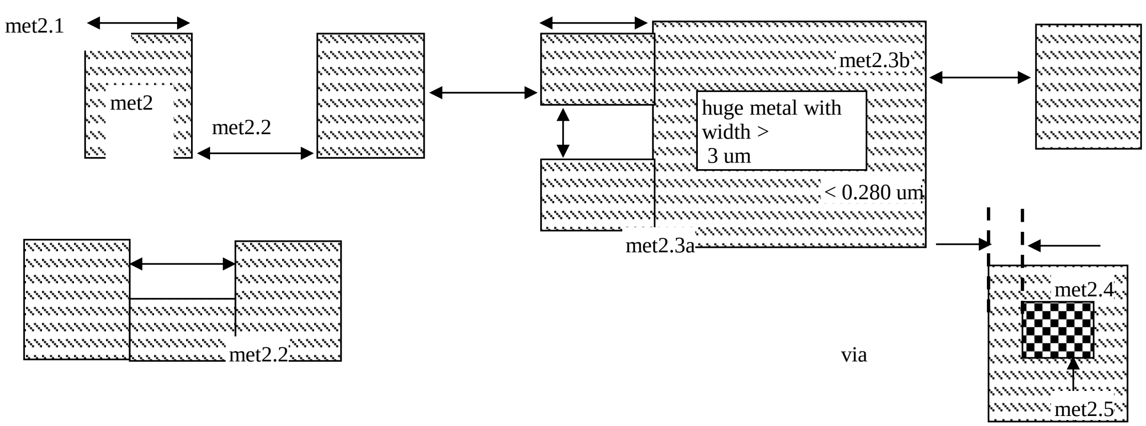


Rule	Section G2: Design Rules for S8*	Use	

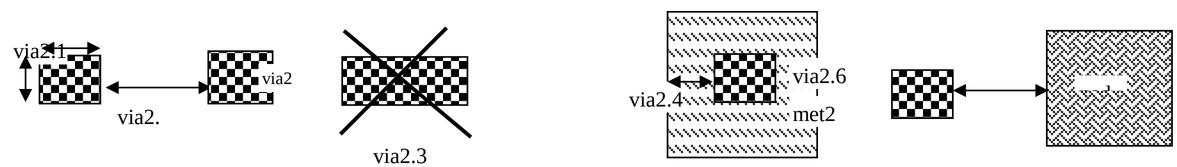
Rule	Section G2: Design Rules for S8*	Use								
(via.-)	Via	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Defines contact between met1 and met2									
1a	Min and max L and W of via outside areaid.mt	MR, AI	0.150	0.150	0.150	0.150	0.150	0.150	0.150	0.150
1b	Three sizes of square Vias allowed inside areaid:mt: 0.150um, 0.230um and 0.280um	AI								
2	Spacing of via to via	MR, AI	0.170	0.170	0.170	0.170	0.170	0.170	0.170	0.170
3	Only min. square vias are allowed except die seal ring where vias are (Via CD)*L	MR	0.2*L	0.2*L	0.2*L	0.2*L	0.2*L	0.2*L	0.2*L	0.2*L
4a	0.150 um Via must be enclosed by Met1 by at least …	MR	0.055	0.055	0.055	0.055	0.055	0.055	0.055	0.055
4b	Inside areaid.mt, min enclosure of 0.230 um Via by met1	AI	0.030	0.030	0.030	0.030	0.030	0.030	0.030	0.030
4c	Inside areaid.mt, min enclosure of 0.280 um Via by met1	AI	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
5a	Min enclosure of 0.150 um Via by Met1 on one of two adjacent sides	MR	0.085	0.085	0.085	0.085	0.085	0.085	0.085	0.085
5b	Inside areaid.mt, min enclosure of 0.230 um Via by met1 on one of two adjacent sides	AI	0.060	0.060	0.060	0.060	0.060	0.060	0.060	0.060
5c	Inside areaid.mt, min enclosure of 0.280 um Via by met1 on one of two adjacent sides	AI	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000



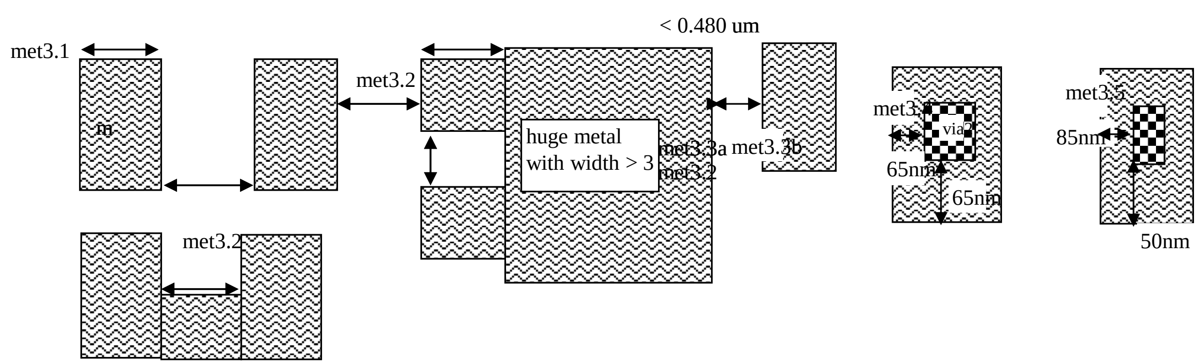
Rule	Section G2: Design Rules for S8*	Use								
(m2.-)	**Metal 2**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines second level of metal interconnects, buses etc									
.X.1	Algorithm should flag errors, for met2, if ANY of the following is true: An entire 700x700 window is covered by cmm2 waffleDrop, and metX PD < 70% for same window. 80-100% of 700x700 window is covered by cmm2 waffleDrop, and metX PD < 65% for same window. 60-80% of 700x700 window is covered by cmm2 waffleDrop, and metX PD < 60% for same window. 50-60% of 700x700 window is covered by cmm2 waffleDrop, and metX PD < 50% for same window. 40-50% of 700x700 window is covered by cmm2 waffleDrop, and metX PD < 40% for same window. 30-40% of 700x700 window is covered by cmm2 waffleDrop, and metX PD < 30% for same window. Exclude cells whose area is below 40Kum2. Required for IP, Recommended for Chip-level.	**RC**								
1	Width of metal 2	**MR**	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
2	Spacing of metal 2 to metal 2	**MR**	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
3a	Min. spacing of features attached to or extending from huge_met2 for a distance of up to 0.280 um to metal2 (rule not checked over non-huge met2 features)	**MR**	0.280	0.280	0.280	0.280	0.280	0.280	0.280	0.280
3b	Min. spacing of huge_met2 to metal2 excluding features checked by m2.3a	**MR**	0.280	0.280	0.280	0.280	0.280	0.280	0.280	0.280
3c	Min spacing between *floating_met2* with *AR_met2_A* >= 0.05 and *AR_met2_B* =< 0.032, outside areaid:sc must be greater than	**RR**	0.145	0.145	0.145	0.145	0.145	0.145	0.145	0.145
4	Min enclosure of Via by met2	**P, Al**	0.055	0.055	0.055	0.055	0.055	0.055	0.055	0.055
5	Min enclosure of Via by Met2 on one of two adjacent sides	**MR, Al**	0.085	0.085	0.085	0.085	0.085	0.085	0.085	0.085
6	Min metal2 area [um^2]	**MR**	0.0676	0.0676	0.0676	0.0676	0.0676	0.0676	0.0676	0.0676
7	Min area of metal2 holes [um^2]	**MR**	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
pd.1	Min *MM2_oxide_Pattern_density*	**RR**	70%	70%	70%	70%	70%	70%	70%	70%
pd.2a	Size of square regions used to check Rule m2.pd.1	**A**	700	700	700	700	700	700	700	700
pd.2b	Stepping distance of sampling method used to check Rule m2.pd.1	**A**	70	70	70	70	70	70	70	70



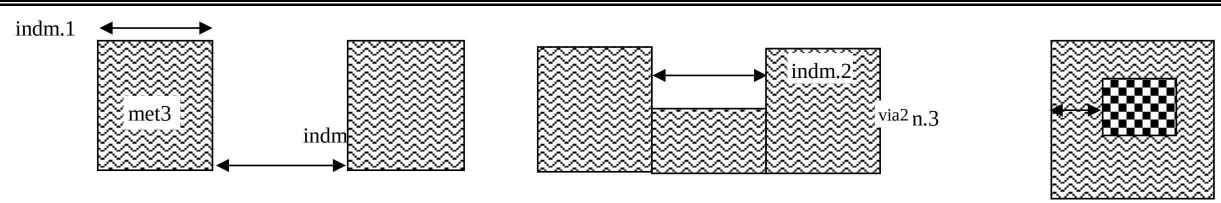
Rule	Section G2: Design Rules for S8*	Use								
(via2.-)	**Via2**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Via2 connects met2 to met3 in the S8T*/S8P*/SP8Q/SP8P* flows and met2/capm to met3 in the S8DI* flow.									
X.1	Via2 connects met2 to met3 in the S8T*/S8P*/SP8Q/SP8P* flow and met2/capm to met3 in the S8DI* flow.									
1a	Min and max L and W of via2 (except for rule via2.1b/1c/1d/1e/1f)	**MR, Al**	1.200	0.280	0.800	0.200	0.200	0.200	0.200	0.200
1b	Three sizes of square Vias allowed inside areaid:mt: 0.280um, 1.2 um and 1.5 um	**Al**	N/A		N/A	N/A	N/A	N/A	N/A	N/A
1c	Two sizes of square Vias allowed inside areaid:mt: 1.2 um and 1.5 um	**Al**		N/A	N/A	N/A	N/A	N/A	N/A	N/A
1d	Four sizes of square Vias allowed inside areaid:mt: 0.2um, 0.280um, 1.2 um and 1.5 um	**Al**	N/A	N/A	N/A					
1e	Three sizes of square Vias allowed inside areaid:mt: 0.8um, 1.2 um and 1.5 um	**Al**	N/A	N/A		N/A	N/A	N/A	N/A	N/A
1f	Two sizes of square Vias allowed outside areaid:mt: 0.8um and 1.2 um	**Al**	N/A	N/A		N/A	N/A	N/A	N/A	N/A
2	Spacing of via2 to via2	**MR, Al**	1.270	0.280	0.800	0.200	0.200	0.200	0.200	0.200
3	Only min. square via2s are allowed except die seal ring where via2s are (Via2 CD)*L	**MR, Al**	1.2*L	0.28*L	0.8*L	0.2*L	0.2*L	0.2*L	0.2*L	0.2*L
4	Min enclosure of Via2 by met2	**MR, Al**	0.190	0.095	0.190	0.040	0.040	0.040	0.040	0.040
4a	Inside areaid.mt, min enclosure of 1.5 um Via2 by met2		0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
5	Min enclosure of Via2 by Met2 on one of two adjacent sides	**MR, Al**	N/A	N/A	N/A	0.085	0.085	0.085	0.085	0.085



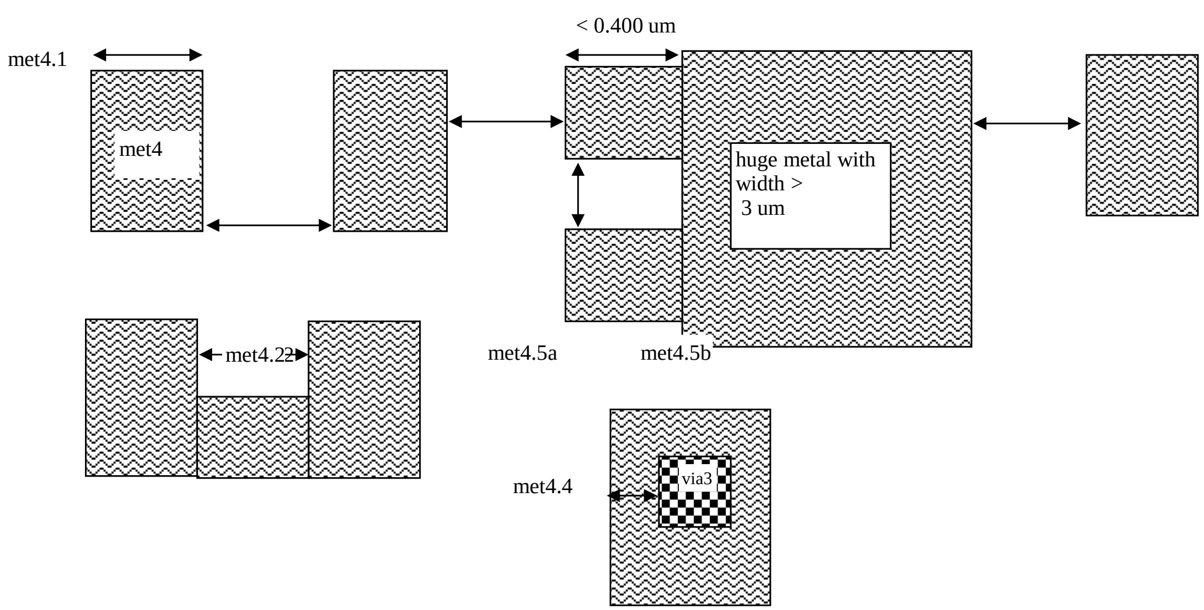
Rule	Section G2: Design Rules for S8*	Use								
(m3.-)	**Metal 3**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines third level of metal interconnects, buses etc									
.X.1	Algorithm should flag errors, for met3, if ANY of the following is true: An entire 700x700 window is covered by cmm3 waffleDrop, and metX PD < 70% for same window. 80-100% of 700x700 window is covered by cmm3 waffleDrop, and metX PD < 65% for same window. 60-80% of 700x700 window is covered by cmm3 waffleDrop, and metX PD < 60% for same window. 50-60% of 700x700 window is covered by cmm3 waffleDrop, and metX PD < 50% for same window. 40-50% of 700x700 window is covered by cmm3 waffleDrop, and metX PD < 40% for same window. 30-40% of 700x700 window is covered by cmm3 waffleDrop, and metX PD < 30% for same window. Exclude cells whose area is below 40Kum2. NOTE: Required for IP, Recommended for Chip-level.	RC								
1	Width of metal 3	MR	N/A	0.360	0.800	0.300	0.300	0.300	0.300	0.300
2	Spacing of metal 3 to metal 3	MR	N/A	0.360	0.800	0.300	0.300	0.300	0.300	0.300
3a	Min. spacing of features attached to or extending from huge_met3 for a distance of up to 0.480 um to metal3 (rule not checked over non-huge met3 features)		N/A	0.480	N/A	N/A	N/A	N/A	N/A	N/A
3b	Min. spacing of huge_met3 to metal3 excluding features checked by m3.3a		N/A	0.480	N/A	N/A	N/A	N/A	N/A	N/A
3c	Min. spacing of features attached to or extending from huge_met3 for a distance of up to 0.400 um to metal3 (rule not checked over non-huge met3 features)	MR	N/A	N/A	N/A	0.400	0.400	0.400	0.400	0.400
3d	Min. spacing of huge_met3 to met3 excluding features checked by m3.3a	MR	N/A	N/A	N/A	0.400	0.400	0.400	0.400	0.400
4	Min enclosure of Via2 by met3	MR, AI	N/A	0.045	0.310	0.065	0.065	0.065	0.065	0.065
5	Min enclosure of Via2 by Met3 on one of two adjacent sides		N/A	0.070	N/A	N/A	N/A	N/A	N/A	N/A
5a	Min enclosure of Via2 by Met3 on all sides, when either of rules m3.4 and m3.5 are not met		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
6	Min area of metal3		N/A	0.3066	4.000	0.240	0.240	0.240	0.240	0.240
pd.1	Min *MM3_oxide_Pattern_density*	RR	N/A	N/A	N/A	70%	70%	70%	70%	70%
pd.2a	Size of square regions used to check Rule m3.pd.1	A	N/A	N/A	N/A	700	700	700	700	700
pd.2b	Stepping distance of sampling method used to check Rule m3.pd.1	A	N/A	N/A	N/A	70	70	70	70	70



Rule	Section G2: Design Rules for S8*	Use								
(via3.-)	**Via3**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Via3 connects met3 to met4 in the S8Q*/S8P*/SP8Q/SP8P* flow									
1	Min and max L and W of via3 (except for rule via3.1a)	MR, AI	N/A	N/A	N/A	0.200	0.200	0.200	0.200	0.200
1a	Two sizes of square via3 allowed inside areaid.mt: 0.200um and 0.800um	AI	N/A	N/A	N/A					
2	Spacing of via3 to via3	MR, AI	N/A	N/A	N/A	0.200	0.200	0.200	0.200	0.200
3	Only min. square via3s are allowed except die seal ring where via3s are (Via3 CD)*L		N/A	N/A	N/A	0.2*L	0.2*L	0.2*L	0.2*L	0.2*L
4	Min enclosure of Via3 by Met3	MR, AI	N/A	N/A	N/A	0.060	0.060	0.060	0.060	0.060
5	Min enclosure of Via3 by Met3 on one of two adjacent sides	MR, AI	N/A	N/A	N/A	0.090	0.090	0.090	0.090	0.090
(nsm.-)	**Nitride Seal Mask**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
1	Min. width of nsm	MR	3.000	3.000	3.000	3.000	3.000	3.000	3.000	3.000
2	Min. spacing of nsm to nsm	MR	4.000	4.000	4.000	4.000	4.000	4.000	4.000	4.000
3	Min spacing, no overlap, between *NSM_keepout* to diff.dg, tap.dg, fom.dy, cfom.dg, cfom.mk, poly.dg, p1m.mk, li1.dg, cli1m.mk, metX.dg (X=1 to 5) and cmmX.mk (X=1 to 5). Exempt the following from the check: (a) cell name "nikon*" and (b) diff ring inside areaid.sl	AI	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
3a	Min enclosure of diff.dg, tap.dg, fom.dy, cfom.dg, cfom.mk, poly.dg, p1m.mk, li1.dg, cli1m.mk, metX.dg (X=1 to 5) and cmmX.mk (X=1 to 5) by areaid.ft. Exempt the following from the check: (a) cell name "s8Fab_crntic*" (b) blankings in the frame (rule uses areaid.dt for exemption)		3.000	3.000	3.000	3.000	3.000	3.000	3.000	3.000
3b	Min spacing between areaid.dt to diff.dg, tap.dg, fom.dy, cfom.dg, cfom.mk, poly.dg, p1m.mk, li1.dg, cli1m.mk, metX.dg (X=1 to 5) and cmmX.mk (X=1 to 5). Exempt the following from the check: (a) blankings in the frame (rule uses areaid.dt for exemption)		3.000	3.000	3.000	3.000	3.000	3.000	3.000	3.000
(indm.-)	**Inductor metal**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines third level of metal interconnects, buses and inductor; *top_indmMetal* is met3 for S8D* flows; Similarly *top_padVia* is Via2 for S8D*									
1	Min width of *top_indmMetal*	MR	2.500	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2	Min spacing between two *top_indmMetal*	MR	2.500	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3	*top_padVia* must be enclosed by *top_indmMetal* by at least	MR	0.310	N/A	N/A	N/A	N/A	N/A	N/A	N/A
4	Min area of *top_indmMetal*	MR	15.000	N/A	N/A	N/A	N/A	N/A	N/A	N/A



Rule	Section G2: Design Rules for S8*	Use								
(m4.-)	Metal 4	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Defines Fourth level of metal interconnects;									
.X.1	Algorithm should flag errors, for met4, if ANY of the following is true: An entire 700x700 window is covered by cmm4 waffleDrop, and metX PD < 70% for same window. 80-100% of 700x700 window is covered by cmm4 waffleDrop, and metX PD < 65% for same window. 60-80% of 700x700 window is covered by cmm4 waffleDrop, and metX PD < 60% for same window. 50-60% of 700x700 window is covered by cmm4 waffleDrop, and metX PD < 50% for same window. 40-50% of 700x700 window is covered by cmm4 waffleDrop, and metX PD < 40% for same window. 30-40% of 700x700 window is covered by cmm4 waffleDrop, and metX PD < 30% for same window. Exclude cells whose area is below 40Kum2. Required for IP, Recommended for Chip-level.	RC								
1	Min width of met4	MR	N/A	N/A	N/A	0.300	0.300	0.300	0.300	0.300
2	Min spacing between two met4	MR	N/A	N/A	N/A	0.300	0.300	0.300	0.300	0.300
3	Min enclosure of via3 by met4	MR, AI	N/A	N/A	N/A	0.065	0.065	0.065	0.065	0.065
4	Min area of met4 (rule exempted for probe pads which are exactly 1.42um by 1.42um)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
4a	Min area of met4	MR	N/A	N/A	N/A	0.240	0.240	0.240	0.240	0.240
5a	Min. spacing of features attached to or extending from huge_met4 for a distance of up to 0.400 um to metal4 (rule not checked over non-huge met4 features)	MR	N/A	N/A	N/A	0.400	0.400	0.400	0.400	0.400
5b	Min. spacing of huge_met4 to metal4 excluding features checked by m4.5a	MR	N/A	N/A	N/A	0.400	0.400	0.400	0.400	0.400
pd.1	Min *MM4_oxide_Pattern_density*	RR	N/A	N/A	N/A	70%	70%	70%	70%	70%
pd.2a	Size of square regions used to check Rule m4.pd.1	A	N/A	N/A	N/A	700	700	700	700	700
pd.2b	Stepping distance of sampling method used to check Rule m4.pd.1	A	N/A	N/A	N/A	70	70	70	70	70

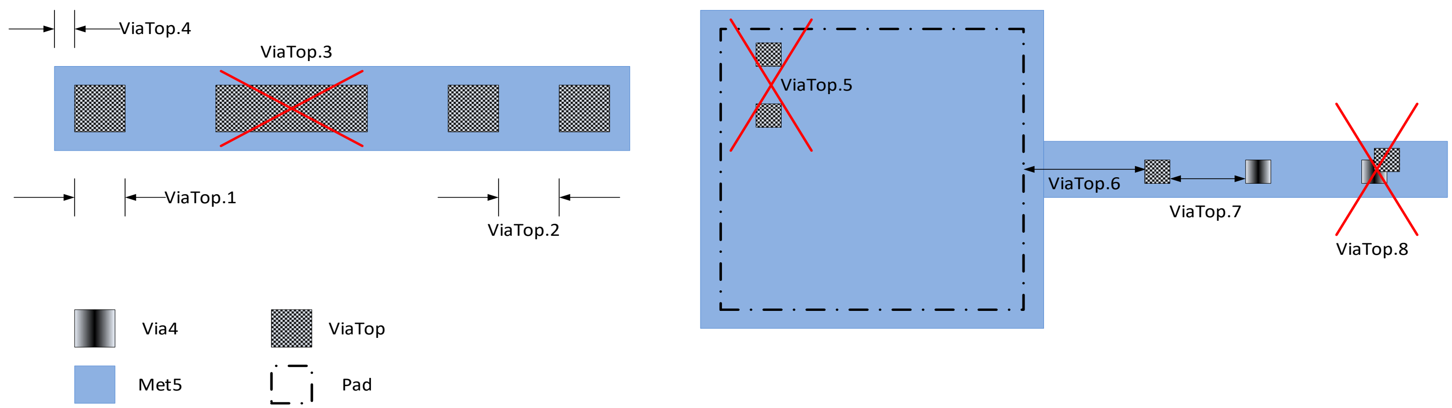


Rule	Section G2: Design Rules for S8*	Use								
(via4.-)	**Via4**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Via4 connects met4 to met5 in the S8P*/SP8P* flow									
1	Min and max L and W of via4	**MR**	N/A	N/A	N/A	0.800	0.800	0.800	0.800	0.800
2	Spacing of via4 to via4	**MR**	N/A	N/A	N/A	0.800	0.800	0.800	0.800	0.800
3	Only min. square via4s are allowed except die seal ring where via4s are (Via4 CD)*L	**MR**	N/A	N/A	N/A	0.8*L	0.8*L	0.8*L	0.8*L	0.8*L
4	Min enclosure of Via4 by Met4	**MR**	N/A	N/A	N/A	0.190	0.190	0.060	0.060	0.060

(m5.-)	Metal 5	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Defines Fifth level of metal interconnects;									
1	Min width of met5	**MR**	N/A	N/A	N/A	0.800	1.600	1.600	1.600	1.600
2	Min spacing between two met5	**MR**	N/A	N/A	N/A	0.800	1.600	1.600	1.600	1.600
3	Min enclosure of via4 by met5	**MR**	N/A	N/A	N/A	0.310	0.310	0.310	0.310	0.310
4	Min area of met5 (For all flows except S8PIR*/S8PF*, the rule is exempted for probe pads which are exactly 1.42um by 1.42um)	**MR**	N/A	N/A	N/A	4.000	4.000	4.000	4.000	4.000

(pad.-)	Pad	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Opens the passivation									
2	Min spacing of pad:dg to pad:dg	**MR**	N/A	N/A	N/A	N/A	1.270	N/A	N/A	N/A
3	Max area of *hugePad* NOT *top_metal*		30000	30000	30000	30000	30000	30000	30000	30000

Rule	Section G2: Design Rules for S8*	Use								
(viatop.-)	**ViaTop**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Opens topside vias to enable wafer-wafer or die-die bonding									
1	Min and max L and W of ViaTop	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.000
2	Min spacing of ViaTop to ViaTop	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.200
3	Only min. square ViaTop's are allowed	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
4	Minimum enclosure of ViaTop by met5		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.400
5	ViaTop must not be placed over Pad	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
6	Min spacing of ViaTop to Pad		N/A	N/A	N/A	N/A	N/A	N/A	N/A	10.000
7	ViaTop must not be placed over Via4	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
8	Min spacing of ViaTop to Via4		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.200

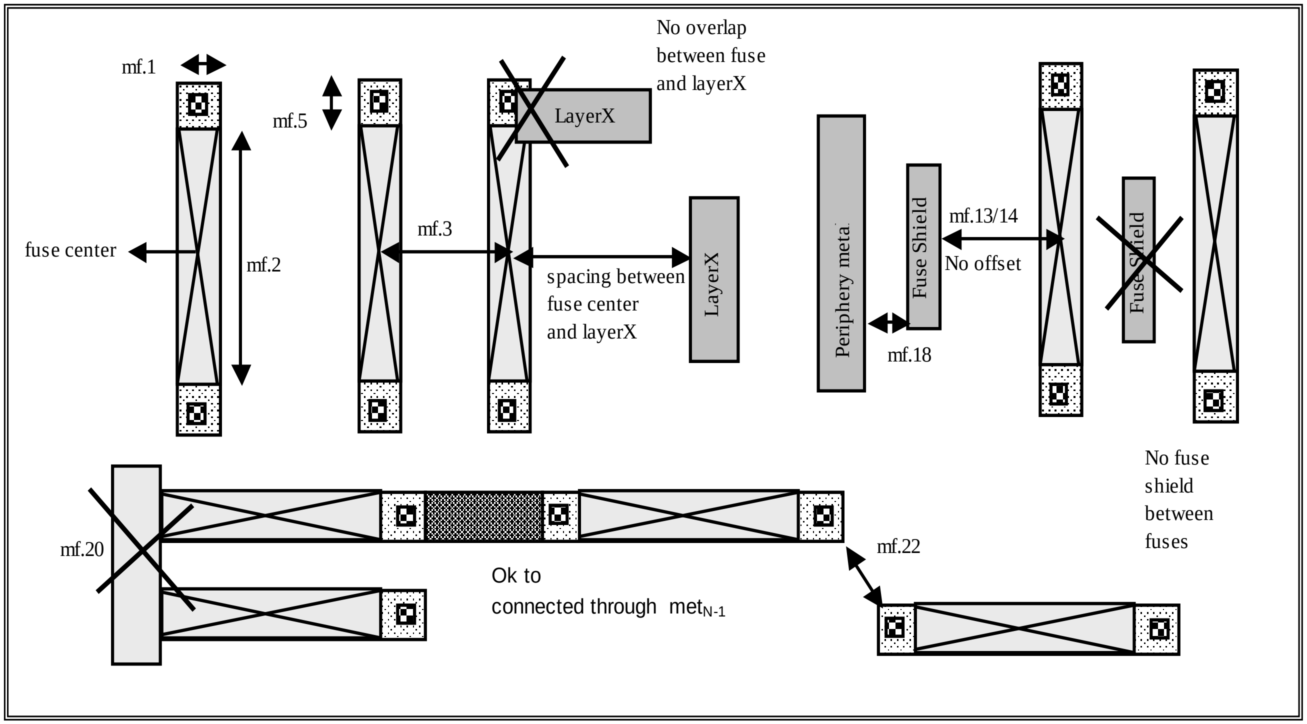


(mtest.-_	**Metal-Test**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Test Metal layer used only for etest verification of ViaTop. Not for customer use.									
1	Min width of Mtest	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.000
2	Min spacing of Mtest to Mtest	MR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.000
3	Minimum enclosure of ViaTop by Mtest		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.400
4	MTest is not to be used on any product wafers which must be shipped to a customer.		N/A	N/A	N/A	N/A	N/A	N/A	N/A	

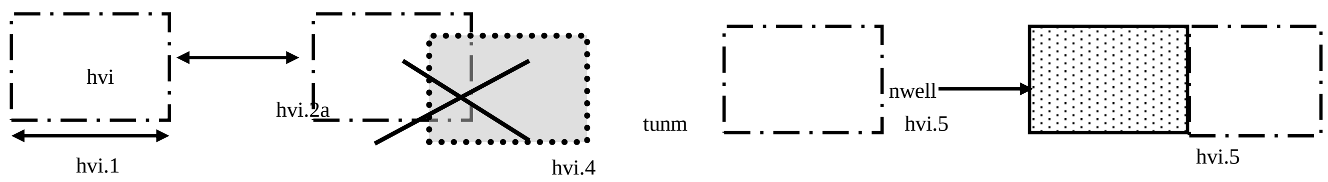
(rdl.-)	**Cu Inductor**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines the Cu Inductor. Connects to met5 through the pad opening									
1	Min width of rdl	MR	N/A	N/A	N/A	N/A	10	N/A	N/A	N/A
2	Min spacing between two rdl	MR	N/A	N/A	N/A	N/A	10	N/A	N/A	N/A
3	Min enclosure of pad by rdl, except rdl interacting with bump		N/A	N/A	N/A	N/A	10.750	N/A	N/A	N/A
4	Min spacing between rdl and outer edge of the seal ring		N/A	N/A	N/A	N/A	15.000	N/A	N/A	N/A
5	(rdl OR ccu1m.mk) must not overlap areaid.ft. Exempt the following from the check: (a) blankings in the frame (rule uses areaid.dt for exemption)		N/A	N/A	N/A	N/A		N/A	N/A	N/A
6	Min spacing of rdl to pad, except rdl interacting with bump		N/A	N/A	N/A	N/A	19.660	N/A	N/A	N/A

Rule	Section G2: Design Rules for S8*	Use								
(mf.-)	**Metal Fuse**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines metal fuses									
	For S8D* and S8TM* CADflow use MM2 for Metal Fuse									
	For SP8P*/S8P* (PLM) CADflow use MM4 for Metal Fuse									
1	Min. and max width of fuse		0.800	0.800	0.800	0.800	0.800	0.800	0.800	0.800
2	Length of fuse		7.200	7.200	7.200	7.200	7.200	7.200	7.200	7.200
3	Spacing between centers of adjacent fuses		2.760	2.760	2.760	2.760	2.760	2.760	2.760	2.760
4	Spacing between center of fuse and *fuse_metal* (fuse shields are exempted)		3.300	3.300	3.300	3.300	3.300	3.300	3.300	3.300
5	Max. extension of *fuse_metal* beyond fuse boundary		0.830	0.830	0.830	0.830	0.830	0.830	0.830	0.830
6	Spacing (no overlapping) between fuse center and Metal1		3.300	3.300	3.300	3.300	3.300	3.300	3.300	3.300
7	Spacing (no overlapping) between fuse center and LI		3.300	3.300	3.300	3.300	3.300	3.300	3.300	3.300
8	Spacing (no overlapping) between fuse center and poly		2.660	2.660	2.660	2.660	2.660	2.660	2.660	2.660
9	Spacing (no overlapping) between fuse center and tap		2.640	2.640	2.640	2.640	2.640	2.640	2.640	2.640
10	Spacing (no overlapping) between fuse center and diff		3.250	3.250	3.250	3.250	3.250	3.250	3.250	3.250
11	Spacing (no overlapping) between fuse center and nwell		3.320	3.320	3.320	3.320	3.320	3.320	3.320	3.320
12	Size of *fuse_shield*		0.500 x 2.400	0.500 x 2.400	0.500 x 2.400	0.500 x 2.400	0.500 x 2.400	0.500 x 2.400	0.500 x 2.400	0.500 x 2.400
13	Min. spacing of center of fuse to *fuse_shield*		2.200	2.200	2.200	2.200	2.200	2.200	2.200	2.200
14	Max. spacing of center of fuse to *fuse_shield*		3.300	3.300	3.300	3.300	3.300	3.300	3.300	3.300
15	*Fuse_shields* are only placed between periphery metal (i.e., without fuse:dg) and non-isolated edges of fuse as defined by mf.16									
16	The edge of a fuse is considered *non-isolated* if wider than or equal to mf.2 and spaced to *fuse_metal* by less than …		4.000	4.000	4.000	4.000	4.000	4.000	4.000	4.000
17	Offset between *fuse_shields* center and fuse center	**NC**	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
18	Min and max space between *fuse_shield* and *fuse_metal* (opposite edges). **Rule checked within 1 gridpoint**.		0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600
19	Spacing (no overlapping) between fuse center and Metal2		N/A	3.300	N/A	3.300	3.300	3.300	3.300	3.300
20	Only one fuse per metal line allowed									
21	Min spacing , no overlap, between metal3 and fuse center		3.300	3.300	3.300	3.300	3.300	3.300	3.300	3.300
22	Min spacing between *fuse_contact* to *fuse_contact*		1.960	1.960	1.960	1.960	1.960	1.960	1.960	1.960
23	Spacing (no overlapping) between fuse center and Metal4		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
24	Spacing (no overlapping) between fuse center and Metal5		N/A	N/A	N/A	3.300	3.300	3.300	3.300	3.300

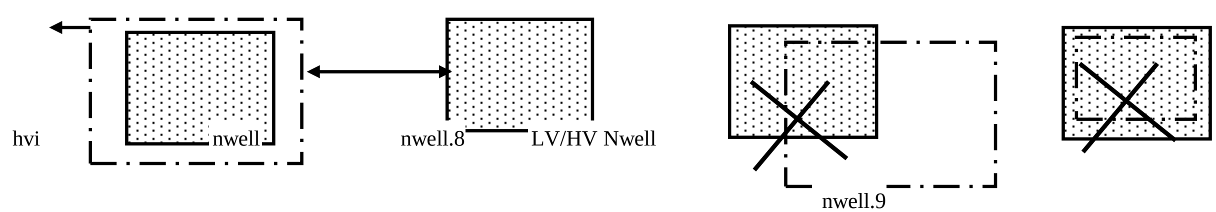
Rule	Section G2: Design Rules for S8*	Use	



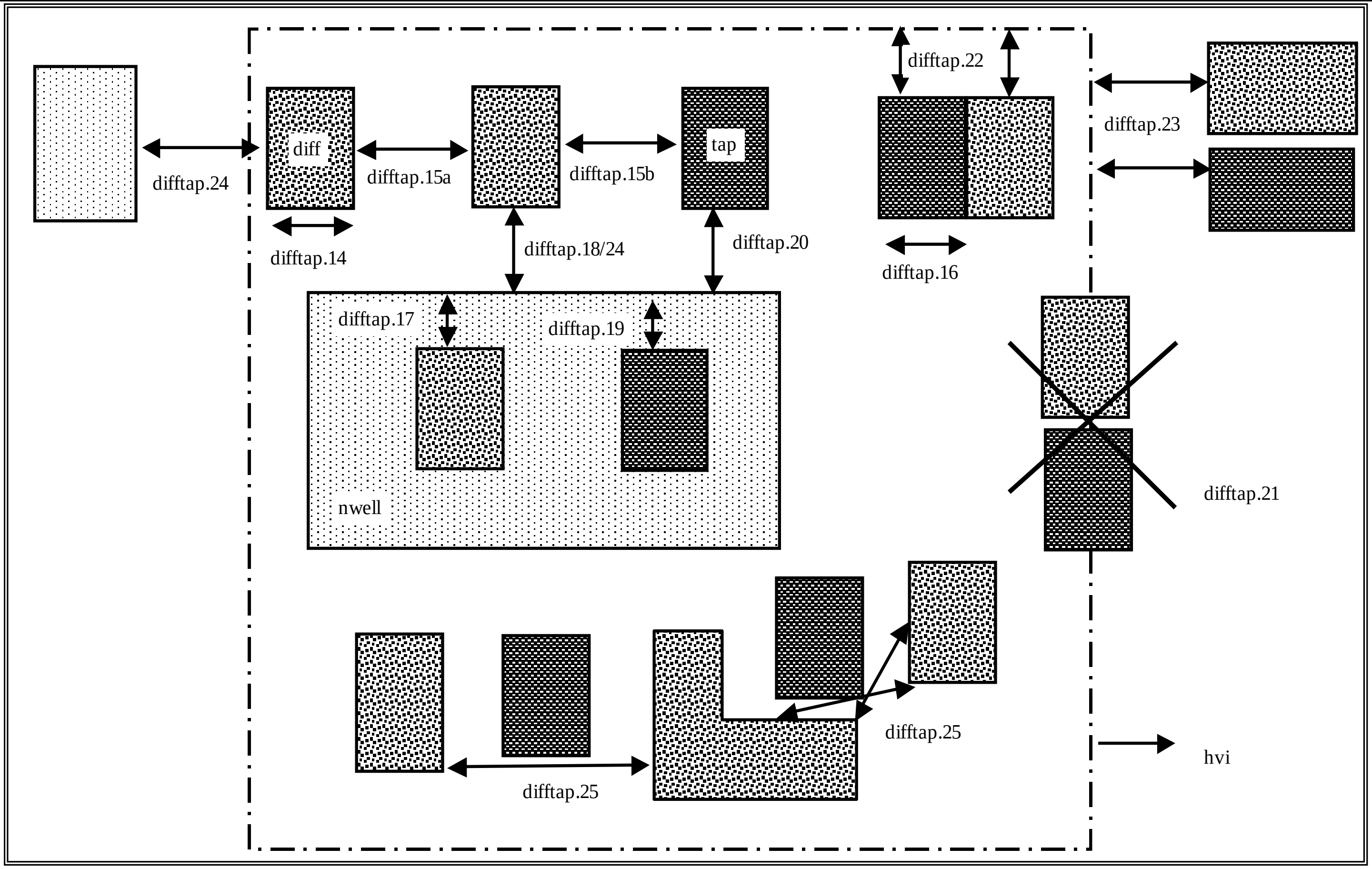
Rule	Section G2: Design Rules for S8*	Use								
	Section G2b: Rules for HV devices									
(hvi.-)	**HVI**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines thick oxide for high voltage devices									
1	Min width of Hvi	**MR, P**	0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600
2a	Min spacing of Hvi to Hvi	**MR, P**	0.700	0.700	0.700	0.700	0.700	0.700	0.700	0.700
2b	Manual merge if space is below minimum									
4	Hvi must not overlap tunm									
5	Min space between hvi and nwell (exclude coincident edges)		0.700	0.700	0.700	0.700	0.700	0.700	0.700	0.700



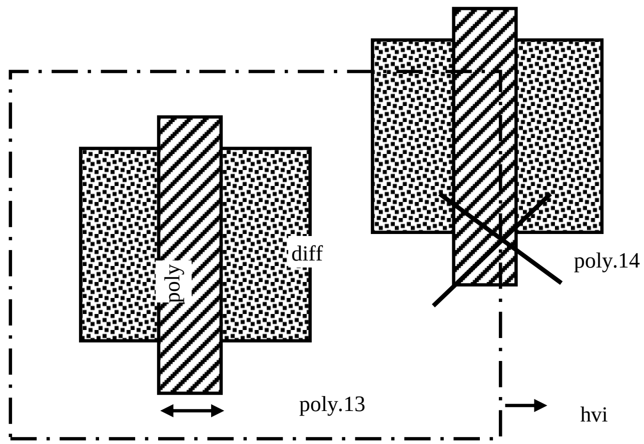
(nwell.-)	**High Voltage Nwell**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines rules for HV nwell; All nwell connected to voltages greater than 1.8V must be enclosed by hvi; Nets connected to LV nwell or nwell overlapping hvi but connected to LV voltages (i.e 1.8V) should be tagged "lv_net" using text.dg; This tag should be only on Li layer									
8	Min space between *HV_nwell* and any nwell on different nets		2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
9	(Nwell overlapping hvi) must be enclosed by hvi									
10	*LVnwell* and *HnWell* should not be on the same net (for the purposes of this check, short the connectivity through resistors); Exempt *HnWell* with li nets tagged "lv_net" using text.dg and *Hnwell* connected to nwell overlapping areaid.hl	**TC**								
11	Nwell connected to the nets mentioned in the "Power_Net_Hv" field of the latcup GUI must be enclosed by hvi (exempt nwell inside areaid.hl). Also for the purposes of this check, short the connectivity through resistors. The rule will be checked in the latchup run and exempted for cells "s8tsg5_tx_ibias_gen" and "s8bbcnv_psoc3p_top_18", "rainier_top, indus_top*", "rainier_top, manas_top, ccg3_top"									



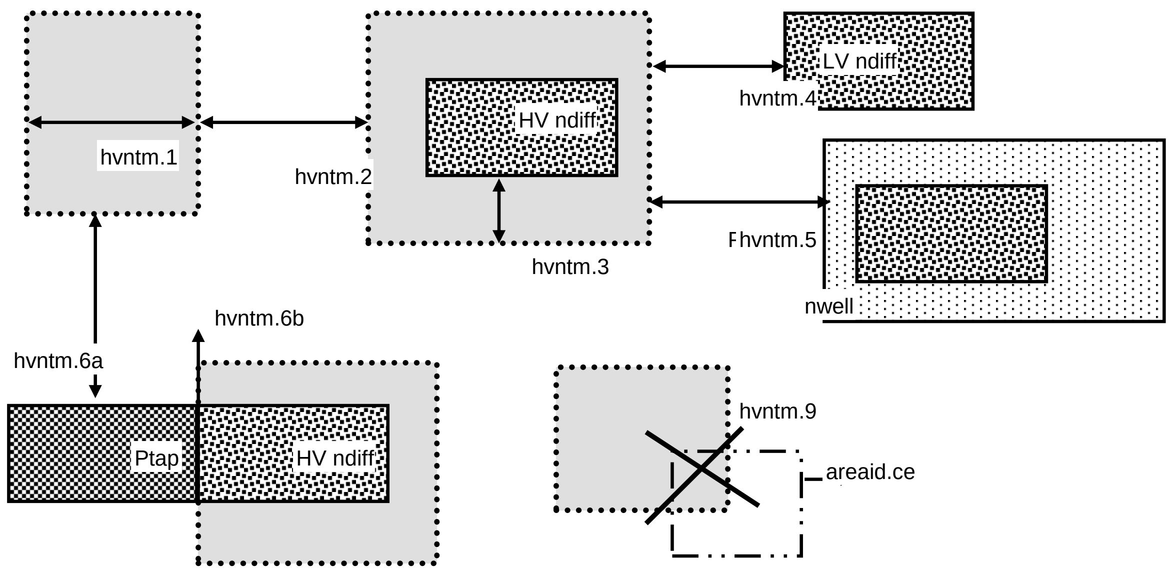
Rule	Section G2: Design Rules for S8*	Use								
(difftap.-)	High Voltage Diff/Tap	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Defines rules for HV diff/tap									
14	Min width of diff inside Hvi, except HV *Pdiff* resistors (difftap.14a)	P	0.290	0.290	0.290	0.290	0.290	0.290	0.290	0.290
14a	Min width of diff inside Hvi, HV Pdiff resistors only	P	0.150	0.150	0.150	0.150	0.150	0.150	0.150	0.150
15a	Min space of *Hdiff* to *Hdiff*	P	0.300	0.300	0.300	0.300	0.300	0.300	0.300	0.300
15b	Min space of n+diff to non-abutting p+tap inside Hvi	P	0.370	0.370	0.370	0.370	0.370	0.370	0.370	0.370
16	Min width tap butting diff on one or two sides inside Hvi (rule exempted inside UHVI)		0.700	0.700	0.700	0.700	0.700	0.700	0.700	0.700
17	P+ *Hdiff* or Pdiff inside areaid:hvnwell must be enclosed by *Hv_nwell* by at least ….[Rule exempted inside UHVI]	DE, NE	0.330	0.330	0.330	0.330	0.330	0.330	0.330	0.330
18	Spacing of N+ diff to *HV_nwell* (rule exempted inside UHVI)	DE, NE	0.430	0.430	0.430	0.430	0.430	0.430	0.430	0.430
19	N+ *Htap* must be enclosed by *Hv_nwell* by at least …Rule exempted inside UHVI.	NE	0.330	0.330	0.330	0.330	0.330	0.330	0.330	0.330
20	Spacing of P+ tap to *HV_nwell* (Exempted for p+tap butting pwell.rs; rule exempted inside UHVI)		0.430	0.430	0.430	0.430	0.430	0.430	0.430	0.430
21	Diff or tap cannot straddle Hvi	P								
22	Min enclosure of *Hdiff* or *Htap* by Hvi. Rule exempted inside UHVI.	P	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180
23	Space between diff or tap outside Hvi and Hvi	P	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180
24	Spacing of nwell to N+ *Hdiff* (rule exempted inside UHVI)	DE, NE	0.430	0.430	0.430	0.430	0.430	0.430	0.430	0.430
25	Min space of N+ *Hdiff* inside HVI across non-abutting P+_tap	NC	1.070	1.070	1.070	1.070	1.070	1.070	1.070	1.070
26	Min spacing between pwbm to difftap outside UHVI		N/A	N/A	N/A	N/A	N/A	0.500	N/A	N/A

Rule	Section G2: Design Rules for S8*	Use	
			

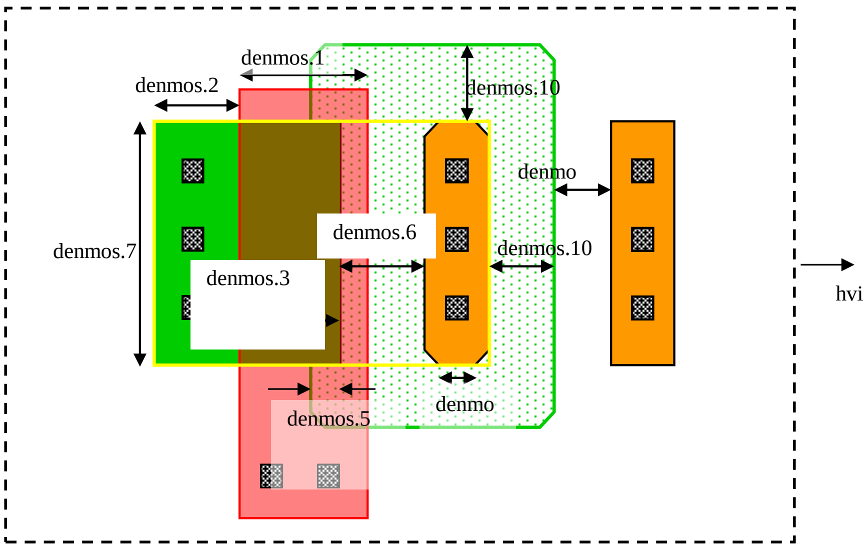
Rule	Section G2: Design Rules for S8*	Use								
(poly.-)	High Voltage Poly	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Defines rules for HV poly									
13	Min width of poly over diff inside Hvi	P	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500
14	(poly and diff) cannot straddle Hvi									



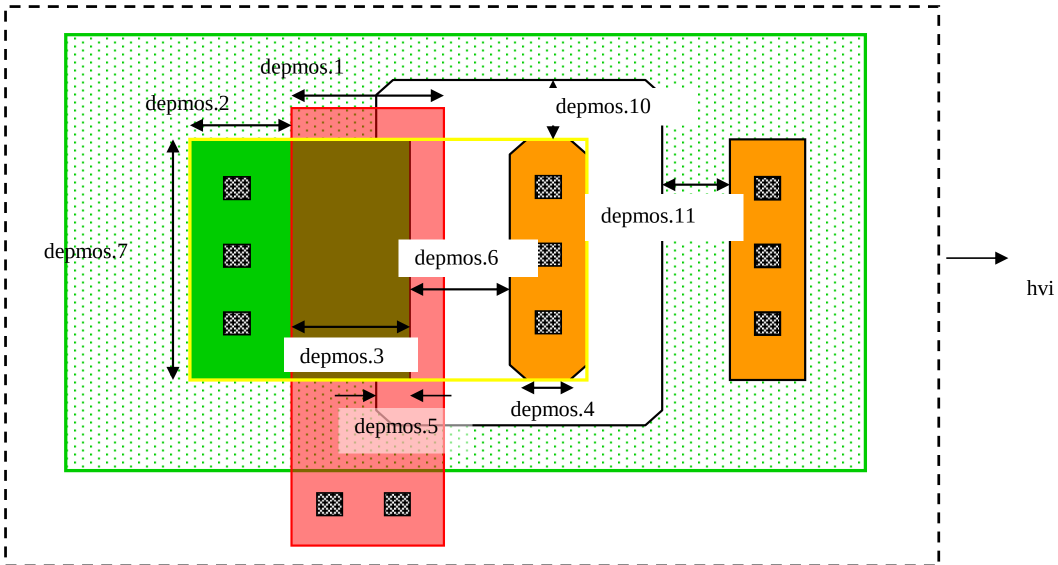
Rule	Section G2: Design Rules for S8*	Use								
(hvntm.-)	Hvntm	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Defines tip implants for the HV NMOS									
X.1	Hvntm can be drawn inside HVI. Drawn layer will be OR-ed with the CL and rechecked for CLDRC									
1	Width of hvntm	MR, P	0.700	0.700	0.700	0.700	0.700	0.700	0.700	0.700
2	Spacing of hvntm to hvntm	MR, P	0.700	0.700	0.700	0.700	0.700	0.700	0.700	0.700
3	Min. enclosure of (n+_diff inside Hvi) but not overlapping areaid.ce by hvntm	P	0.185	0.185	0.185	0.185	0.185	0.185	0.185	0.185
4	Space, no overlap, between n+_diff outside Hvi and hvntm	P	0.185	0.185	0.185	0.185	0.185	0.185	0.185	0.185
5	Space, no overlap, between p+_diff and hvntm	P, DE	0.185	0.185	0.185	0.185	0.185	0.185	0.185	0.185
6a	Space, no overlap, between p+_tap and hvntm (except along the diff-butting edge)	P	0.185	0.185	0.185	0.185	0.185	0.185	0.185	0.185
6b	Space, no overlap, between p+_tap and hvntm along the diff-butting edge	P	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
7	hvntm must enclose ESD_nwell_tap inside hvi by at least	P	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
9	Hvntm must not overlap areaid.ce									
10	Hvntm must overlap hvi									



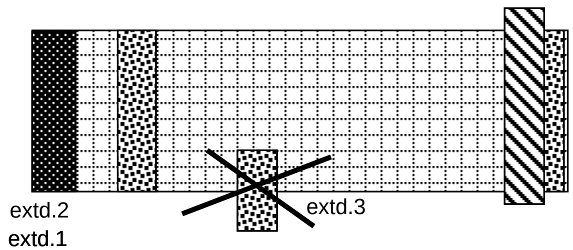
Rule	Section G2: Design Rules for S8*	Use							
(denmos.-)	Denmos	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Defines rules for the 16V Drain extended NMOS devices									
1	Min width of *de_nFet_gate*		1.055	1.055	1.055	1.055	1.055	1.055	1.055	1.055
2	Min width of *de_nFet_source* not overlapping poly		0.280	0.280	0.280	0.280	0.280	0.280	0.280	0.280
3	Min width of *de_nFet_source* overlapping poly		0.925	0.925	0.925	0.925	0.925	0.925	0.925	0.925
4	Min width of the *de_nFet_drain*		0.170	0.170	0.170	0.170	0.170	0.170	0.170	0.170
5	Min/Max extension of *de_nFet_source* over nwell		0.225	0.225	0.225	0.225	0.225	0.225	0.225	0.225
6	Min/Max spacing between *de_nFet_drain* and *de_nFet_source*		1.585	1.585	1.585	1.585	1.585	1.585	1.585	1.585
7	Min channel width for *de_nFet_gate*		5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000
8	90 degree angles are not permitted for nwell overlapping *de_nFET_drain*									
9a	All bevels on nwell are 45 degree, 0.43 um from corners		**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**
9b	All bevels on *de_nFet_drain* are 45 degree, 0.05 um from corners		**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**
10	Min enclosure of *de_nFet_drain* by nwell		0.660	0.660	0.660	0.660	0.660	0.660	0.660	0.660
11	Min spacing between p+ tap and (nwell overlapping *de_nFet_drain*)		0.860	0.860	0.860	0.860	0.860	0.860	0.860	0.860
12	Min spacing between nwells overlapping *de_nFET_drain*		2.400	2.400	2.400	2.400	2.400	2.400	2.400	2.400
13	*de_nFet_source* must be enclosed by nsdm by		0.130	0.130	0.130	0.130	0.130	0.130	0.130	0.130
14	nvhv FETs must be enclosed by areaid.mt					N/A	N/A	N/A	N/A	N/A



Rule	Section G2: Design Rules for S8*	Use								
(depmos.-)	Depmos	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Defines rules for the 16V Drain extended NMOS devices									
1	Min width of *de_pFet_gate*		1.050	1.050	1.050	1.050	1.050	1.050	1.050	1.050
2	Min width of *de_pFet_source* not overlapping poly		0.280	0.280	0.280	0.280	0.280	0.280	0.280	0.280
3	Min width of *de_pFet_source* overlapping poly		0.920	0.920	0.920	0.920	0.920	0.920	0.920	0.920
4	Min width of the *de_pFet_drain*		0.170	0.170	0.170	0.170	0.170	0.170	0.170	0.170
5	Min/Max extension of *de_pFet_source* beyond nwell		0.260	0.260	0.260	0.260	0.260	0.260	0.260	0.260
6	Min/Max spacing between *de_pFet_drain* and *de_pFet_source*		1.190	1.190	1.190	1.190	1.190	1.190	1.190	1.190
7	Min channel width for *de_pFet_gate*		5.000	5.000	5.000	5.000	5.000	5.000	5.000	5.000
8	90 degree angles are not permitted for nwell hole overlapping *de_pFET_drain*									
9a	All bevels on nwell hole are 45 degree, 0.43 um from corners		**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**
9b	All bevels on *de_pFet_drain* are 45 degree, 0.05 um from corners		**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**
10	Min enclosure of *de_pFet_drain* by nwell hole		0.860	0.860	0.860	0.860	0.860	0.860	0.860	0.860
11	Min spacing between n+ tap and (nwell hole enclosing *de_pFET_drain*)		0.660	0.660	0.660	0.660	0.660	0.660	0.660	0.660
12	*de_pFet_source* must be enclosed by psdm by		0.130	0.130	0.130	0.130	0.130	0.130	0.130	0.130
13	pvhv fets(except those with W/L = 5.0/0.66) must be enclosed by areaid.mt					N/A	N/A	N/A	N/A	N/A

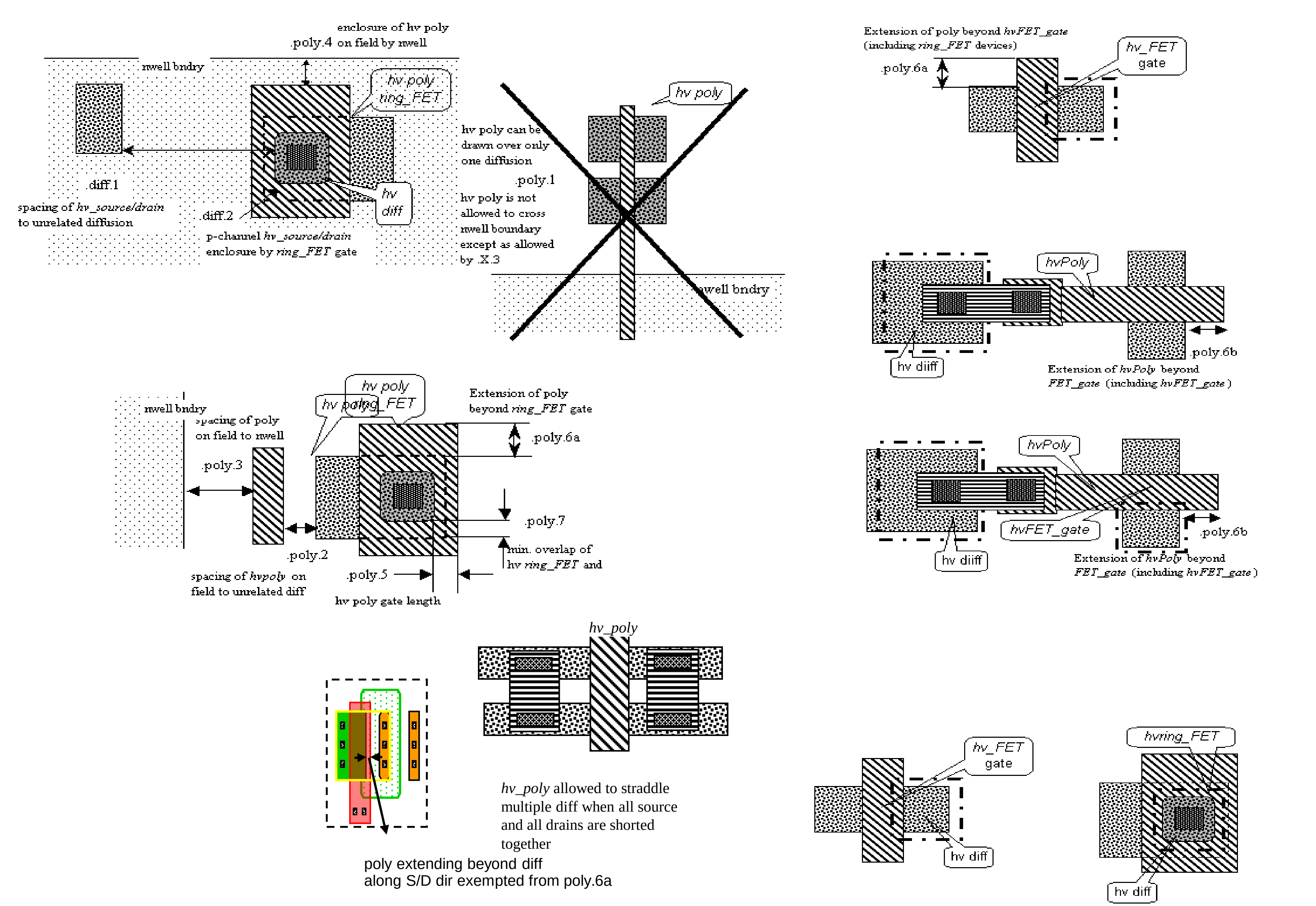


Rule	Section G2: Design Rules for S8*	Use								
(extd.-)	**Extended Drain**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Defines rules areaid.en									
1	Difftap cannot straddle areaid:en									
2	DiffTap must have 2 or 3 coincident edges with areaid:en if enclosed by areaid:en									
3	Poly must not be entirely overlapping difftap in areaid:en									
4	Only cell name "s8rf_n20vhv1*" is a valid cell name for n20vhv1 device (LVS check)		N/A	N/A	N/A	N/A	N/A		N/A	N/A
5	Only cell name "s8rf_n20vhviso1" is a valid cell name for n20vhviso1 device (LVS check)		N/A	N/A	N/A	N/A	N/A		N/A	N/A
6	Only cell name "s8rf_p20vhv1" is a valid cell name for p20vhv1 device (LVS check)		N/A	N/A	N/A	N/A	N/A		N/A	N/A
7	Only cell name "s8rf_n20nativevhv1*" is a valid cell name for n20nativevhv1 device (LVS check)		N/A	N/A	N/A	N/A	N/A		N/A	N/A
8	Only cell name "s8rf_n20zvtvhv1*" is a valid cell name for n20zvtvhv1 device (LVS check)		N/A	N/A	N/A	N/A	N/A		N/A	N/A



Rule	Section G2: Design Rules for S8*	Use								
(hv.-.-)	**High Voltage Rules**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
Note	High voltage rule apply for an operating voltage range of 5.5 - 12V; Nodes switching between 0 to 5.5V do not need to follow these rules									
.X.1	High voltage source/drain regions must be tagged by diff:hv									
.X.3	High voltage poly can be drawn over multiple diff regions that are ALL reverse-biased by at least 300 mV (existence of reverse-bias is not checked by the CAD flow). It can also be drawn over multiple diffs when all sources and all drain are shorted together. In these case, the high voltage poly can be tagged with the text:dg label with a value “hv_bb”. Exceptions to this use of the hv_bb label must be approved by technology. Under certain bias conditions, high voltage poly tagged with hv_bb can cross an nwell boundary. The poly of the drain extended device crosses nwell by construction and can be tagged with the "hv_bb" label. Use of the hv_bb label on high voltage poly crossing an nwell boundary must be approved by technology. All high voltage poly tagged with hv_bb will not be checked to hv.poly.1, hv.poly.2, hv.poly.3 and hv.poly.4.									
.X.4	Any piece of layout that is shorted to *hv_source/drain* becomes a high voltage feature.									
.X.5	In cases where an hv poly gate abuts only low voltage source and drain, the poly gate can be tagged with the text:dg label with a value "hv_lv". In this case, the "hv_lv" tagged poly gate and its extensions will not be checked to hv.poly.6, but is checked by rules in the **poly.-.-** section. The use of the hv_lv label must be approved by technology.									
.X.6	Nwell biased at voltages >= 7.2V must be tagged with text "shv_nwell"	**NC**								
.nwell.1	Min spacing of nwell tagged with text "shv_nwell" to any nwell on different nets		2.500	2.500	2.500	2.500	2.500	2.500	2.500	2.500
.diff.1a	Minimum *hv_source/drain* spacing to diff for edges of *hv_source/drain* and diff not butting tap		0.300	0.300	0.300	0.300	0.300	0.300	0.300	0.300
.diff.1b	Minimum spacing of (n+/p+ diff resistors and diodes) connected to hv_source/drain to diff		0.300	0.300	0.300	0.300	0.300	0.300	0.300	0.300
.diff.2	Minimum spacing of nwell connected to hv_source/drain to n+ diff	**DE**	0.430	0.430	0.430	0.430	0.430	0.430	0.430	0.430
.diff.3a	Minimum n+ hv_source/drain spacing to nwell		0.550	0.550	0.550	0.550	0.550	0.550	0.550	0.550
.diff.3b	Minimum spacing of (n+ diff resistors and diodes) connected to hv_source/drain to nwell		0.550	0.550	0.550	0.550	0.550	0.550	0.550	0.550
.poly.1	Hv poly feature *hvPoly* (including hv poly resistors) can be drawn over only one diff region and is not allowed to cross nwell boundary except (1) as allowed in rule .X.3 and (2) nwell hole boundary in depmos									
.poly.2	Min spacing of hvPoly (including hv poly resistor) on field to diff (diff butting hvPoly are excluded)		0.300	0.300	0.300	0.300	0.300	0.300	0.300	0.300
.poly.3	Min spacing of hvPoly (including hv poly resistor) on field to n-well (exempt poly stradding nwell in a denmos/depmos)		0.550	0.550	0.550	0.550	0.550	0.550	0.550	0.550
.poly.4	Enclosure of *hvPoly* (including hv poly resistor) on field by n-well (exempt poly stradding nwell in a denmos/depmos)		0.300	0.300	0.300	0.300	0.300	0.300	0.300	0.300
.poly.6a	Min extension of poly beyond *hvFET_gate* (exempt poly extending beyond diff along the S/D direction in a denmos/depmos)		0.160	0.160	0.160	0.160	0.160	0.160	0.160	0.160
.poly.6b	Extension of hv poly beyond *FET_gate* (including *hvFET_gate;* exempt poly extending beyond diff along the S/D direction in a denmos/depmos)		0.160	0.160	0.160	0.160	0.160	0.160	0.160	0.160
.poly.8	Any poly gate abutting *hv_source/drain* becomes a *hvFET_gate*									

Rule	Section G2: Design Rules for S8*	Use	

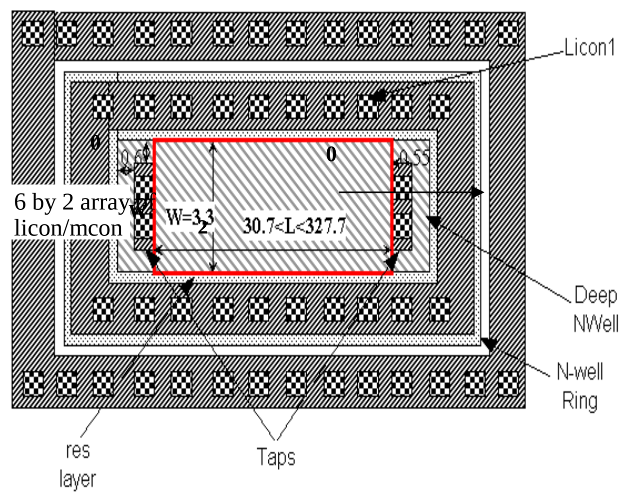


Rule	Section G2: Design Rules for S8*	Use								
(vhvi.-.-)	**VHVI - Very HV ID and Rules**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Identify nets working between 12-16V									
.vhv.1	Terminals operating at nominal 12V (maximum 16V) bias must be tagged as Very-High-Voltage (VHV) using vhvi:dg layer		**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**
.vhv.2	A source or drain of a drain-extended device can be tagged by vhvi:dg. A device with either source or drain (not both) tagged with vhvi:dg serves as a VHV propagation stopper		**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**
.vhv.3	Any feature connected to *VHVSourceDrain* becomes a very-high-voltage feature		**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**
.vhv.4	Any feature connected to *VHVPoly* becomes a very-high-voltage feature		**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**
.vhv.5	Diffusion that is not a part of a drain-extended device (i.e., diff not areaid:en) must not be on the same net as *VHVSourceDrain.* Only diffusion inside areaid.ed and LV diffusion tagged with vhvi:dg are exempted.									
.vhv.6	Poly resistor can act as a VHV propagation stopper. For this, it should be tagged with text "vhv_block"		**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**	**NC**
1	Min width of vhvi:dg		0.020	0.020	0.020	0.020	0.020	0.020	0.020	0.020
2	Vhvi:dg cannot overlap areaid:ce									
3	*VHVGate* must overlap hvi:dg									
4	Poly connected to the same net as a *VHVSourceDrain* must be tagged with vhvi:dg layer									
5	Vhvi:dg cannot straddle *VHVSourceDrain*									
6	Vhvi:dg overlapping *VHVSourceDrain* must not overlap poly									
7	Vhvi:dg cannot straddle *VHVPoly*									
8	Min space between nwell tagged with vhvi:dg and deep nwell, nwell, or n+diff on a separate net (except for n+diff overlapping nwell tagged with vhvi:dg).		11.240	11.240	11.240	11.240	11.240	11.240	11.240	11.240

(uhvi.-.-)	**UHVI - Ultra HV ID and Rules**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Identify nets working between 20V									
1	diff/tap can not straddle UHVI		N/A	N/A	N/A	N/A	N/A		N/A	N/A
2	poly can not straddle UHVI		N/A	N/A	N/A	N/A	N/A		N/A	N/A
3	pwbm.dg must be enclosed by UHVI (exempt inside areaid.lw)		N/A	N/A	N/A	N/A	N/A		N/A	N/A
4	dnw.dg can not straddle UHVI		N/A	N/A	N/A	N/A	N/A		N/A	N/A
5	UHVI must enclose areaid.ext		N/A	N/A	N/A	N/A	N/A		N/A	N/A
6	UHVI must enclose dnwell		N/A	N/A	N/A	N/A	N/A		N/A	N/A
10	natfet.dg layer is not allowed		N/A	N/A	N/A	N/A	N/A		N/A	N/A

(ulvt-.-)	**areaid.low_vt for UHV Diodes**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Identify dnwdiodehv_Psub(BV~60V)									
			NA	NA	NA	NA	NA		NA	N/A
1	areaid.low_vt must enclose dnw for the UHV dnw-psub diode texted "condiodeHvPsub"		NA	NA	NA	NA	NA		NA	N/A
2	areaid.low_vt must enclose pwbm.dg for the UHV dnw-psub diode texted "condiodeHvPsub"		NA	NA	NA	NA	NA		NA	N/A
3	areaid.low_vt can not straddle UHVI		NA	NA	NA	NA	NA		NA	N/A

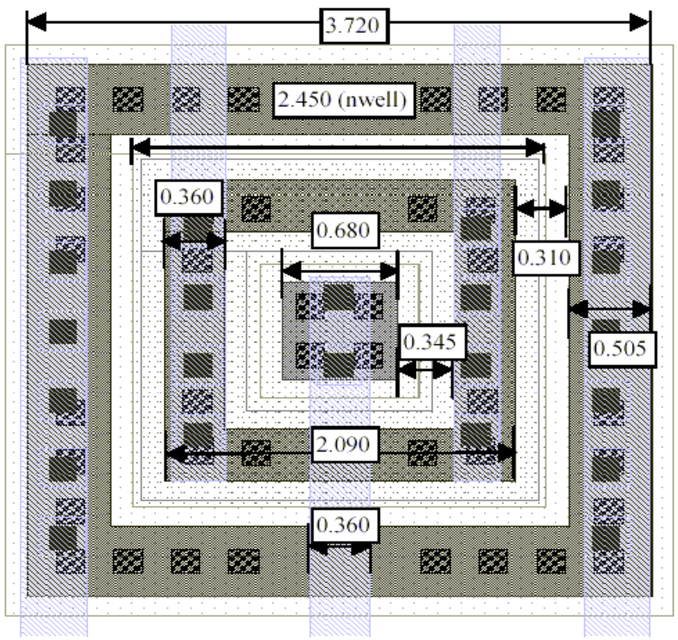
Rule	Section G2: Design Rules for S8*	Use								
(pwres.-.-)	**Pwell resistor**	**Use**	**s8di-5r***	**s8tnv-5r***	**s8tma-5r***	**s8p-10r***	**s8phirs-10r**	**s8pfn-20r***	**s8pfhd-10r***	**s8phrc-20r**
	Function: Identify pwell resistors									
1	Pwell resistor has to be enclosed by the res layer	**NC**								
2	Min/Max width of pwell resistor		2.650	2.650	2.650	2.650	2.650	2.650	2.650	2.650
3	Min length of pwell resistor		26.500	26.500	26.500	26.500	26.500	26.500	26.500	26.500
4	Max length of pwell resistor		265.00	265.00	265.00	265.00	265.00	265.00	265.00	265.00
5	Min/Max spacing of tap inside the pwell resistor to nwell		0.220	0.220	0.220	0.220	0.220	0.220	0.220	0.220
6	Min/Max width of tap inside the pwell resistor		0.530	0.530	0.530	0.530	0.530	0.530	0.530	0.530
7a	Every *pwres_terminal* must enclose 12 licon1									
7b	Every *pwres_terminal* must enclose 12 mcons if routed through metal1									
8	Diff or poly is not allowed in the pwell resistor.									
9	Nwell surrounding the pwell resistor must have a full ring of contacted tap strapped with metal.									
10	The res layer must abut *pwres_terminal* on opposite and parallel edges									
11	The res layer must abut nwell on opposite and parallel edges not checked in Rule pwres.10									



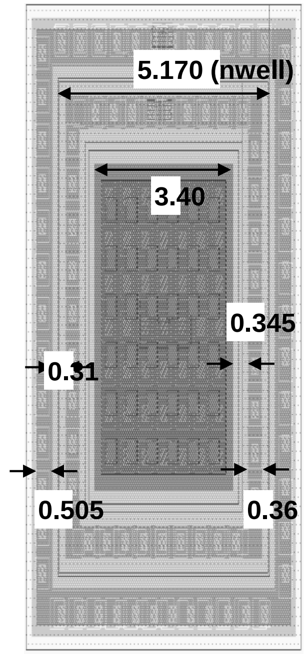
Rule	Section G2: Design Rules for S8*	Use								
(rfdiode.-.-)	Areaid.re for RF Diodes	Use	s8di-5r*	s8tnv-5r*	s8tma-5r*	s8p-10r*	s8phirs-10r	s8pfn-20r*	s8pfhd-10r*	s8phrc-20r
	Function: Identify RF diodes; Used for RCX									
1	Only 90 degrees allowed for areaid.re									
2	areaid.re must be coincident with nwell for the rf nwell diode									
3	areaid.re must be coincident with inner edge of the nwell ring for the rf pwell-deep nwell diode									

Allowed PNP layout

Layout: pnppar



Layout: pnppar5x

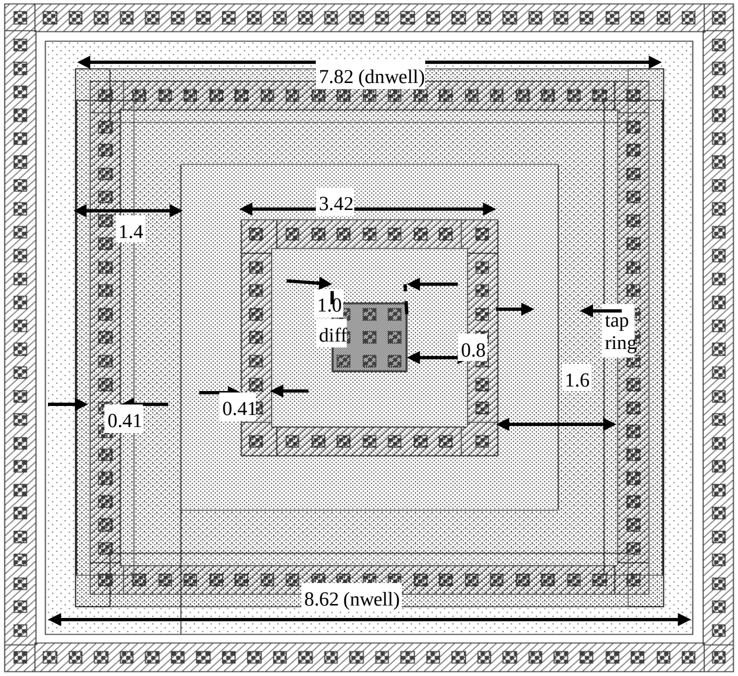


Notes:
- Li overlaps all diff and taps
- Nsdm/Psdm enclose diff/tap at min. enclosure

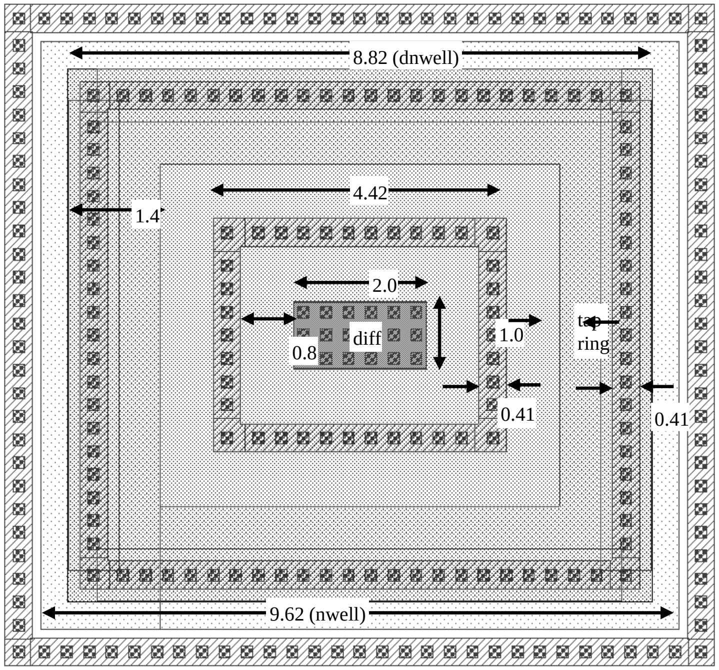
Rule	Section G2: Design Rules for S8*	Use	

Allowed NPN layout

Layout: npnpar1x1



Layout: npnpar1x2



(s.-)	Section G2d. Final mask rules for all layers		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
0	Mask DRC would be run on all mask layers to be streamed out, that are defined in Table L differently than:									
1a	Minimum mask width of cmm1 and cmm2 (incl. sizing, mask add and drop features)		0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080
1b	Minimum mask width for all other layers other than those included in s.1a and s.1c (incl. sizing, mask add and drop features)		0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100
1c	Minimum mask width of cp1m (incl. sizing, mask add and drop features)		0.055	0.055	0.055	0.055	0.055	0.055	0.055	0.055
2a	Minimum mask spacing for cvim (including sizing, mask add and drop features); Exempt violation between masks and the corresponding blanking inside areaid.dt		0.070	0.070	0.070	0.070	0.070	0.070	0.070	0.070
2b	Minimum mask spacing of chrome for cmm1 and cmm2 (including sizing, mask add and drop features); Exempt violation between cmm* and the corresponding blanking inside areaid.dt		0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080
2c	Minimum mask spacing for all other layers than those included in s.2a, s.2b. s.2d, s.2i and s.2j (including sizing, mask add and drop features); Exempt violation between masks and the corresponding blanking inside areaid.dt		0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100
2d	Minimum mask spacing of chrome for cp1m (including sizing, mask add and drop features); Exempt violation between cp1m and corresponding blanking inside areaid.dt		0.055	0.055	0.055	0.055	0.055	0.055	0.055	0.055
2e	Minimum enclosure of cvim (including sizing, mask add and drop features) by areaid.ft (frame cut); Exempt cvim edges coincident with frame cut (areaid.ft)		0.070	0.070	0.070	0.070	0.070	0.070	0.070	0.070
2f	Minimum enclosure of cmm1/cmm2 (including sizing, mask add and drop features) by areaid.ft (frame cut); Exempt cmm1/cmm2 edges coincident with frame cut (areaid.ft)		0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080
2g	Minimum enclosure of all other layers than those included in s.2e, s.2f. s.2h (including sizing, mask add and drop features) by areaid.ft (frame cut); Exempt edges coincident with frame cut (areaid.ft)		0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100
2h	Minimum enclosure of cp1m (including sizing, mask add and drop features) by areaid.ft (frame cut); Exempt cp1m edges coincident with frame cut (areaid.ft)		0.055	0.055	0.055	0.055	0.055	0.055	0.055	0.055
2i	Minimum mask spacing of chrome for cnwm (including sizing, mask add and drop features); Exempt violation between cnwm and corresponding blanking inside areaid.dt		0.070	0.070	0.070	0.070	0.070	0.070	0.070	0.070
2j	Minimum mask spacing of chrome for cnwm for parallel opposite edges (including sizing, mask add and drop features); Exempt violation between cnwm and corresponding blanking inside areaid.dt		0.100	0.100	0.100	0.100	0.100	0.100	0.100	0.100
3	Smoothing of all sized layers is done via opc tool and/or CAD code when needed. Smooth edge length should be min Width value unless negative sizing is enforced in which case smoothLength = max(minWidth,minSpace). Smoothing when hammerHeads are added is done with min									

Use	Explanation
P	Rule applies to periphery and between core and periphery. A corresponding core rule may or may not exist.
A	Rule documents are functionality implemented in CL algorithms and may not be checked by DRC.
NE	Rule not checked for *esd_nwell_tap* . There are no corresponding rules for *esd_nwell_tap* .
CO	Code only rule: Not required by any active flow but kept in the code for easy turn-on if needed
S	Rule used for CAD sizing, not checked by DRC

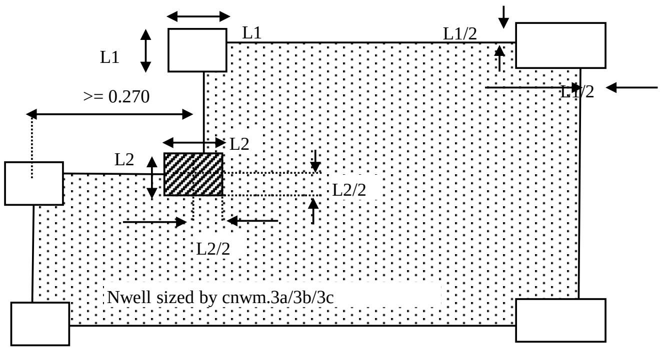
(cfom.-)	Cfom		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
.x.1	The CLDRC creates a layer fomPatternDensity that shows the pattern density on the whole layout.	A								
.x.2	Pattern Density rules are checked inside the seal ring and in the frame (excluding areaid.dt)	A								
1a	Width of cfom	MR	0.170	0.170	0.170	0.170	0.170	0.170	0.170	0.170
1b	Width of cfom	P	0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180
2	Spacing of cfom to cfom	MR	0.190	0.190	0.190	0.190	0.190	0.190	0.190	0.190
3a	Global sizing of diff or tap by cfom in core	S	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015
3b	Global sizing of diff or tap by cfom for W <= 0.200	S, P	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025
3c	Global sizing of diff or tap by cfom for W > 0.200	S, P	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015
3d	removed (N/A for SkyWater)	S, P	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3e	removed (N/A for SkyWater)	S, P	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3f	removed (N/A for SkyWater)	S, P	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3g	removed (N/A for SkyWater)	S, P	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
4	Line-end extension of diff or tap by cfom for W <= 0.200	S, P	0.040	0.040	0.040	0.040	0.040	0.040	0.040	0.040
5	Min enclosure of *pnp_emitter* by cfom (before global sizing)	S, P	0.030	0.030	0.030	0.030	0.030	0.030	0.030	0.000
.pd.1a	Max FOM Pattern_density range (after waffling)		30.00%	30.00%	30.00%	30.00%	30.00%	30.00%	30.00%	30.00%
.pd.1c	Min FOM Pattern_density (after waffling); Target used by waffle algorithm to reduce PD range (default);	A	40.00%	40.00%	40.00%	40.00%	40.00%	40.00%	40.00%	40.00%
.pd.1d	Min FOM Pattern_density (after waffling)	MR	28.00%	28.00%	28.00%	28.00%	28.00%	28.00%	28.00%	28.00%
.pd.1e	Max FOM pattern_density (after waffling)	MR	62.00%	62.00%	62.00%	62.00%	62.00%	62.00%	62.00%	62.00%
.pd.1f	Min FOM Pattern_density (after waffling) using switch "fom_30pd"; Target used by waffle algorithm in the frame to help deal with die + frame PD issues	A	30.00%	30.00%	30.00%	30.00%	30.00%	30.00%	30.00%	30.00%
.pd.1g	Min FOM Pattern_density (after waffling) using switch "fom_35pd"; Target used by waffle algorithm in the frame to help deal with die + frame PD issues	A	35.00%	35.00%	35.00%	35.00%	35.00%	35.00%	35.00%	35.00%
.pd.2a	Size of square region used to check rules cfom.pd.1a/1b/1c/1d	A	700	700	700	700	700	700	700	700
.pd.2b	Step size for checking rules cfom.pd.1a/1b/1c/1d	A	70	70	70	70	70	70	70	70
.pd.3a	Max FOM Pattern_density range (after waffling)		15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%	15.0%
.pd.3b	Size of square region used to check rule cfom.pd.3a	A	2000	2000	2000	2000	2000	2000	2000	2000
.pd.3c	Step size for checking rule cfom.pd.3a	A	200	200	200	200	200	200	200	200
waffle.1	Min spacing of FOM_fill to any other cfom	P	0.400	0.400	0.400	0.400	0.400	0.400	0.400	0.400
waffle.2	FOM_fill waffles may not touch (difftap or FOMmk)	P								
waffle.2a	FOM_fill waffles may not touch (poly or P1Mmk or P1M fill)	P								

(cdnm.-)	Cdnm		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
3	Enclosure of dnwell:dg by cdnm		-0.160	-0.160	-0.160	-0.160	-0.160	-0.160	-0.160	-0.100

(cpwbm.-)	Cpwbm		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
1	Enclosure of pwbm:dg by cpwbm		N/A	N/A	N/A	N/A	N/A	0.000	N/A	N/A

(cpwdem.-)	Cpwdem		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
1	Enclosure of pwdem:dg by cpwdem		N/A	N/A	N/A	N/A	N/A	0.000	N/A	N/A

(cnwm.-)	Nwell mask		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
3a	Min enclosure of (nwell NOT hvi) by cnwm; Exempt (nwell hole overlapping pwell.rs) from cnwm.3a sizing		0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035
3b	Min enclosure of (*nwell_all* AND hvi) by cnwm; Exempt ((nwell hole inside hvi) overlapping (areaid.en or pwell.rs)) from cnwm.3b sizing		0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015
3c	Space, no overlap, between ((empty nwell hole inside hvi) overlapping areaid.en) and cnwm	A	0.065	0.065	0.065	0.065	0.065	0.065	0.065	0.065
3d	Min/Max length of the add serif (L1) added to convex corner of nwell sized by cnwm.3a/3b/3c (Do not add serif when convex corner is < 0.270um from another concave corner on the nwell sized by cnwm.3a/3b/3c)	A	0.220	0.220	0.220	0.220	0.220	0.220	0.220	0.220
3e	Min/Max length of the drop serif (L2) added to convex corner (Do not drop serif when convex corner is < 0.270um from another concave corner on the nwell sized by cnwm.3a/3b/3c)	A	0.120	0.120	0.120	0.120	0.120	0.120	0.120	0.120
3f	Min extension of cnwm beyond nwell edge straddling *de_nFet_source* (45 degree edges are retained). This is done before final hv nwell sizing (cnwm.3b) is applied.		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A



(chvtpm.-)	Chvtpm		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
X.1	Hvtpm is drawn for the High Vt LV PMOS, high Vt varactor and PMOS core FET; Created over (LV nwell not overlapping varactors) NOT lvtn or (hvtp AND (LV nwell overlapping varactor))	NC								
1	Min width of chvtpm	MR	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
2a	Min spacing of chvtpm to chvtpm	MR	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
3	Min enclosure of ((*LVnwell* not overlapping *Var_channel*) NOT lvtn) by chvtpm	A	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
4	Min enclosure of ((*LVnwell* overlapping *Var_channel*) AND hvtp) by chvtpm	A	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

(clvtnm.-)	Clvtnm		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
X.1	Clvtnm is drawn for the LV low-Vt NMOS, HV native NMOS, Flash npass, SONOS FETs, low Vt ndiode, HV pdiode and native ndiode core over Nmos. It is created in nwell overlapping (hvtp OR areaid.ce or *Var_channel*); lvtnm can be manually drawn	NC								
1	Min width of clvtnm	P, MR	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
2a	Min spacing of clvtnm to clvtnm	MR	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
2b	Merge if space is below minimum	A								
3	Min enclosure of (nwell AND (hvtp OR areaid.ce)) by clvtnm	A	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
4	Min enclosure of (*LVnwell* overlapping *Var_channel*) by clvtnm	A	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

(ctunm.-)	Tunm mask		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
3	Minimum enclosure of tunm by ctunm		-0.010	-0.010	-0.010	-0.010	-0.010	-0.010	-0.010	-0.010

(clonom.-)	Clonom		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
1	Min width of clonom	MR	0.480	0.480	0.480	0.480	0.480	0.480	0.480	0.480
2	Min spacing of clonom to clonom (merge tunm if smaller space)	MR	0.430	0.430	0.430	0.430	0.430	0.430	0.430	0.430
3	Minimum enclosure of tunm by clonom		0.035	0.035	0.035	0.035	0.035	0.035	0.035	0.035
4	Min spacing of clonom to (poly and diff) outside of clonom		0.055	0.055	0.055	0.055	0.055	0.055	0.055	0.055
5	Minimum extension of clonom beyond (poly and diff)		0.105	0.105	0.105	0.105	0.105	0.105	0.105	0.105

(clvom.-)	Clvom		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
1	Min width of clvom	MR	0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.700
2	Min spacing of clvom to clvom	MR	0.700	0.700	0.700	0.700	0.700	0.700	0.700	0.600
3	Enclosure of cllvom by clvom		0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.050

(cp1m.-)	Cp1m		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
.x.1	Larger P1M PD range check for designs with photo array is enabled by using switch "photo_pd";	NC								
1	Width of cp1m (checked before mask add/drop and hammerhead addition)	C	0.155	0.155	0.155	0.155	0.155	0.155	0.155	0.155
2a	Spacing of cp1m to cp1m (parallel edges only, checked before mask add/drop and hammerhead addition) except for cp1m.2c	C	0.160	0.160	0.160	0.160	0.160	0.160	0.160	0.160
2b	Spacing of cp1m to cp1m except for cp1m.2c (checked before mask add/drop and hammerhead addition)	C	0.155	0.155	0.155	0.155	0.155	0.155	0.155	0.155
2c	Core poly gap: spacing between two exactly opposite poly line ends where the width of both poly lines is = poly.1	C	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
3a	Enclosure of poly by cp1m for width W >= 0.200	S,C	0.005	0.005	0.005	0.005	0.005	0.005	0.005	0.005
3b	Enclosure of poly by cp1m for W < 0.200 and 0.210 <= S < 0.280	S,C	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010
3c	Enclosure of poly by cp1m for W < 0.200 and 0.280 <= S < 0.320	S,C	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015
3d	Enclosure of poly by cp1m for W < 0.200 and 0.320 <= S < 0.420	S,C	0.020	0.020	0.020	0.020	0.020	0.020	0.020	0.020
3e	Enclosure of poly by cp1m for W < 0.200 and 0.420 <= S < 0.670	S,C	0.020	0.020	0.020	0.020	0.020	0.020	0.020	0.020
3f	Enclosure of poly by cp1m for W < 0.200 and S => 0.670	S,C	0.015	0.015	0.015	0.015	0.015	0.015	0.015	0.015
8	Sizing post-OPC for *cp1m_HV* except (*flare_gates* and resistors) outside nwell when W = 0.200 (geomAnd(cp1m geomAndNot(HVI nwell)))	A,C	-0.010	-0.010	-0.010	-0.010	-0.010	-0.010	-0.010	-0.010
mb.1	Mask manufacturability rules are checked on final mask data.	NC	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
mb.2	Pre-OPC notch cleanup is applied to concave jogs smaller than …	A, P	0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080
mb.3	Pre-OPC notch cleanup is applied to convex jogs smaller than …	A, P	0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080
mb.4	Post-OPC smoothing is applied to remove notches smaller than (note: use notch fill)…	A, P	0.055	0.055	0.055	0.055	0.055	0.055	0.055	0.055
mb.5	Min width of pseudo-ORC image	P	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
mb.6	Min space of pseudo-ORC image	P	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
.pd.1a	Max P1M Pattern_density range (after waffling)	MR	40.00%	40.00%	40.00%	40.00%	40.00%	40.00%	40.00%	40.00%
.pd.1b	Max P1M Pattern_density range (after waffling; when switch "photo_pd" is enabled)		52.00%	52.00%	52.00%	52.00%	52.00%	52.00%	52.00%	52.00%
.pd.1c	Min P1M Pattern_density (after waffling); Target used by waffle algorithm to reduce PD range	A	30.00%	30.00%	30.00%	30.00%	30.00%	30.00%	30.00%	30.00%
.pd.2a	Size of square region used to check rules cp1m.pd.1a/1b	A	700	700	700	700	700	700	700	700
.pd.2b	Step size used for checking rules cp1m.pd.1a/1b	A	70	70	70	70	70	70	70	70
waffle.1	Min spacing of cp1m waffles to any other poly	P	0.360	0.360	0.360	0.360	0.360	0.360	0.360	0.360
waffle.2a	cp1m waffles may not touch (difftap or FOMmk or FOM fill)	P								

(crpm.-)	Crpm		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
1	Min width of crpm		N/A	N/A	N/A	N/A	N/A	N/A	N/A	1.370
2	Min spacing of crpm to crpm		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.740
3	Min enclosure of rpm:dg by clrpm by …	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.050

(crrpm.-)	Crrpm		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
	Crrpm defines implants for high-res poly (300 ohm/sq)									
1	Min width of crrpm		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.600
2	Min spacing of crrpm to crrpm		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.700
3	Enclosure of (poly.rs and rpm) by crrpm (on sides of resistors)	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.200
3a	Enclosure on endcaps of (poly.rs and rpm) by crrpm on endcaps	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.150
4	Min spacing of crrpm to non-overlapping poly		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.200

(curpm.)	Curpm		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
	Curpm defines implants for ultra-high resistance poly (2K ohm/sq)									
1	Min width of curpm		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.600
2	Min spacing of curpm to (curpm or crrpm)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.700
3	Enclosure of (poly.rs and urpm) by curpm (on sides of resistors)	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.200
3a	Enclosure on endcaps of (poly.rs and urpm) by curpm on endcaps	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.150
4	Min spacing of curpm to non-overlapping poly		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.200
5	Min enclosure of urpm:dg by clrpm		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.050

(cnpcm.-)	Cnpcm (s8phrc* only)		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
1	Enclosure of slot_licon by npc except for the side extending into poly.rs	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.175
1a	Enclosure of slot_licon by npc for the side extending into poly.rs	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.100
1b	Enclosure of slot_licon by npc if poly.rs < 0.500 um length		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.050
3	Min width of npc to npc		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.270
4	Min spacing of npc to npc, merge if less than minimum		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.270
5	Spacing (no overlap) of npc to Gate		N/A	N/A	N/A	N/A	N/A	N/A	N/A	0.090
6	slot_licon must abut poly.rs		N/A	N/A	N/A	N/A	N/A	N/A	N/A	

(cntm.-)	Cntm/Ldntm		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
1	Width of cntm	MR	0.840	0.840	0.840	0.840	0.840	0.840	0.840	0.940
2*	Spacing of cntm to cntm; for *-fab25 flows: opposite parallel edges only		0.700	0.700	0.700	0.700	0.700	0.700	0.700	0.600
2b	Minimum spacing, merge if less than minimum (prior to final bias sizing of cntm.3, 4a, 7)	A	0.700	0.700	0.700	0.700	0.700	0.700	0.700	0.700
2c	Min spacing of cntm to cntm after the merge		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3	Min enclosure of nwell:dg or clntm:dg by cntm	A	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.050
4a	Min enclosure of (Hvi outside areaid.ce) by cntm	A	0.000	N/A	0.000	0.000	0.000	0.000	0.000	0.050
4b	Min enclosure of Hvi by cntm	A	N/A	0.000	N/A	N/A	N/A	N/A	N/A	N/A
7	Min enclosure of ldntm by cntm	A	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.050
8	Min enclosure of rpm by cntm	A	N/A	N/A	0.000	N/A	0.000	N/A	N/A	0.000

(chvntm.-)	Chvntm		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
X.1	Chvntm will be created and OR-ed with hvntm:dg; checked for min spacing rule									
1	Width of chvntm	MR	0.700	0.700	0.700	0.700	0.700	0.700	0.700	0.700
2a	Spacing of chvntm to chvntm	MR	0.700	0.700	0.700	0.700	0.700	0.700	0.700	0.700
2b	Merge if space is below minimum	A								
3	Min. enclosure of ((n+_diff NOT *de_pFet_source*) inside Hvi) not overlapping areaid.ce by chvntm;	A	0.185	0.185	0.185	0.185	0.185	0.185	0.185	0.185
4	Space, no overlap, between *n+_diff* outside Hvi and chvntm		0.185	0.185	0.185	0.185	0.185	0.185	0.185	0.185
5	Space, no overlap, between *p+_diff* and chvntm	DE	0.185	0.185	0.185	0.185	0.185	0.185	0.185	0.185
6a	Space, no overlap, between *p+_tap* and chvntm (except along the diff-butting edge)		0.185	0.185	0.185	0.185	0.185	0.185	0.185	0.185
6b	Space, no overlap, between *p+_tap* and chvntm along the diff-butting edge		0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
7	Chvntm must enclose *ESD_nwell_tap* inside hvi by at least	A	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
8	Min space, no overlap, between *prec_resistor* and chvntm		N/A	N/A	0.185	N/A	0.185	N/A	N/A	0.185

(clicm1.-)	Clicm1		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
1	Min width of clicm1 (except for rule clicm1.1a)	MR	0.220	0.220	0.220	0.220	0.220	0.220	0.220	0.220
1a	Min width of clicm1 for slotted_licon in prec_resistor terminals where slotted_licon > 1		N/A	N/A	0.190	N/A	0.190	N/A	N/A	0.190
2a	Spacing of clicm1 to clicm1 (parallel edges only)		0.115	0.115	0.115	0.115	0.115	0.115	0.115	0.115
2b	Spacing of clicm1 to clicm1	MR	0.095	0.095	0.095	0.095	0.095	0.095	0.095	0.095
3a	Enclosure of licon1 by clicm1 when S <= 0.330	A	0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025
3b	Enclosure of licon1 by clicm1 when S > 0.330 ; Exception: slotted_licon_edge1 for clicm.1a receives zero sizing	A	0.030	0.030	0.030	0.030	0.030	0.030	0.030	0.030
3c	Enclosure of licon1 by clicm1 for *slotted_licon* overlapping hrpoly and rpm	A	N/A	N/A	0.000	N/A	0.000	N/A	N/A	0.000
4	removed (N/A for SkyWater)	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

(cli1m.-)	Cli1m		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
1a	Width of cli1m	MR	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
1b	Width of cli1m (except cli1m.1c)	P	0.170	0.170	0.170	0.170	0.170	0.170	0.170	0.170
1c	Width of cli1m inside of capacitor.dg	P	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
2a	Spacing of cli1m to cli1m (parallel edges only)		0.135	0.135	0.135	0.135	0.135	0.135	0.135	0.135
2b	Spacing of cli1m to cli1m	MR	0.110	0.110	0.110	0.110	0.110	0.110	0.110	0.110
3a	Enclosure of li by cli1m in the core	S	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3aa	Enclosure of li by cli1m in core for 0.14 <= W <= 0.17 and S > 0.30	S	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3ab	Enclosure of li by cli1m in core for 0.14 <= W <= 0.17, L < 1.00, and 0.185 =< S < 0.25, with min sized length of the sized line >= 0.165	S	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3b	Enclosure of li by cli1m for W< 0.30 AND S < 0.25	S, P	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010
3c	Enclosure of li by cli1m for W > = 0.30 AND S < 0.25	S, P	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3d	Enclosure of li by cli1m for lines S >= 0.25	S, P	0.030	0.030	0.030	0.030	0.030	0.030	0.030	0.030
4	Min clear area of cli1m		N/A	N/A	N/A	35%	35%	35%	35%	35%
5	Max clear area of cli1m		N/A	N/A	N/A	70%	70%	70%	70%	70%
waffle.1	Min spacing of LI1M_fill waffle to any other cli1m	P	0.500	0.500	0.500	0.500	0.500	0.500	0.500	0.500
waffle.2a	LI1M_fill waffles may not touch FOM waffles or any cp1m	P								

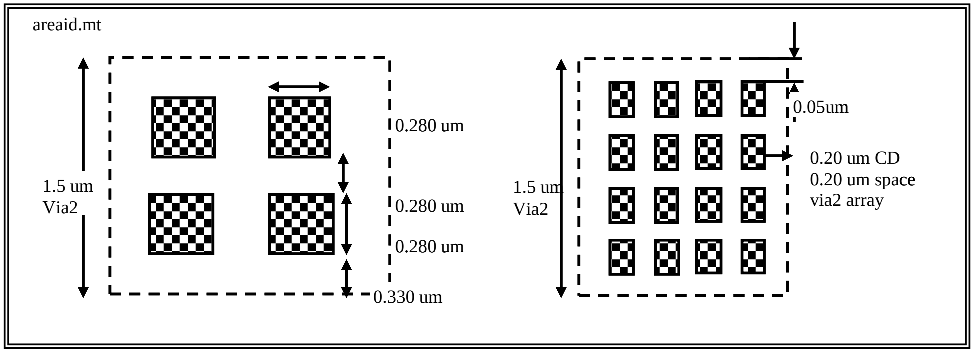
(cctm1.-)	Cctm1		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
1	Sizing of squared contacts 0.17 x 0.17 (per side)		0.025	0.025	0.025	0.025	0.025	0.025	0.025	0.025

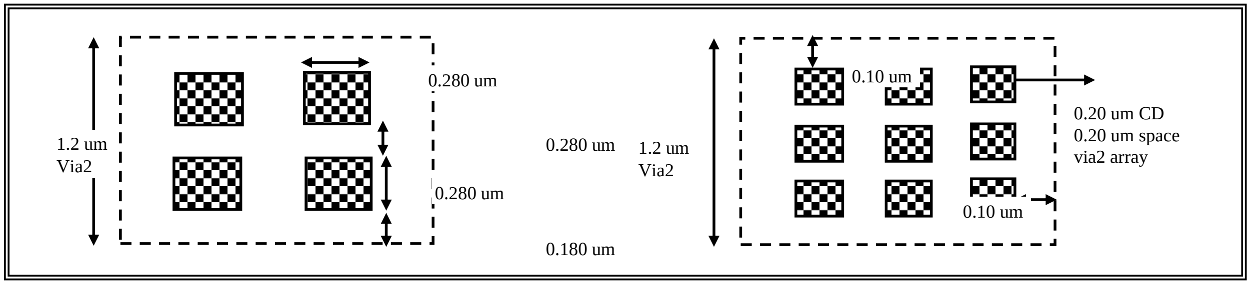
(cmm1.-)	Cmm1		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
.X.1	The CLDRC creates a layer mm1PatternDensity that shows the pattern density on the whole layout.	A								
.X.2	Pattern Density rules are checked only inside the seal ring	A								
.X.3	*MM1_oxide* is defined as mm1 sized up by …	A	0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600
.X.4	Grid size metal1 mask data (cmm1:md/mp/mk)		0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
.X.5	*photoArray* should be excluded from pattern density checks	A								
3a	Enclosure of met1 by cmm1 for W<= 0.160 AND S >= 0.250	A	0.010	0.010	0.010	0.010	0.010	0.010	0.010	0.010
3b	Enclosure of met1 by cmm1 for W >= 0.500 AND S < 0.150	A	-0.005	-0.005	-0.005	-0.005	-0.005	-0.005	-0.005	-0.005
pd.1	Max *MM1_oxide_Pattern_density* range (after waffling)	MR	30%	30%	30%	30%	30%	30%	30%	30%
pd.1b	Min *MM1_oxide_Pattern_density* (after waffling); Target used by waffle algorithm to reduce PD range (default);	A	80%	80%	80%	80%	80%	80%	80%	80%
pd.1c	Min *MM1_oxide_Pattern_density* (after waffling) using switch "mm1_70opd"; Target used by waffle algorithm in the frame to help deal with die + frame PD issues;	A	70%	70%	70%	70%	70%	70%	70%	70%
pd.2a	Size of square region used to check rule cmm1.pd.1	A	700	700	700	700	700	700	700	700
pd.2b	Step size for checking rule cmm1.pd.1	A	70	70	70	70	70	70	70	70
pd.3	Min clear area of cmm1	MR	N/A	N/A	N/A	35%	35%	35%	35%	35%
pd.4	Max clear area of cmm1	MR	N/A	N/A	N/A	70%	70%	70%	70%	70%
waffle.1	Min space of MM1fill to any met1	MR	N/A	N/A	N/A	0.200	0.200	0.200	0.200	0.200
waffle.2	MM1fill may not touch met1	MR	N/A	N/A	N/A					
waffle.3	45 degree angles not allowed on waffles									
waffle.4.1	Exact length and width of met1 waffle (tile) preference 1	A	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
waffle.4.2	Exact length and width of met1 waffle (tile) preference 2	A	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
waffle.4.3	Exact length and width of met1 waffle (tile) preference 3	A	0.580	0.580	0.580	0.580	0.580	0.580	0.580	0.580
waffle.4.4	Exact length and width of met1 waffle (tile) preference 4	AD	0.300	0.300	0.300	0.300	0.300	0.300	0.300	0.300
mb.1	Mask manufacturability rules are checked on final mask data.	NC								
mb.2	Pre-OPC notch cleanup is applied to concave jogs smaller than …	A	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
mb.3	Pre-OPC notch cleanup is applied to convex jogs smaller than …	A	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
mb.4	Post-OPC smoothing is applied to remove notches smaller than (note: use notch fill)…	A	0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080
mb.5	Min width of pseudo-ORC image		0.120	0.120	0.120	0.120	0.120	0.120	0.120	0.120
mb.6	Min space of pseudo-ORC image		0.120	0.120	0.120	0.120	0.120	0.120	0.120	0.120

(cvim.-)	Cvim		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
1	Min/Max width of cvim (after sizing)	**MR, AI**	0.150	0.150	0.150	0.150	0.150	0.150	0.150	0.150
3a	Sizing of squared vias 0.150um (per side)		0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3b	Sizing of squared vias 0.230um inside areaid.mt (per side)		-0.040	-0.040	-0.040	-0.040	-0.040	-0.040	-0.040	-0.040
3c	Sizing of squared vias 0.280um inside areaid.mt (per side)		-0.065	-0.065	-0.065	-0.065	-0.065	-0.065	-0.065	-0.065
mb.1	Minimum width of pseudo-ORC image	**AI**	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
mb.2	Maximum width of pseudo-ORC image		0.160	0.160	0.160	0.160	0.160	0.160	0.160	0.160
mb.3	Minimum space of pseudo-ORC image	**AI**	0.160	0.160	0.160	0.160	0.160	0.160	0.160	0.160

(cmm2.-)	Cmm2		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
.X.1	The CLDRC creates a layer mm2PatternDensity that shows the pattern density on the whole layout.	**A**								
.X.2	Pattern Density rules are checked only inside the seal ring	**A**								
.X.3	*MM2_oxide* is defined as mm2 sized up by …	**A**	0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600
.X.4	Grid size metal2 mask data (cmm2:md/mp/mk)		0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
.X.5	*photoArray* should be excluded from pattern density checks	**A**								
pd.1	Max *MM2_oxide_Pattern_density* range (after waffling)	**MR**	30%	30%	30%	30%	30%	30%	30%	30%
pd.1b	Min *MM2_oxide_Pattern_density* (after waffling); Target used by waffle algorithm to reduce PD range (default);	**A**	80%	80%	80%	80%	80%	80%	80%	80%
pd.1c	Min *MM2_oxide_Pattern_density* (after waffling) using switch "mm2_70opd"; Target used by waffle algorithm in the frame to help deal with die + frame PD issues;	**A**	70%	70%	70%	70%	70%	70%	70%	70%
pd.2a	Size of square region used to check rule cmm2.pd.1	**A**	700	700	700	700	700	700	700	700
pd.2b	Step size for checking rule cmm2.pd.1	**A**	70	70	70	70	70	70	70	70
pd.3	Min clear area of cmm2	**MR**	N/A	N/A	N/A	35%	35%	35%	35%	35%
pd.4	Max clear area of cmm2	**MR**	N/A	N/A	N/A	70%	70%	70%	70%	70%
waffle.1	Min space of MM2fill to any met2	**MR**	N/A	N/A	N/A	0.200	0.200	0.200	0.200	0.200
waffle.2	MM2fill may not touch met2	**MR**	N/A	N/A	N/A					
waffle.3	45 degree angles not allowed on waffles									
waffle.4.1	Exact length and width of met2 waffle (tile) preference 1	**A**	2.000	2.000	2.000	2.000	2.000	2.000	2.000	2.000
waffle.4.2	Exact length and width of met2 waffle (tile) preference 2	**A**	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
waffle.4.3	Exact length and width of met2 waffle (tile) preference 3	**A**	0.580	0.580	0.580	0.580	0.580	0.580	0.580	0.580
waffle.4.4	Exact length and width of met2 waffle (tile) preference 4	**AD**	0.300	0.300	0.300	0.300	0.300	0.300	0.300	0.300
mb.1	Mask manufacturability rules are checked on final mask data.	**NC**								
mb.2	Pre-OPC notch cleanup is applied to concave jogs smaller than …	**A**	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
mb.3	Pre-OPC notch cleanup is applied to convex jogs smaller than …	**A**	0.140	0.140	0.140	0.140	0.140	0.140	0.140	0.140
mb.4	Post-OPC smoothing is applied to remove notches smaller than (note: use notch fill)…	**A, AI**	0.080	0.080	0.080	0.080	0.080	0.080	0.080	0.080
mb.5	Min width of pseudo-ORC image		0.120	0.120	0.120	0.120	0.120	0.120	0.120	0.120
mb.6	Min space of pseudo-ORC image		0.120	0.120	0.120	0.120	0.120	0.120	0.120	0.120

(cvim2.-)	Cvim2		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
3b	Sizing for square 1.5 um via2 inside areaid.mt (per side)		-0.150	N/A	-0.350	N/A	N/A	N/A	N/A	N/A
3e	Sizing for square 0.28 um via2 inside areaid.mt (per side)		N/A	N/A	N/A	-0.040	-0.040	-0.040	-0.040	-0.040
3f	Sizing for square vias 0.200um (per side). Sizing is done after larger via2 are sized to 0.2um		N/A	N/A	N/A	0.020	0.020	0.020	0.020	0.020
3g	Sizing of square 1.2um via2 (per side)		N/A	N/A	-0.200	N/A	N/A	N/A	N/A	N/A

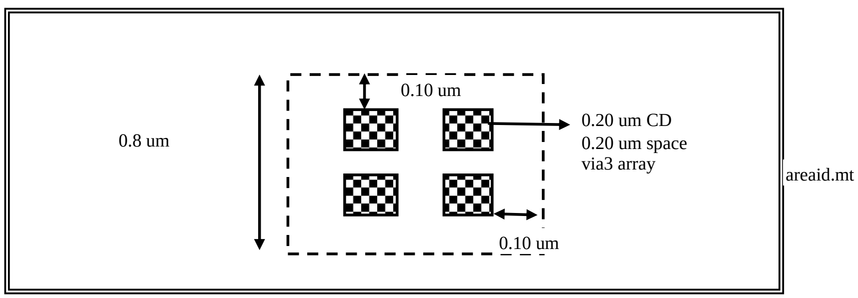




(cmm3.-)	Cmm3		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
.X.1	The CLDRC creates a layer mm3PatternDensity that shows the pattern density on the whole layout.	A	N/A	N/A	N/A					
.X.2	Pattern Density rules are checked only inside the seal ring	A	N/A	N/A	N/A					
.X.3	*MM3_oxide* is defined as mm3 sized up by …	A	N/A	N/A	N/A	1.150	1.150	1.150	1.150	1.150
3	Min enclosure of met3 by cmm3	AI	N/A	N/A	0.015	0.010	0.010	0.010	0.010	0.010
3a	Min enclosure of met3 of area <= 0.3364 um2 by cmm3		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3b	Min enclosure of met3 of area > 0.3364 um2 by cmm3	AI	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
pd.1	Max *MM3_oxide_Pattern_density* range (after waffling)	MR, AI	N/A	N/A	N/A	30%	30%	30%	30%	30%
pd.1b	Min *MM3_oxide_Pattern_density* (after waffling); Target used by waffle algorithm to reduce PD range;	A	N/A	N/A	N/A	80%	80%	80%	80%	80%
pd.1c	Min *MM3_oxide_Pattern_density* (after waffling) using switch "mm3_70opd"; Target used by waffle algorithm in the frame to help deal with die + frame PD issues (default);	A	N/A	N/A	N/A	70%	70%	70%	70%	70%
pd.2a	Size of square region used to check rule cmm3.pd.1	A	N/A	N/A	N/A	700	700	700	700	700
pd.2b	Step size for checking rule cmm3.pd.1	A	N/A	N/A	N/A	70	70	70	70	70
pd.3	Min clear area of cmm3	MR, AI	N/A	N/A	N/A	35%	35%	35%	35%	35%
pd.4	Max clear area of cmm3	MR, AI	N/A	N/A	N/A	70%	70%	70%	70%	70%
waffle.1	Min space of MM3fill to any met3	MR	N/A	N/A	N/A	0.300	0.300	0.300	0.300	0.300
waffle.2	MM3fill may not touch met3	MR	N/A	N/A	N/A					
waffle.3	45 degree angles not allowed on waffles		N/A	N/A	N/A					
waffle.4.1	Exact length and width of met3 waffle (tile) preference 1	A, AI	N/A	N/A	N/A	2.000	2.000	2.000	2.000	2.000
waffle.4.2	Exact length and width of met3 waffle (tile) preference 2	A, AI	N/A	N/A	N/A	1.000	1.000	1.000	1.000	1.000
waffle.4.3	Exact length and width of met3 waffle (tile) preference 3	A, AI	N/A	N/A	N/A	0.580	0.580	0.580	0.580	0.580
waffle.4.4	Exact length and width of met3 waffle (tile) preference 4	AD	N/A	N/A	N/A	0.400	0.400	0.400	0.400	0.400

(cindm.-)	Cindm		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
3	Min enclosure of met3.dg by cindm		0.000	N/A	N/A	N/A	N/A	N/A	N/A	N/A

(cvim3.-)	Cvim3		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
3	Squared 0.200um via3s must be sized by		N/A	N/A	N/A	0.020	0.020	0.020	0.020	0.020
3a	Square 0.8 um via3 inside areaid.mt must be replaced by 2x2 array of 0.2 um Via3 spaced at 0.200um apart and centered inside the larger Via3 (see figure); This is done before cvim3.3 sizing		N/A	N/A	N/A					



(cmm4.-)	Cmm4		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
.X.1	The CLDRC creates a layer mm4PatternDensity that shows the pattern density on the whole layout.	A	N/A	N/A	N/A					
.X.2	Pattern Density rules are checked only inside the seal ring	A	N/A	N/A	N/A					
.X.3	*MM4_oxide* is defined as mm4 sized up by …	A	N/A	N/A	N/A	1.150	1.150	1.150	1.150	1.150
3	Min enclosure of met4 by cmm4	AI	N/A	N/A	N/A	0.010	0.010	0.010	0.010	0.010
3a	Min enclosure of met4 of area <= 0.3364 um2 by cmm4	AI	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3b	Min enclosure of met4 of area > 0.3364 um2 by cmm4	AI	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
pd.1	Max *MM4_oxide_Pattern_density* range (after waffling)	MR	N/A	N/A	N/A	30%	30%	30%	30%	34%
pd.1b	Min *MM4_oxide_Pattern_density* (after waffling); Target used by waffle algorithm to reduce PD range (default);	A	N/A	N/A	N/A	80%	80%	80%	80%	80%
pd.1c	Min *MM4_oxide_Pattern_density* (after waffling) using switch "mm4_70opd"; Target used by waffle algorithm in the frame to help deal with die + frame PD issues;	A	N/A	N/A	N/A	70%	70%	70%	70%	70%
pd.2a	Size of square region used to check rule cmm4.pd.1	A	N/A	N/A	N/A	700	700	700	700	700
pd.2b	Step size for checking rule cmm4.pd.1	A	N/A	N/A	N/A	70	70	70	70	70
pd.3	Min clear area of cmm4	MR	N/A	N/A	N/A	35%	35%	35%	35%	35%
pd.4	Max clear area of cmm4	MR	N/A	N/A	N/A	70%	70%	70%	70%	70%
waffle.1	Min space of MM4fill to any met4	MR	N/A	N/A	N/A	0.300	0.300	0.300	0.300	0.300
waffle.2	MM4fill may not touch met4	MR	N/A	N/A	N/A					
waffle.3	45 degree angles not allowed on waffles		N/A	N/A	N/A					
waffle.4.1	Exact length and width of met4 waffle (tile) preference 1	A, AI	N/A	N/A	N/A	2.000	2.000	2.000	2.000	2.000
waffle.4.2	Exact length and width of met4 waffle (tile) preference 2	A, AI	N/A	N/A	N/A	1.000	1.000	1.000	1.000	1.000
waffle.4.3	Exact length and width of met4 waffle (tile) preference 3	A, AI	N/A	N/A	N/A	0.580	0.580	0.580	0.580	0.580
waffle.4.4	Exact length and width of met4 waffle (tile) preference 4	AD, AI	N/A	N/A	N/A	0.400	0.400	0.400	0.400	0.400

(cmm5.-)	Cmm5		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
3	Min enclosure of met5 by cmm5		N/A	N/A	N/A	0.015	0.015	0.015	0.015	0.015
4	Min clear area of cmm5		N/A	N/A	N/A	19%	19%	19%	19%	N/A
5	Max clear area of cmm5 (use switch waffle_mm5 if bigger that max)		N/A	N/A	N/A	60%	60%	60%	60%	N/A
waffle.1	Min space of MM5fill to any met5	MR	N/A	N/A	N/A	1.600	1.600	1.600	1.600	1.600
waffle.2	MM5fill may not touch met5	MR								
waffle.4.1	Exact length and width of met5 waffle (tile), when using switch waffle_mm5	A	N/A	N/A	N/A	3.000	3.000	3.000	3.000	3.000

(cnsm.-)	NSM		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
3	Min enclosure of nsm by cnsm		0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000

(cncm.-)	Cncm		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
X.1	Ncm is drawn in the core. It is created in periphery ; NCM will created and then OR'ed with ncm.dg inside areaid.ce before space check; NCM rules will be checked in all flows to ensure IP compatibility but will be taped out only in S8TNV-5R flow	NC								
X.2	Ncm overlapping areaid:ce is checked for core rules only	NC								
1	Width of cncm	MR	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
2a	Spacing of cncm to cncm	MR	0.380	0.380	0.380	0.380	0.380	0.380	0.380	0.380
2b	Merge ncm if space is below minimum	A, P								
3a	Min enclosure of [((*LVnwell* not overlapping *Var_channel*) NOT lvtn)] by cncm	A, P, NC	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3b	Min enclosure of ((*LVnwell* overlapping *Var_channel*) AND hvtp) by cncm	A, P, NC	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
3c	Min enclosure of (P+diff outside (hvi OR areaid.ce)) not overlapping lvtn by cncm		0.180	0.180	0.180	0.180	0.180	0.180	0.180	0.180

(cpdm.-)	Cpdm (Pad Mask)		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
4	Min enclosure of (cpmm:dg AND areaid.mt) by cpdm	A	N/A	0.000	N/A	N/A	0.000	N/A	0.000	0.000
5	Min spacing of cpdm:mk to areaid:dt		3.000	3.000	3.000	3.000	3.000	3.000	3.000	3.000

(cpdmm.-)	Cpdmm (Pad&Polyimide Mask for Std WB products only)		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
1	Min. width of cpdmm		5.000	N/A	5.000	5.000	N/A	N/A	N/A	N/A
2	Min. spacing of cpdmm to cpdmm; Merge if space is less than min		15.000	N/A	15.000	15.000	N/A	N/A	N/A	N/A
3	Cpdmm must not overlap *fuse:dg* or *photoArray*			N/A			N/A	N/A	N/A	N/A
4	*NoScribePMM* switch is not allowed for flows using PDMM			N/A			N/A	N/A	N/A	N/A
5	Min enclosure of pad by cpdmm outside *uPads,* exc. merged shapes	A	-3.000	N/A	-3.000	-3.000	N/A	N/A	N/A	N/A
6	Min enclosure of pad by cpdmm for *uPads,* except merged shapes	A	-3.250	N/A	-3.250	-3.250	N/A	N/A	N/A	N/A
7	Min width of cpmm inside center of scribe, with scribe 65 =< W < 400 um	A	50.000	N/A	50.000	50.000	N/A	N/A	N/A	N/A
7a	Min. spacing between *Diecut_pmm* to cpdmm, further than 65.5 um from areaid:ft	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
7b	Min width of cpmm inside center of scribe, with scribe W < 65 um	A	35.000	N/A	35.000	35.000	N/A	N/A	N/A	N/A
8	Min spacing between areaid:mt and cpdmm, in cells named s8*_md* except 45 deg corners, *uPads*, and merged shapes	A	7.500	N/A	7.500	7.500	N/A	N/A	N/A	N/A
9	Min spacing of cpdmm to areaid:ft exc. cells named s8*_md*	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

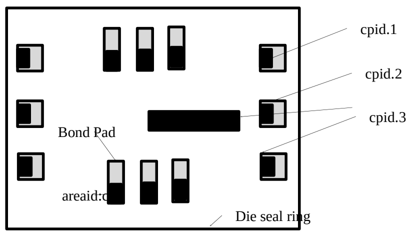
(cpmm.-)	Cpmm (Polyimide Mask)		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
X.1	cpmm.dg is used in the S8 RF pads only									
X.2	cpmm will be created and OR-ed with pmm.dg; Then checked for min spacing rule; pmm.dg can be drawn to fix cldrc errors									
1	Min. width of cpmm		N/A	5.000	N/A	N/A	N/A	5.000	5.000	5.000
1a	removed (N/A for SkyWater)	CO	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2	Min. spacing of cpmm to cpmm; Merge if space is less than minimum		N/A	15.000	N/A	N/A	N/A	15.000	15.000	15.000
3	Min. enclosure of *fuse:dg* by cpmm	A	N/A	12.000	N/A	N/A	N/A	12.000	12.000	12.000
4	Min. enclosure of *bondPad* by cpmm (Rule does not apply to pad.dg inside "M5RDL" pcell and for pad.dg inside inductor.dg)	A	N/A	0.500	N/A	N/A	N/A	0.500	0.500	0.500
4a	removed (N/A for SkyWater)	A, CO	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
5	Min. enclosure of *scribe_line* by cpmm (enabled only when "NoScribePMM" switch is on)	A	N/A	0.000	N/A	N/A	N/A	0.000	0.000	0.000
5a	Min. spacing between *Diecut_pmm* to cpmm (For flows other than s8pir-10r, rule not checked when "NoScribePMM" switch is turned on)	A	N/A	7.500	N/A	N/A	N/A	7.500	7.500	7.500
7	Min. enclosure of (cpmm:dg and areaid:mt) in the die by cpmm	A	N/A	30.000	N/A	N/A	N/A	30.000	30.000	30.000
9	Min enclosure of *photoArray* by cpmm	A	N/A	0.000	N/A	N/A	N/A	0.000	0.000	0.000
10	Min enclosure of (areaid.mt outside areaid.ft) by cpmm; Exclude cell name "*_tech_CD_top*"	A	N/A	0.000	N/A	N/A	N/A	0.000	0.000	0.000

(cpbo.-)	Cpbo (PBO Mask for QFN)		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
X.2	cpbo will be created and OR-ed with pmm.dg; Then checked for min spacing rule; pmm.dg can be drawn to fix cldrc errors		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
1	Min. width of cpbo		N/A	N/A	N/A	N/A	13.000	N/A	N/A	N/A
2	Min. spacing of cpbo to cpbo; Merge if space is less than minimum		N/A	N/A	N/A	N/A	15.000	N/A	N/A	N/A
3	Min. enclosure of *fuse:dg* by cpbo	A	N/A	N/A	N/A	N/A	12.000	N/A	N/A	N/A
4	Min. enclosure of *bondPad* by cpbo (Rule does not apply to pad.dg inside "M5RDL" pcell)	A	N/A	N/A	N/A	N/A	7.500	N/A	N/A	N/A
4a	Min/Max enclosure of 5x5 pad.dg by cpbo	A	N/A	N/A	N/A	N/A	4.000	N/A	N/A	N/A
7	Min. enclosure of cpdm:mk in the die by cpbo	A	N/A	N/A	N/A	N/A	30.000	N/A	N/A	N/A
9	Min enclosure of *photoArray* by cpbo	A	N/A	N/A	N/A	N/A	0.000	N/A	N/A	N/A
10	Min enclosure of (areaid.mt outside areaid.ft) by cpbo; Exclude cell name "*_tech_CD_top*"	A	N/A	N/A	N/A	N/A	0.000	N/A	N/A	N/A
11	cpbo must be enclosed by (ccu1m or cpmm2)		N/A	N/A	N/A	N/A		N/A	N/A	N/A
11a	Min enclosure of cpbo by cpmm2 (Rule checked outside areaid.ft)		N/A	N/A	N/A	N/A	5.000	N/A	N/A	N/A
12	Min space of cpbo to rdl:dg		N/A	N/A	N/A	N/A	7.360	N/A	N/A	N/A

(ccu1m.-)	Cu inductor Mask		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
NOTE	Refer to JNET-80 for best practices to draw CU1M dummy pattern (not checked by DRC).									
3	Min pattern density of ccu1m across full chip		N/A	N/A	N/A	N/A	35.0%	N/A	N/A	N/A
4	Max pattern density of ccu1m across full chip		N/A	N/A	N/A	N/A	45.0%	N/A	N/A	N/A
5	cpbo by ccu1m must be enclosed by at least		N/A	N/A	N/A	N/A	6.750	N/A	N/A	N/A

(cpmm2.-)	Polyimide2 Mask (QFN only)		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
X.1	cpmm2 will be created and OR-ed with pmm2.dg; Then checked for min spacing rule; pmm2.dg can be drawn to fix cldrc errors									
1	Min. width of cpmm2. Rule excludes violations caused by cpmm2.7 inside areaid.sl		N/A	N/A	N/A	N/A	20.000	N/A	N/A	N/A
2	Min. spacing of cpmm2 to cpmm2; Merge if space is less than min		N/A	N/A	N/A	N/A	20.000	N/A	N/A	N/A
3	Min. enclosure of (*bondPad* NOT (inductor.dg OR pad.dg inside "M5RDL" pcell)) by cpmm2	A	N/A	N/A	N/A	N/A	12.500	N/A	N/A	N/A
4	Min spacing between cpmm2 to RDL (Rule is exempted for CPMM2 inside of areaid.rdlprobepad)	A	N/A	N/A	N/A	N/A	6.160	N/A	N/A	N/A
5	Min enclosure of (areaid.mt outside areaid.ft) by cpmm2; Exclude cell name "*_tech_CD_top*"	A	N/A	N/A	N/A	N/A	12.500	N/A	N/A	N/A
6	Min spacing, no overlap, between cpmm2 and Inductor.dg		N/A	N/A	N/A	N/A	6.160	N/A	N/A	N/A
7	Min enclosure of areaid.sl by cpmm2	A	N/A	N/A	N/A	N/A	0.000	N/A	N/A	N/A

(cpid.-)	Copper pillar ID		s8di-5r	s8tnv-5r	s8tma-5r	s8p-10r	s8phirs-10r	s8pfn-20r	s8*phr-10r	s8phrc*
1	Copper pillar ID (areaid:cr) is a mark with W=L of…	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2	In tapeout view, pad:dg in the top die etec cell must enclose areaid:cr on the left or bottom edge on pads larger than 39 um x 39 um with text label CU-PILLAR-PAD, by	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2a	Pad:dg in the top die etec cell must enclose areaid:cr on 39 um x 39 um padswith text label CU-PILLAR-PAD, by	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
3	Min enclosure of pad:dg for pads larger than 39 um x 39 um in the die by areaid:cr on one out of 2 adjacent sides	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A



Main Die tapeout view

Section G3: High Voltage Methodology

High Voltage is defined as a voltage outside the range of GND to Vcc. Any device that is subjected to a voltage outside the range of GND to Vcc is considered a high voltage device. These devices are subjected to special design rules and biasing conditions. The biasing conditions of these high voltage devices are detailed in the ETD

A. Failure Mechanisms in High Voltage Devices

The TDR have a special rules section for the layout and DRC of the high voltage (hv) device. These rules are framed so as to prevent the following failure mechanisms in circuits that use these devices:

1. Transistor Performance Degradation under HV Gate Stress (Section 2.2.2 of EDR): The maximum voltage across the gate oxide (gate to channel voltage) is restricted to:
(a) Any HV NMOS device: 7.3 V @ 25C.
(b) Any HV PMOS device: 8.1 V @ 25C.
These voltages are not operating voltages, but points of failure. They should not be exceeded in any circuit at any time.

2. Junction Leakage/breakdown: The maximum source/drain to substrate junction voltages are restricted to the following:
(a) Any HV NMOS device: 11.0 V @ 25C.
(b) Any HV PMOS device: 11.0 V @ 25C.
These voltages are not operating voltages, but points of failure. They should not be exceeded in any circuit at any time.

3. Gated-Diode Leakage/Breakdown: All high voltage devices use 110A gate oxide thickness just like low voltage (0 to Vcc) devices. The maximum gate-to-junction voltage differentials should be not exceed the voltage criteria set by conditions (1) and (2) above.

4. Source to Drain Punch-through: To prevent punch-through, the hv devices have expanded channel lengths:
(a) HV NMOS/PMOS device channel length = 0.500 um final.

5. Parasitic Isolation Field Leakage: HV poly is prohibited from forming gates with adjacent hv diffusions, and from crossing well boundaries. Exceptions to this rule are made only in cases where the bulk of the isolation device formed is back-biased by at least 300 mV. The presence of the back bias cannot be checked by the CAD flow at this time. Exceptions pass clean through DRC with the presence of the “hv_bb” tag on the hv poly. The usage of the “hv_bb” tag is subject to approval by technology.

6. Sub-threshold EndCap Leakage: The extension of poly forming a high voltage gate onto field to prevent subthreshold leakage due to line-end shortening of the poly/field oxide endcap.

B. High Voltage Implementation Methodology

Following are the features of the high voltage rules:

High Voltage Diffusion (*hvSRCDRN*):

- The source of high voltage is a diffusion(source/drain) tagged within the high voltage identification layer diff:hv. The whole diffusion feature need to be completely enclosed by the diff:hv layer.
- The source/drains that are tagged with the diff:hv layer are called *taggedhvSRCDRN* within the CAD flow code. The propagation of the high voltage property within the tagged piece of diffusion stops at a gate, i.e.. if tagging is done on the drain side, the source does not become a high voltage feature.
- Beginning with *taggedhvSRCDRN*, high voltage propagates through standard interconnect to other *SRCDRN* or poly. Any *SRCDRN* derives the high voltage property through electrically shorting to a *taggedhvSRCDRN* is called a *derivedhvSRCDRN* within the CAD flow.
  Hence by definition, within the CAD flow,

  hvSRCDRN = OR(*taggedhvSRCDRN derivedhvSRCDRN*)

- Rule **hv.X.1** (high voltage source/drain regions must be tagged by diff:hv) will check the presence of the diff:hv tag and flag on all *derivedhvSRCDRN*. When the layout is finally clean of all hv.diff.1 errors, *hvSRCDRN* will consist of only *taggedhvSRCDRN* (all *derivedhvSRCDRN* will need to be tagged with diff:hv to remove errors). This is shown in **Fig.1**.
- Rule **hv.diff.1** (Minimum *hv_source/drain* spacing to diff for edges of *hv_source/drain* and diff not butting tap) prevents adjacent diffusions from punching through. Note that this rule specifies the spacing of a *hvSRCDRN* to any diffusion – be it another *hvSRCDRN* or a normal diffusion. This rule also applies to N+/P+ resistors that become hv by propagation. This is shown schematically in **Fig.2**.

r **High Voltage Poly (*hvPoly*):**

l A high voltage poly feature (*hvPoly*) is defined as a poly feature which is electrically shorted to *hvSRCDRN*, or to another high voltage feature (like another *hvPoly*) through an interconnect. The whole poly feature becomes high voltage feature.

l *hvpoly* propagates the high voltage property to other features which are electrically shorted (through licon1 & li1); but does not act as a source of high voltage. This means that hvpoly does not make underlying diffusions or wells high voltage – it acts as a "conductor" and propagates the high voltage property to other electrically connected features.
hvpoly cannot form parasitic field isolation devices, unless this device is back-biased. Hence, the following rules are in place:
hv.poly.1: Hv poly feature can be drawn over only one diff region and is not allowed to cross nwell boundary except as allowed in rule hv.X.3. Please refer to **Fig.4**.

hv.X.3: High voltage poly can be drawn over multiple diff regions that are ALL reverse-biased by at least 300 mV (existence of reverse-bias is not checked by the CAD flow). In this case, the high voltage poly can be tagged with the text:dg label with a value "hv_bb". Exceptions to this use of the hv_bb label must be approved by technology. Under certain bias conditions, high voltage poly tagged with hv_bb can cross an nwell boundary. Use of the hv_bb label on high voltage poly crossing an nwell boundary must be approved by technology. This is shown in **Fig.5**. All high voltage poly tagged with hv_bb will not be checked to hv.poly.1, hv.poly.2, hv.poly.3 and hv.poly.4.
hv.poly.2: Spacing of hv poly on field to unrelated diff (**Fig.6**).
hv.poly.3: Spacing of hv poly on field to n-well (**Fig.6**).
hv.poly.4: Enclosure of hv poly on field by n-well (**Fig.6**).

l Poly resistors can become high voltage features if the poly is electrically shorted to *hvSRCDRN*, or to another high voltage feature. Nevertheless, these devices cannot act as sources of hv, and the hv propagation stops at the edge of this device.

r **High Voltage Poly Gate (*hvFET_gate*):**

l A high voltage poly gate (*hvFET_gate*) is a gate (PolyAndDiff) abutting *hvSRCDRN*. This is specified in rule hv.poly.8 (Any poly gate abutting *hv_source/drain* becomes a high voltage poly gate). Note that this is the only definition of a *hvFET_gate* and the only way a gate can become a *hvFET_gate*. This is shown schematically in **Fig.7**.

l The high voltage property of the *hvFET_gate* is limited to the gate only – the whole poly feature does not become a *hvPoly*.

l The following rules are in place for *hvFET_gates* (please refer to **Fig.12**):
hv.poly.5: Hv poly gate length (which is bigger than a normal gate length)
hv.poly.6: Extension of poly forming an *hvFET_gate* beyond hv diffusion

r **Stoppers to High Voltage Propagation:**

The following act as stoppers for hv propagation (shown in **Fig.10**):

l For a *hv_source/drain* tagged with diff:hv, the high voltage property terminates at the intersection of this hv diff with a poly, i.e. at the gate edge. This means that one side of the device can have a hv diff, while the other side of the gate can remain low voltage.

l N+/P+ diffusion resistors are allowed per the **Allowed Resistors** table in the TDR. These resistors do not originate high voltage. They also do not propagate high voltage, although the device itself becomes a high voltage device. The hv rule **hv.diff.1** needs to be checked for these devices.

l Diodes do not originate high voltage. Nevertheless, they propagate high voltage and become high voltage devices when high voltage is propagated to them. The hv rule **hv.diff.1 needs to be checked for these devices.**

l A poly forming a poly resistor can become *hvPoly* by virtue of shorting to a *hv_source/drain* or shorting to another high voltage feature through an interconnect. The high voltage propagation stops at a poly resistor, although the device itself becomes high voltage. This device will be checked to the following hv rules: **hv.poly.1**, **hv.poly.2**, **hv.poly.3,** and **hv.poly.4**. These rule checks can be exempted by the use of the "hv_bb" tag with the approval of technology.

l The high voltage propagation also stops at a P-Well resistor. The device becomes a hv device. There are no specific rule checks for this hv device.

r **Summary of High Voltage Propagation:**

The high voltage propagation methodology is summarized below in **Table 1**. A test case utilizing the outlined methodology is shown in **Fig.12**.

Table 1. Truth table for high voltage generation, propagation and retention.

Node Type	Originates HV?	Propagates HV?	Becomes HV? (when HV Propagates to this node)	Notes
Deep N-Well	No	N/A	N/A	1,7
P-Well	No	N/A	N/A	1,7
P-Well Resistor	No	No	Yes	2
N-Well	No	N/A	N/A	1,7
waffle_chip	No	Yes	Yes	
P+ Diffusion	No	Yes	Yes	
N+ Diffusion Resistor	No	No	Yes	3
P+ Diffusion Resistor	No	No	Yes	3
HV Diffusion	Yes	Yes	Yes	
Diodes	No	Yes	Yes	3
Poly	No	Yes	Yes	
Poly Resistor	No	No	Yes	4
GATE	No	No	No	
HvFET_gate (GATE abutting hv Diff)	No	No	Yes	5
Licon1	No	Yes	N/A	6
Li1	No	Yes	N/A	6
Mcon	No	Yes	N/A	6
Met1	No	Yes	N/A	6
Via	No	Yes	N/A	6
Met2	No	Yes	N/A	6

1) Deep N-Wells, N-Wells and P-Wells cannot be used as routing layers.
2) No hv rule checks for this device.
3) For N+ and P+ diffusion resistors and diodes, rule hv.diff.1 (spacing to unrelated diff) needs to be checked.
4) Need to be checked for hv.poly.1, hv.poly.2, hv.poly.3, hv.poly.4. Needs technology approval for use of hv.X.3.
5) The hv property is localized to the hvgate and its extensions.
6) Interconnect and contacts propagate hv, and are hv devices internal to the CAD flow only.
7) "N/A" implies that there are no special hv rules for these layers.

Figure 1: **<u>hv.X.1:</u> High Voltage source/drains must be tagged by diff:hv**

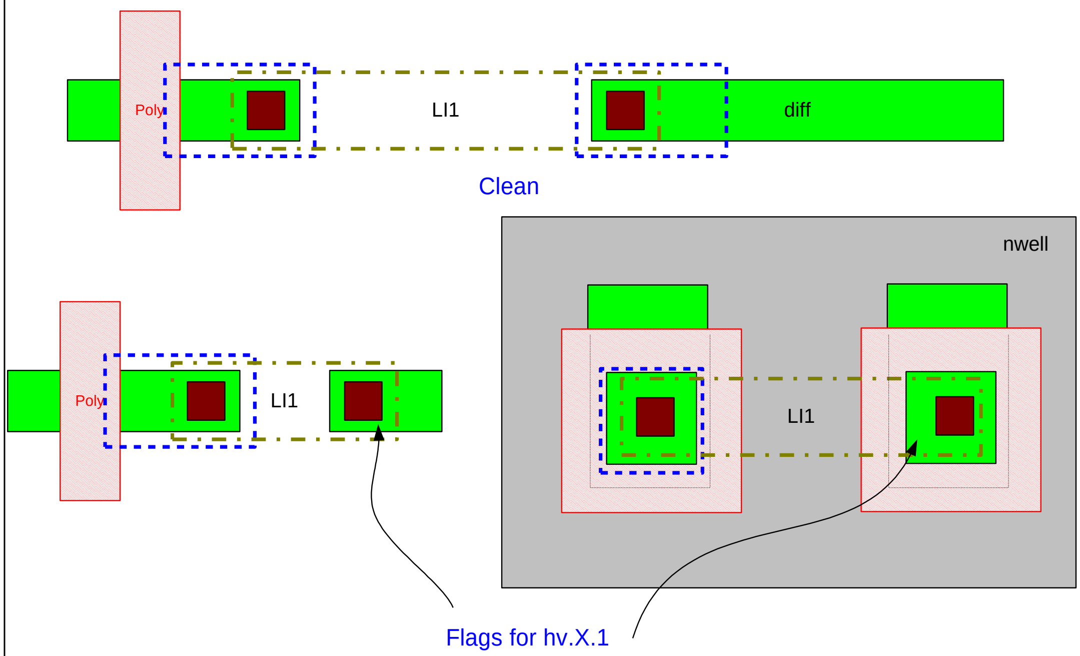


Figure 2: **<u>hv.diff.1:</u> Minimum *hv_source/drain* spacing to diff for edges of *hv_source/drain* not butting tap**

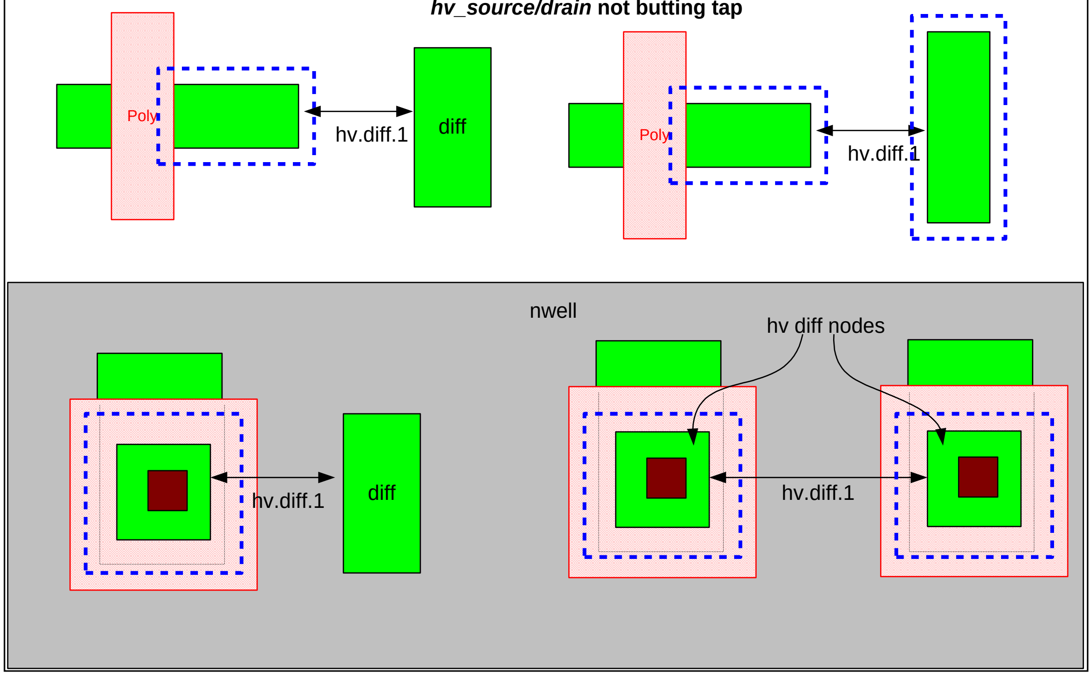


Figure 4: **hv.poly.1:** Hv poly can be drawn over only one diff region and not allowed to cross nwell boundary except when allowed by hv.X.3

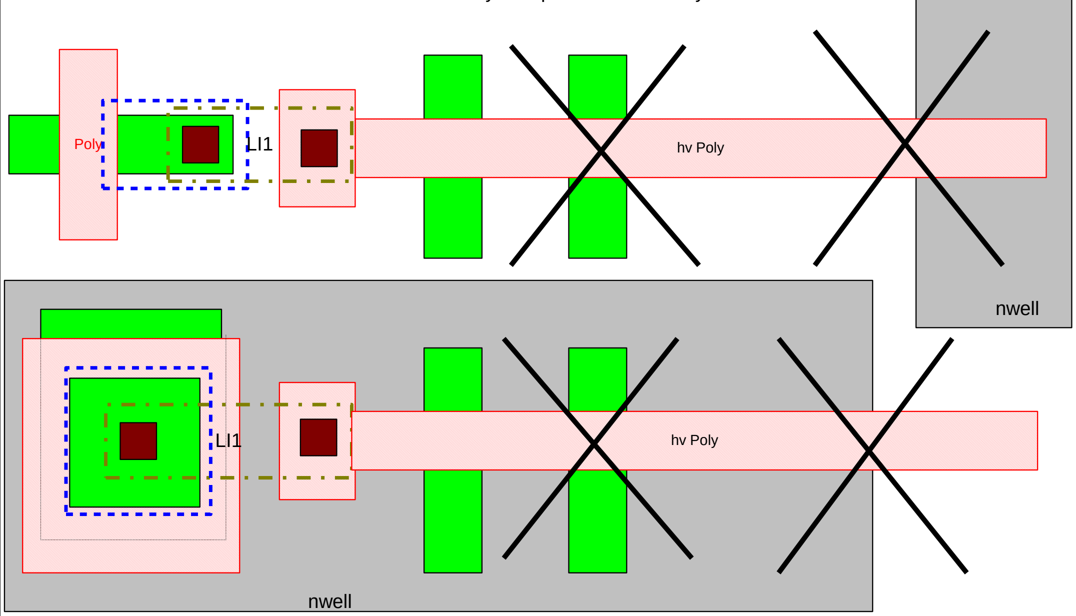


Figure 5: **hv.X.3:** Hv poly can be cross multiple diffusions or cross nwell boundaries if the isolation devices are back-biased. The "hv_bb" label should be used on the hvPoly in this case. hvPoly tagged will not be checked to hv.poly.1, hv.poly.2, hv.poly.3 & hv.poly.4.

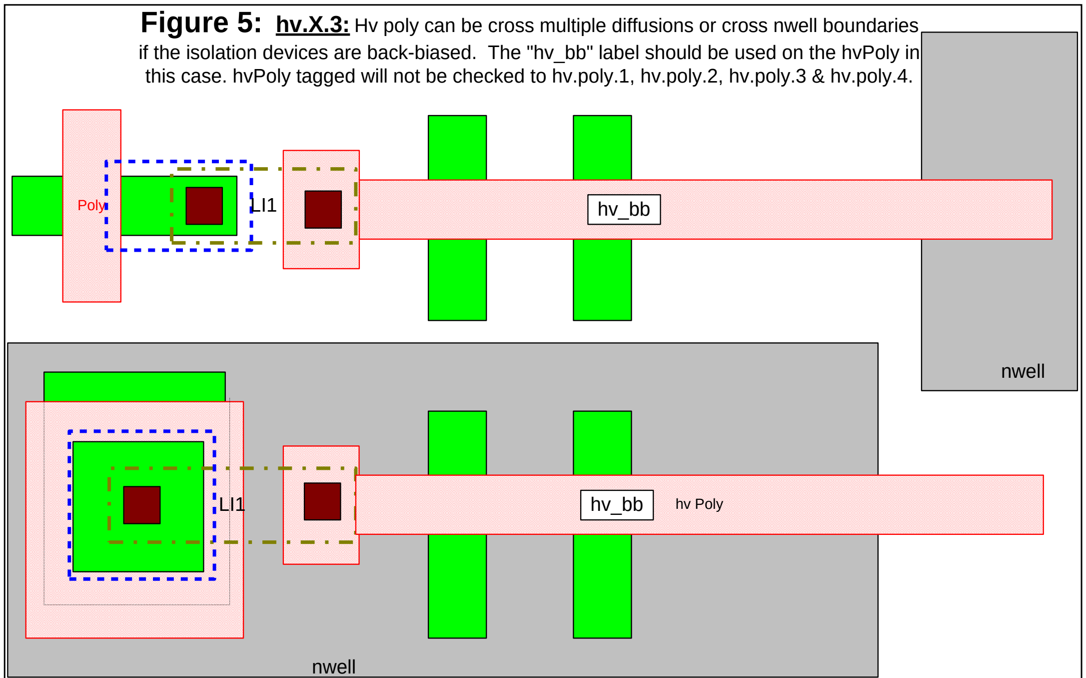


Figure 6: hv.poly.2,3,4: Spacings and enclosures of hvPoly

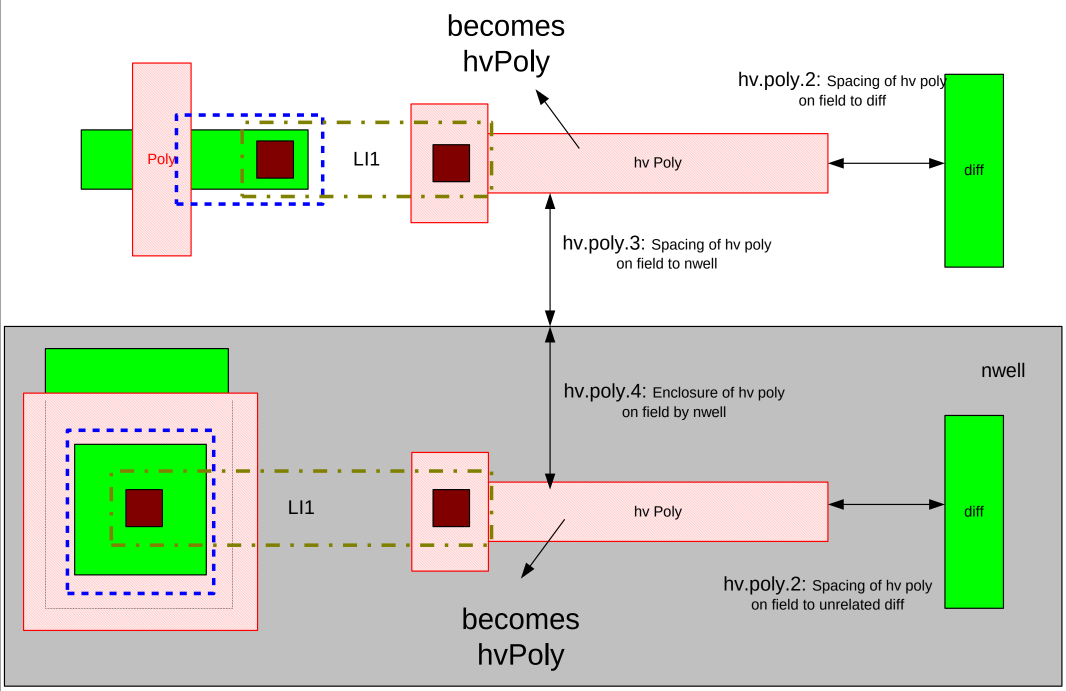


Figure 7: Definition of hvGate = hv.poly.8

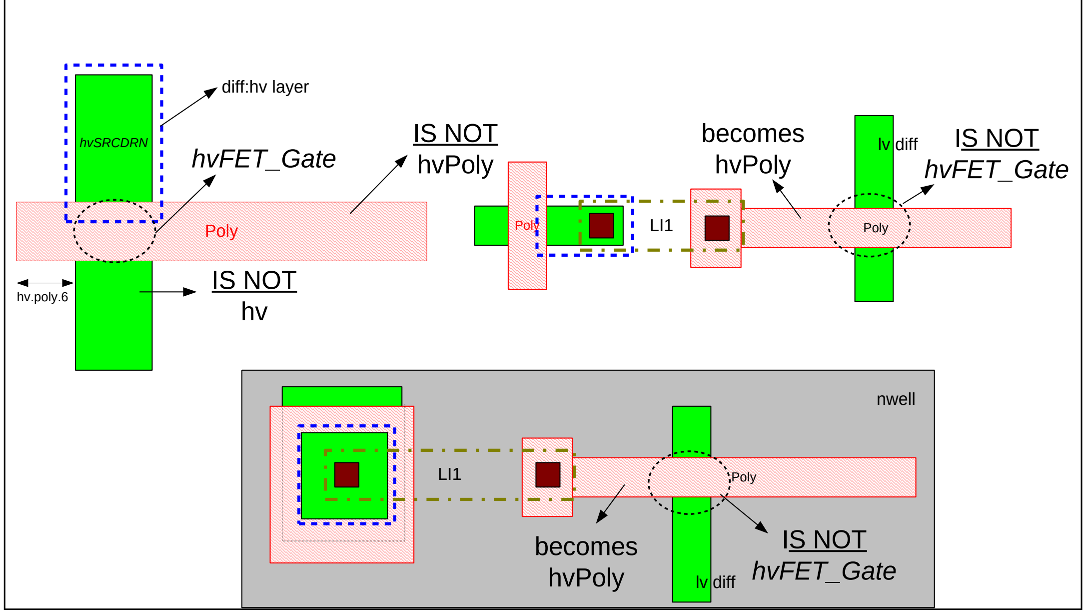


Figure 9: hv.X.4: Any feauture that is shorted to hv_source/drain becomes a hv feature

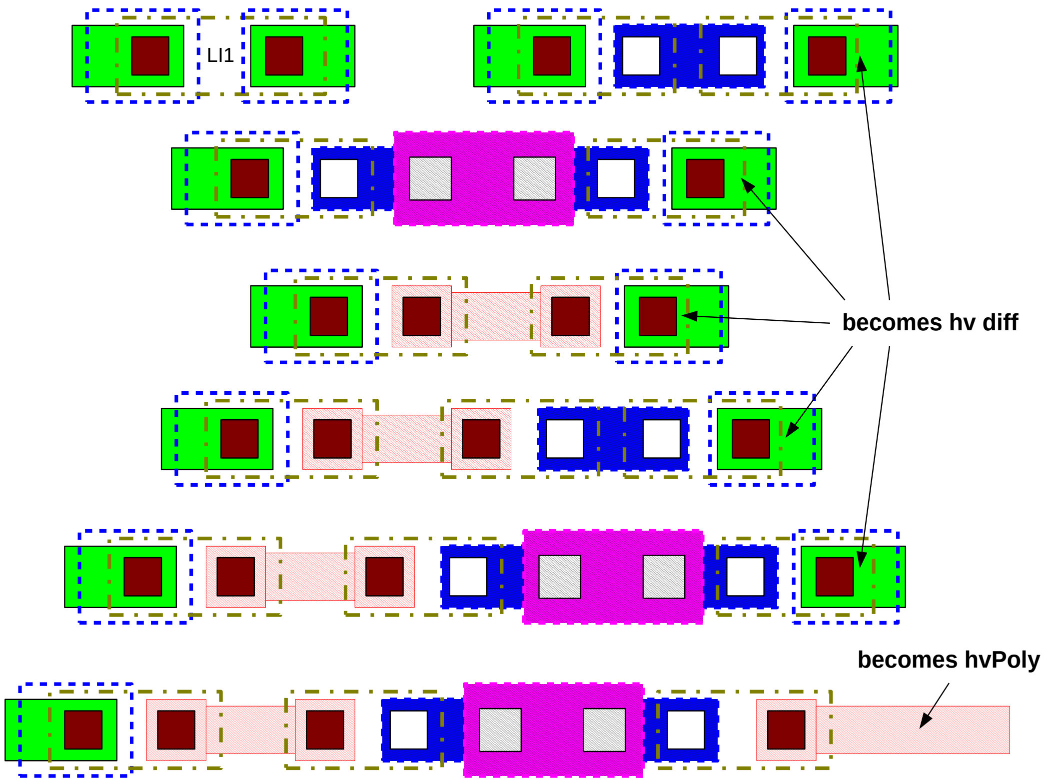


Figure 10: High Voltage Propagation Stoppers

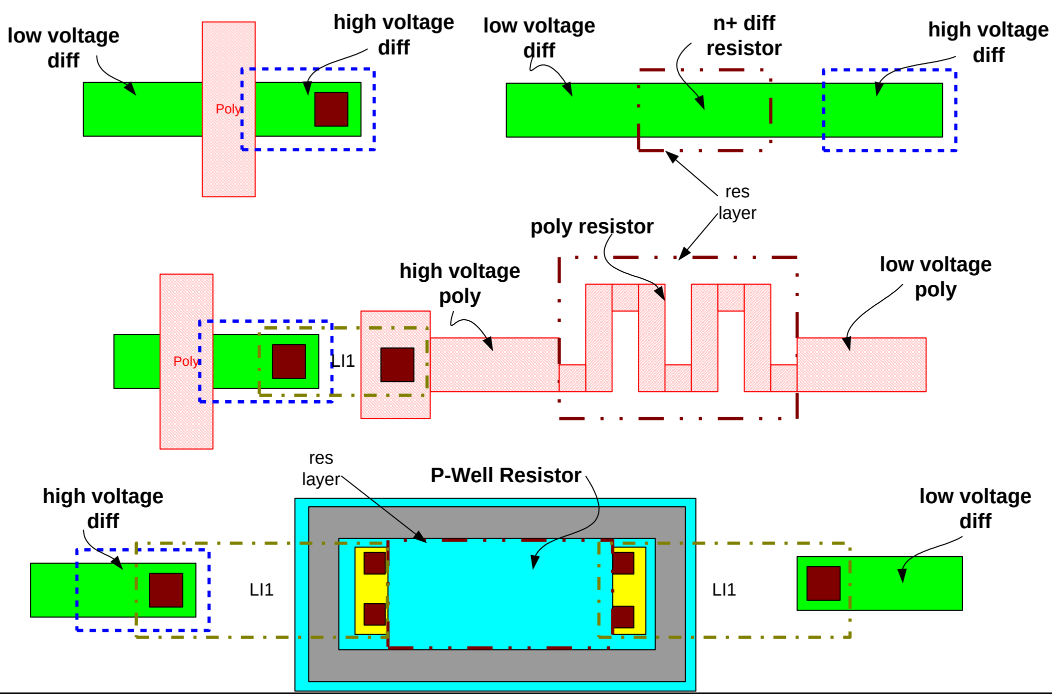


Figure 11: hv.poly.5,6,7: Lenghts, extensions and overlaps of *hvFET_Gate*

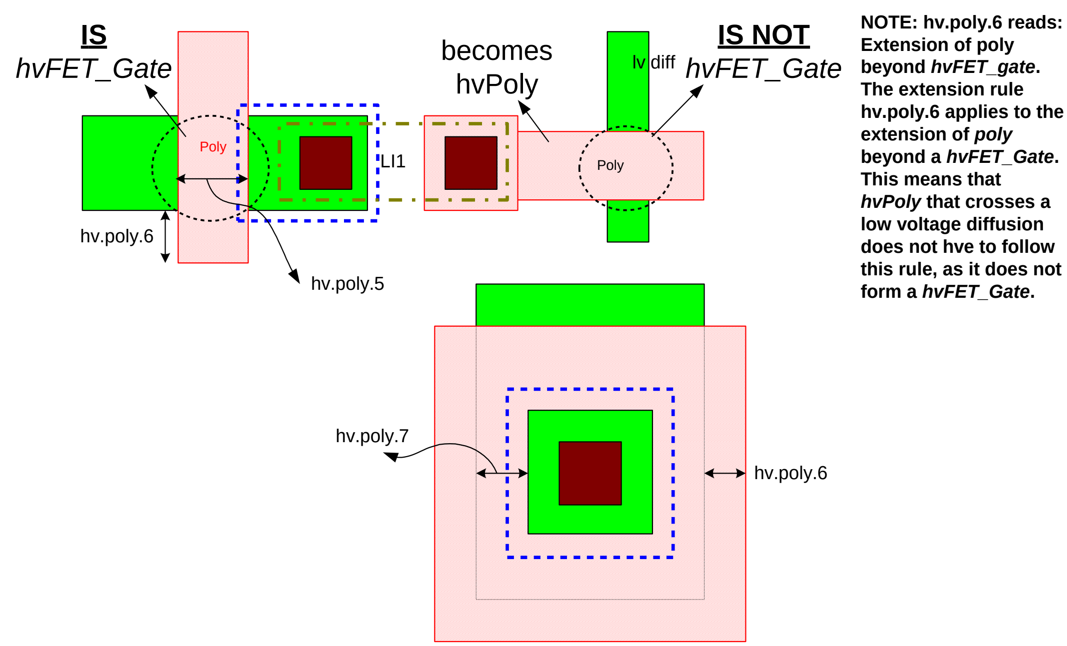
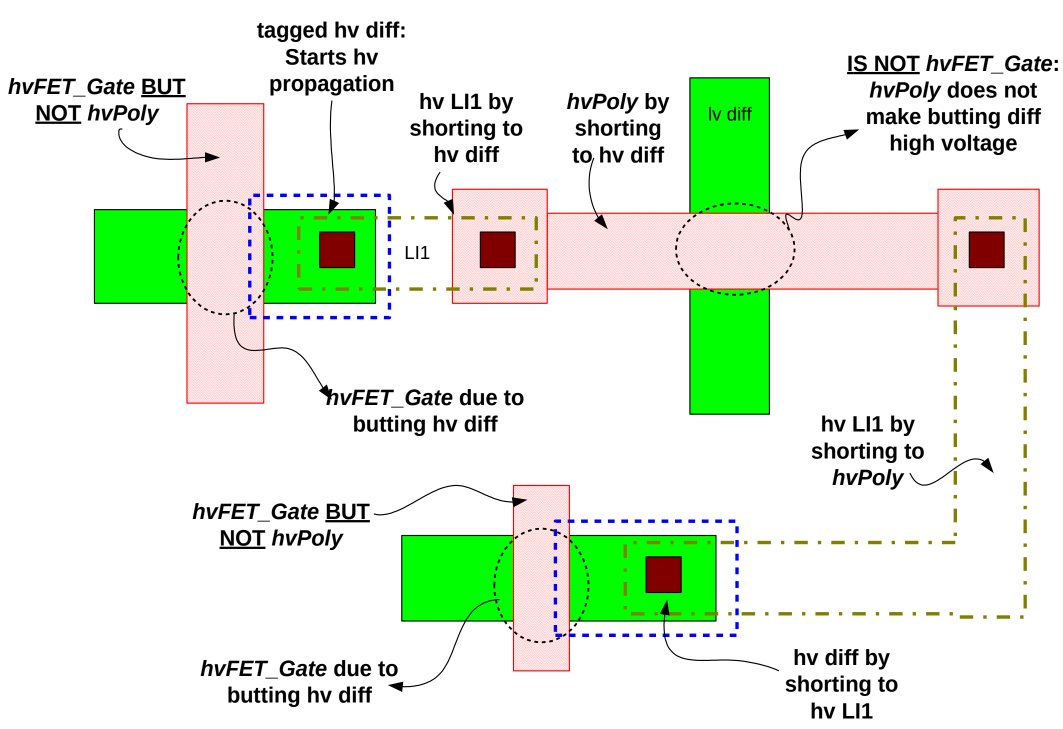


Figure 12: Test Case



Section G4: Very High Voltage Methodology

Very High Voltage is defined as a voltage outside the range of GND to High Voltage (11V). Very high voltage is 16V (12V nominal) Vcc. Any device that is subjected to a voltage outside the range of GND to 11V is considered a Very High Voltage (VHV) device. These devices are subjected to special design rules and biasing conditions.

A. Failure Mechanisms in VHV Devices

The TDR have a special rules section for the layout and DRC of the VHV device. These rules are framed so as to prevent the following failure mechanisms in circuits that use these devices:

1. Transistor Performance Degradation under VHV Gate Stress: the maximum voltage across the gate oxide (gate to channel voltage) is restricted to:
(a) Any VHV NMOS device: 5.5V.
(b) Any VHV PMOS device: 5.5V.

2. Junction Leakage/breakdown: The maximum source/drain to substrate junction voltages are restricted to the following:
(a) Any VHV NMOS device: 16.0V.
(b) Any VHV PMOS device: 16.0V.

3. Gated-Diode Leakage/Breakdown: All VHV devices use 110A gate oxide thickness just like standard 5.0V Vcc devices. The maximum gate-to-junction voltage differentials should not exceed the voltage criteria set by conditions (1) and (2) above. The VHV devices need to be designed with drain extensions (DE) fabricated by lightly doped Nwells and Pwells respectively. Under no circumstances the poly/extended drain overlap and field oxide length should be changed.

4. Source to Drain Punch-through: To prevent punch-through, the VHV devices have expanded channel lengths:
(a) VHV NMOS device channel length = 1.055 um drawn.
(b) VHV PMOS device channel length = 1.050 um drawn.

5. Parasitic Isolation Field Leakage: Poly from a drain extended device is prohibited from forming gates with adjacent hv diffusions.

6. Sub-threshold EndCap Leakage: The extension of poly forming a high voltage gate onto field to prevent subthreshold leakage due to line-end shortening of the poly/field oxide endcap.

7. Reliability performance: In order to preserve the reliability performance of the VHV FETs the Field Oxide (STI) length may not be changed from the values below:
(a) VHV NMOS STI length = 1.585 um
(a) VHV PMOS STI length = 1.190 um

8. A poly gate may never be directly connected to a VHV diffusion region.

9. Poly connecting two VHV nodes over field must be routed through LI or metal.

B. VHV Implementation Methodology

Following are the features of the VHV rules:

* All features operating at 16V (max) voltages can be Very-High-Voltage (VHV)
* Drain or source of the drain-extended device can be tagged with vhvi:dg layer. Device with either drain or source (not both) tagged with vhvi:dg layer serves as propagation stopper
* The *VHVSourceDrain* can be connected to another *VHVSourceDrain* or an output pad. The *VHVSourceDrain* does not propagate the VHV through the device
* All source/drains/gate tagged with vhvi:dg propagate VHV through any interconnects.
* Diff inside areaid.ed on the same net as *VHVSourceDrain* should be tagged with vhvi:dg. They serve as propagation stopper.
* Deep N-well, N-well, P-well, Diff, or Poly cannot be used as routing layers

Table 1. Truth table for high voltage generation, propagation and retention.

Node Type	**Originates VHV?**	**Propagates VHV?**	**Requires tagging with vhvi:dg (flags if not tagged when required)**
Deep N-Well	No	N/A	N/A
P-Well	No	N/A	N/A
P-Well Resistor	No	No (1)	Yes
N-Well	No	N/A	N/A
LV Diffusion	No	Yes	Yes
Diffusion Resistor	No	No (1)	Yes
HV Diffusion	No	Yes	Yes
VHV ESD Diffusion	No	No	Yes
VHVSourceDrain	Yes	No (2)	Yes
Diodes	No	Yes	Yes
Poly	No	N/A	N/A
Poly Resistor	No	No (1)	Yes
VHVPoly	Yes	Yes	Yes
GATE	No	N/A	N/A
de_pFET_gate, de_nFET_gate	No	N/A	N/A
Licon1	No	Yes	No
Li1	No	Yes	No
Mcon	No	Yes	No
Met1	No	Yes	No
Via	No	Yes	No
Met2	No	Yes	No
via2	No	Yes	No
Met3	No	Yes	No

(1) Resistors tagged with text "vhv_block" serve as VHV propagation stopper and it is the duty of the designer to ensure that the resistor can support the required voltage drop. Otherwise components in VHV nets need to be tagged with vhvi:dg layer
(2) If only source or drain is tagged with vhvi:dg layers

IP Integration Enhancement Rules

The following section documents details and the rules required to implement CWR-137 enhancements

Stress requirement

* Critical side stress rules are to be met on all newly designed IP blocks.
* Legacy IP blocks can avoid the stress check using a cadflow switch ("noCritSideReg").
* Allow parts of new IP blocks to meet non-critical side stress rules using an areaid.ncs (non critical side), if following critical side stress rules can lead to die area increase at top-level.

Coupling Protection Requirement

IP block designers can prevent coupling to sensitive lines in an IP block using metX.be (blockage) layer

metblk.-	Metal blockage Rules		
1	Min spacing, no overlap, between MetX blockage layer to another MetX.dg on the same layer		min space of MetX layer
2	Min spacing, no overlap, between Li1 blockage layer to li1.dg		0.17
3	Min spacing between metX blockage layer to routing_terminal in the same layer		min space of MetX layer+ 1 snap grid
4	Min spacing between li1 blockage layer to routing_terminal in li1 layer		0.17 + 1 snap grid
6	prBoundary.boundary not allowed inside layout	**RC**	
7	poly, diff, tap boundary layers not allowed inside layout	**RC**	

Additional rules must be satisfied for the areaid layers

chipint.-	Chip Integration Rules		
1	ESD Diffusion (except diffusion in power to ground clamps specified in spec 001-04469) sized by 50um must not overlap areaid low tap density	**RR**	
2	ESD Diffusion (except diffusion in power to ground clamps specified in spec 001-04469) sized by 150um must not overlap areaid injection	**RR**	

Substrate Noise Isolation Rules

Definitions

1	Victim region: a p-tap ring with a "vic" or "victim" text.drawing label on it, as well as the area the ring encloses
2	Aggressor region: a p-tap ring with a "agr" or "aggressor" text.drawing label on it, as well as the area the ring encloses
3	Both victim and aggressor regions also are enclosed by areaid.substrateCut (see subiso.7)

subiso.-	Rules	Job Type	Value
1	The "local_sub" switch must be used whenever the victim, vic, aggressor, or agr labels are used in the layout	LVS, LU	**N/A**
2	substrate ring for aggressor isolation must connect to net starting with "vsub_agr"	LU	N/A
3	substrate ring for victim isolation must connect to net starting with "vsub_vic"	LU	N/A
4	Victim/aggressor tagged p-tap cannot be outside of victim/aggressor region	LU	N/A
5	Victim and aggressor labels cannot be on the same tap ring	LU	N/A
6	Victim and aggressor rings cannot be in isolated pwell	LU	N/A
7	Minimum enclosure of victim/aggressor tap rings by areaid.substrateCut by number of snap grids	LU	2 um
8	Minimum spacing of victim/aggressor regions	LU	50 um
9	Minimum width of victim and aggressor p-tap rings	LU	10
10	Victim and aggressor p-tap rings must be fully contacted and strapped in li1 and met1	LU	N/A
11	Contacts licon1 and mcon for rings must be at most 5 times the minimum spacing apart	LU	N/A
12	P-tap that is not a ring cannot have victim/aggressor labels on it	LU	N/A
13	areaid.substrateCut inside victim/aggressor regions must not have holes	LU	N/A
14	Victim/aggressor labels must only be on p-tap	LU	N/A
15	Victim substrate p-tap cannot be connected to p-tap in a non-victim region	LU	N/A
16	Aggressor substrate p-tap cannot be connected to a p-tap in a non-victim region	LU	N/A
17	Maximum resistance from victim/aggressor outer guard ring to pin at block level (with block_level switch on)	luRes	5 ohms
18	Maximum resistance from victim/aggressor outer guard ring to pad at full chip level (with block_level switch off)	luRes	10 ohms
19	LVS property tolerance for victim and aggressor diode devices (dnwdiode_psub_victim, nwdiode_victim, dnwdiode_psub_aggressor, nwdiode_aggressor)	LVS	10%

Notes

1 All dnwell/nwell diodes to substrate inside victim/aggressor regions are checked in LVS using the lvsdiode tech element from the tech library
2 The nwell devices are nwdiode_victim and nwdiode_aggressor (lvsdiode tech element)
3 The dnwell devices are dnwdiode_victim and dnwdiode_aggressor (lvsdiode tech element)
4 There is a 10% tolerance for area and perimeter values for the victim/aggressor LVS diodes.
5 Resistance is measured from a 5x5um region at the corner of each victim/aggressor guard ring

Rules for resistance check			
The following section documents details and the rules required for the resistance checks (including li jumper check on pwr/gnd buses)			
res.-	**Resistance rule check (including li jumper)**	**Use**	**Value (um)**
1	Pwr/gnd nets should never have high resistance metals in the path (This is done by doing a li jumper check on all pwr/gnd busses during latchup runs) The algorithm: 1. Identify met1 and li connected to PWR/GND. 2. Disconnect the li to find the floating met1. This will get the initial li_jumper. 3. find the jumper which connected to connected met1 on one side and connected to floating met1 on the other side. 4. Ensure the li_jumper is being used as a distributor by finding multiple discrete contacts are connected to the li_jumper. 5. Find the case where the floating met1 has via on it to verify that it is routing back up. 6. Exemptions: (a) li with area <=0.25 (b) li with aspect ratio w/l of 4:1 The li_jumper can be exempt by using "li_jumper_ok" text on the intentional jumper. Also for the intentional jumper, needs to verify by running RCX with PWR/GND extracted and simulation shows no adverse effect.		
1a	Switched Power net, denoted by a switched_power text label should never have high resistance metals in the path (This is done by doing a li jumper check on all switched power busses during latchup runs) The algorithm: 1. Identify met1 and li connected to switched power net. 2. Disconnect the li to find the floating met1. This will get the initial li_jumper. 3. Find the jumper which connected to connected met1 on one side and connected to floating met1 on the other side. 4. Ensure the li_jumper is being used as a distributor by finding multiple discrete contacts are connected to the li_jumper. 5. Find the case where the floating met1 has via on it to verify that it is routing back up. 6. Exemptions: (a) li with area <=0.25 (b) li with aspect ratio w/l of 4:1 The li_jumper can be exempt by using "li_jumper_ok" text on the intentional jumper. Also for the intentional jumper, needs to verify by running RCX with switched power net extracted and simulation shows no adverse effect.	**RR**	

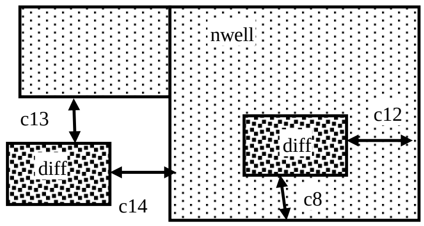
S8 Rule Description — Use — Dimension [mm]

Section H. Core Rules

H1: Core rules for all flows

(difftap.-) Diff/tap

c1	Min. width of diff, tap, or MOSFET channel	**MR**	0.140
c5	Min tap bound by two diffusions		0.380
c8	Enclosure of (p+) diffusion by N-well		0.150
c10	Enclosure of (n+) tap by N-well		0.150
c11	Min width of diff inside hvi		0.150
c12	Min enclosure of (p+) diff by Nwell on one of two adjacent sides		0.180
c13	Min spacing of (n+) diff to Nwell		0.320
c14	Min spacing of (n+) diff to Nwell on one of two adjacent sides		0.340



(hvtp.-) Hvtp

c1	Min/Max enclosure of nwell by hvtp		0.000

(poly.-) Poly

c1	Spacing of poly on field to diff or tap (parallel edges only)		0.030
c2	Core poly gap: spacing between two exactly opposite poly line ends where the width of both lines is = poly.1 (drawn only)	**MR**	0.160
c3	Min poly spacing except poly.c2		0.175

(n/ psd.-) N+/P+ Source/Drain Implants (Nsdm and Psdm)

c1a	Width of nsdm(psdm), opposite, parallel edges only		0.380
c1b	Width of nsdm(psdm)	**MR**	0.290
c2a	Spacing of nsdm(psdm) to nsdm(psdm), opposite, parallel edges only		0.380
c2b	Spacing of nsdm(psdm) to nsdm(psdm)	**MR**	0.290
c5a	Enclosure of *N+tap* by nsdm, except for butting edge		0.130
c5b	Enclosure of *P+tap* by psdm, except for butting edge		0.120

(ldntm.-) Ldntm

c1	Min width of ldntm	**MR**	0.700
c2	Min spacing of ldntm to ldntm	**MR**	0.700
c3a	Enclosure of *N+_diff* by ldntm		0.180
c3b	ldntm can straddle *N+_diff*		
c4	Min enclosure of *nfet* by ldntm		0.125
c5	Ldntm must be enclosed by areaid.ce	**TC**	
c6	Min spacing of ldntm to *P+_diff*	**TC**	0.180

(licon.-)	Licon1		
c1	Spacing of *poly_licon* to diff or tap		0.130
c2	Licon1 can straddle diff	NC	
c3	Spacing of licon to licon	MR	0.165
c4	Psdm cannot overlap (licon And poly)		

(npcon.-)	NPC and contact rules		
c6	NPC must enclose *poly_licon* by…	MR	0.045

(li.-)	LI1		
c1	Min width of LI	MR	0.140
c2	Min space LI to LI	MR	0.140

(ct.-)	Mcon		
c1	Mcon must overlap LI		
c2	Spacing of mcon to mcon	MR	0.190

(m1.-)	Met1		
c1	Mcon must be enclosed by Met1 by, at least, …		0.000

(m2.-)	Met2		
c4	Via must be enclosed by Met2 by at least …	MR	0.040

Rules for all flows except S8TNV-5R

(ncm.-)	Ncm		
c8	Min enclosure of *P+_diff* by ncm	TC	0.235
c9	Min spacing, no overlap, between ncm and *N+_diff*	TC	0.235
c10	Min spacing between ncm to (nwell outside areaid.ce) [exclude coincident edges]		0.380

Rules for S8TNV-5R flow only

(ncm.-)	Ncm		
c4	Min enclosure of *N+_diff* by ncm	TC	0.185
c5	Ncm can straddle *N+_diff*		
c6	Min enclosure of *nfet* by ncm		0.120
c7	Min space of ncm to *nfet*		0.120
c10	Min spacing between ncm to (nwell outside areaid.ce) [exclude coincident edges]		0.380

Section I: Antenna Rules

Antenna rules specify the maximum allowed ratio of interconnect area exposed to plasma etch to active gate poly area that is electrically connected to it when the interconnect is etched. Interconnect areas exposed to plasma etch are:

- bottom areas of Licon, Mcon, Via, Via2
- perimeter areas of connection layers Poly, Li, Met1, Met2, Met3

Two types of checks are introduced in the following tables: vertical for perimeter area and horizontal for contact
The numbers checked for are in the MAX_EGAR column.

Definitions:		
Symbol	***Explanation***	***Unit***
PI	Perimeter of Interconnect	um
FLT	Final Layer thickness	um
W	Width of MOS Transistor	um
L	Length of MOS Transistor	um
A	Area of MOS Transistor gate (= W x L)	um^2
CA	Area of contact or via	um^2
SW	Sidewall area (= PI x FLT)	um^2
EA	Etched area (= CA for horizontal, = SW for vertical areas)	um^2

Antenna rule numbers depend on the connection to the following devices:
pAntennaShort = (tap AndNot poly) AndNot nwell. It is a p+ tap contact used to shortcut to substrate ground buses.
AntennaDiode = (diff OR tap) AndNot (poly OR pAntennaShort). It is a reverse biased diode whose leakage current will discharge the interconnect area.
These devices are not subject to LVS check and must not be reported in the schematic.

Antenna rules are defined for the following ratio:

When a diode is used

EGAR **Etch Gate Area Ratio = (EA / A_gate) - K x (AntennaDiode_area in um^2) - diode_bonus [unitless]**

When diodes are not used

EGAR **Etch Gate Area Ratio = (EA / A_gate) [unitless]**

where:

K is a multiplying factor specified for each layer

AntennaDiode_area is the area of the AntennaDiode used to discharge the interconnect area

The layout should satisfy the condition: EGAR <= MAX_EGAR. The diode_bonus applies only when at least one diode is

Table Ia. Antenna rules (S8D*)

(ar_q.-.-)	Antenna ratios	Area checked: H/V	FLT	MAX_EGAR Max EA/A w/o diode	K	Diode bonus
.poly.1	Poly (poly perimeter area/gate area)	Vertical	0.180	50	n/a	n/a
.licon.1	Licon (licon1 area/gate area)	Horizontal		3	n/a	n/a
.li.1	LI (LI perimeter area/gate area)	Vertical	0.100	75	450	n/a
.mcon.1	Mcon (mcon area/gate area)	Horizontal		3	18	n/a
.met1.1	Met1 (met1 perimeter area/gate area)	Vertical	0.350	400	400	2200
.via.1	Via (via area/gate area)	Horizontal		6	36	n/a
.met2.1	Met2 (met2 perimeter area/gate area)	Vertical	0.350	400	400	2200
.pad.1	pad (via2 area/gate area)	Horizontal		6	36	n/a
.indm.1	INDM (met3 perimeter area/gate area)	Vertical	4.000	400	400	2200
.ar.1	Antenna rules not checked for features connected to a pAntennaShort					

Design limitations due to antenna rules (FET gate = $1um^2$).	**w/o diode**	**with $1um^2$ diode**
Max length of Poly (approx, assumes min width)	135	n/a
Max number of Licons	103	n/a
Max length of LI (approx, assumes min width)	370	2620
Max number of Mcon	103	726
Max length of Met1 (approx, assumes min width)	570	4280
Max number of Via	266	1866
Max length of Met2 (approx, assumes min width)	570	4280
Max number of pad via	4	29
Max length of Met3 (approx, assumes min width)	45	370

Table Ib. Antenna rules (S8TNV-5R)

(ar_q.-.-)	Antenna ratios	Area checked: H/V	FLT	MAX_EGAR		
				Max EA/A w/o diode	K	Diode bonus
.poly.1	Poly (poly perimeter area/gate area)	Vertical	0.180	50	n/a	n/a
.licon.1	Licon (licon1 area/gate area)	Horizontal		3	n/a	n/a
.li.1	LI (LI perimeter area/gate area)	Vertical	0.100	75	450	n/a
.mcon.1	Mcon (mcon area/gate area)	Horizontal		3	18	n/a
.met1.1	Met1 (met1 perimeter area/gate area)	Vertical	0.350	400	400	2200
.via.1	Via (via area/gate area)	Horizontal		6	36	n/a
.met2.1	Met2 (met2 perimeter area/gate area)	Vertical	0.350	400	400	2200
.via2.1	Via2 (via2 area/gate area)	Horizontal		6	36	n/a
.met3.1	Met3 (met3 perimeter area/gate area)	Vertical	0.850	400	400	2200
.ar.1	Antenna rules not checked for features connected to a pAntennaShort					

Design limitations due to antenna rules (FET gate = 1um^2).	w/o diode	with 1um^2 diode
Max length of Poly (approx, assumes min width)	135	n/a
Max number of Licons	103	n/a
Max length of LI (approx, assumes min width)	370	2620
Max number of Mcon	103	726
Max length of Met1 (approx, assumes min width)	570	4280
Max number of Via	266	1866
Max length of Met2 (approx, assumes min width)	570	4280
Max number of Via2	76	535
Max length of Met3 (approx, assumes min width)	230	1760

Table Ic. Antenna rules (S8TM-5R*/S8TMC-5R*/S8TMA-5R*)

(ar_q.-.-)	Antenna ratios	Area checked: H/V	FLT	MAX_EGAR		
				Max EA/A w/o diode	K	Diode bonus
.poly.1	Poly (poly perimeter area/gate area)	Vertical	0.180	50	n/a	n/a
.licon.1	Licon (licon1 area/gate area)	Horizontal		3	n/a	n/a
.li.1	LI (LI perimeter area/gate area)	Vertical	0.100	75	450	n/a
.mcon.1	Mcon (mcon area/gate area)	Horizontal		3	18	n/a
.met1.1	Met1 (met1 perimeter area/gate area)	Vertical	0.350	400	400	2200
.via.1	Via (via area/gate area)	Horizontal		6	36	n/a
.met2.1	Met2 (met2 perimeter area/gate area)	Vertical	0.350	400	400	2200
.via2.1	Via2 (via2 area/gate area)	Horizontal		6	36	n/a
.met3.1	Met3 (met3 perimeter area/gate area)	Vertical	2.000	400	400	2200
.ar.1	Antenna rules not checked for features connected to a pAntennaShort					

Design limitations due to antenna rules (FET gate = 1um^2).	w/o diode	with 1um^2 diode
Max length of Poly (approx, assumes min width)	135	n/a
Max number of Licons	103	n/a
Max length of LI (approx, assumes min width)	370	2620
Max number of Mcon	103	726
Max length of Met1 (approx, assumes min width)	570	4280
Max number of Via	266	1866
Max length of Met2 (approx, assumes min width)	570	4280
Max number of Via2	9	65
Max length of Met3 (approx, assumes min width)	95	740

Table Ie. Antenna rules (S8P-5R/SP8P-5R/S8P-10R*)

(ar_q.-.-)	**Antenna ratios**	Area checked: H/V	**FLT**	**MAX_EGAR** Max EA/A w/o diode	**K**	**Diode bonus**
.poly.1	Poly (poly perimeter area/gate area)	Vertical	0.180	50	n/a	n/a
.licon.1	Licon (licon1 area/gate area)	Horizontal		3	n/a	n/a
.li.1	LI (LI perimeter area/gate area)	Vertical	0.100	75	450	n/a
.mcon.1	Mcon (mcon area/gate area)	Horizontal		3	18	n/a
.met1.1	Met1 (met1 perimeter area/gate area)	Vertical	0.350	400	400	2200
.via.1	Via (via area/gate area)	Horizontal		6	36	n/a
.met2.1	Met2 (met2 perimeter area/gate area)	Vertical	0.350	400	400	2200
.via2.1	Via2 (via2 area/gate area)	Horizontal		6	36	n/a
.met3.1	Met3 (met3 perimeter area/gate area)	Vertical	0.800	400	400	2200
via3.1	Via3 (via3 area/gate area)	Horizontal		6	36	n/a
met4.1	Met4 (met4 perimeter area/gate area)	Vertical	0.800	400	400	2200
via4.1	Via3 (via3 area/gate area)	Horizontal		6	36	n/a
waffle_chip	Met4 (met4 perimeter area/gate area)	Vertical	2.000	400	400	2200
.ar.1	Antenna rules not checked for features connected to a pAntennaShort					

Design limitations due to antenna rules (FET gate = 1um^2).	**w/o diode**	**with 1um^2 diode**
Max length of Poly (approx, assumes min width)	135	n/a
Max number of Licons	103	n/a
Max length of LI (approx, assumes min width)	370	2620
Max number of Mcon	103	726
Max length of Met1 (approx, assumes min width)	570	4280
Max number of Via	266	1866
Max length of Met2 (approx, assumes min width)	570	4280
Max number of Via2	150	1050
Max length of Met3 (approx, assumes min width)	245	1870
Max number of Via3	150	1050
Max length of Met4 (approx, assumes min width)	245	1870
Max number of Via4	9	65
Max length of Met5 (approx, assumes min width)	95	740

Table Ig. Antenna rules (S8P12-10R*/S8PIR-10R/S8PF-10R*)

(ar_q.-.-)	Antenna ratios	Area checked: H/V	FLT	MAX_EGAR Max EA/A w/o diode	MAX_EGAR K	MAX_EGAR Diode bonus
.poly.1	Poly (poly perimeter area/gate area)	Vertical	0.180	50	n/a	n/a
.licon.1	Licon (licon1 area/gate area)	Horizontal		3	n/a	n/a
.li.1	LI (LI perimeter area/gate area)	Vertical	0.100	75	450	n/a
.mcon.1	Mcon (mcon area/gate area)	Horizontal		3	18	n/a
.met1.1	Met1 (met1 perimeter area/gate area)	Vertical	0.350	400	400	2200
.via.1	Via (via area/gate area)	Horizontal		6	36	n/a
.met2.1	Met2 (met2 perimeter area/gate area)	Vertical	0.350	400	400	2200
.via2.1	Via2 (via2 area/gate area)	Horizontal		6	36	n/a
.met3.1	Met3 (met3 perimeter area/gate area)	Vertical	0.800	400	400	2200
via3.1	Via3 (via3 area/gate area)	Horizontal		6	36	n/a
met4.1	Met4 (met4 perimeter area/gate area)	Vertical	0.800	400	400	2200
via4.1	Via3 (via3 area/gate area)	Horizontal		6	36	n/a
met5.1	Met4 (met4 perimeter area/gate area)	Vertical	1.200	400	400	2200
.ar.1	Antenna rules not checked for features connected to a pAntennaShort					

Design limitations due to antenna rules (FET gate = 1um^2).	w/o diode	with 1um^2 diode
Max length of Poly (approx, assumes min width)	135	n/a
Max number of Licons	103	n/a
Max length of LI (approx, assumes min width)	370	2620
Max number of Mcon	103	726
Max length of Met1 (approx, assumes min width)	570	4280
Max number of Via	266	1866
Max length of Met2 (approx, assumes min width)	570	4280
Max number of Via2	150	1050
Max length of Met3 (approx, assumes min width)	245	1870
Max number of Via3	150	1050
Max length of Met4 (approx, assumes min width)	245	1870
Max number of Via4	9	65
Max length of Met5 (approx, assumes min width)	165	1240

SECTION K: PARASITIC LAYOUT EXTRACTION TABLE

This section is relevant for CAD and modeling engineers.

This table list layers and contacts included in SPICE models, and parasitic layers include in the AssuraLayout Extraction.
The **modeled** columns indicate sheets and contacts that are parasitic resistance/capacitance included in the model extraction measurements.
The **CAD** columns indicate sheets and contacts that are parasitics included in the schematic/layout RCX from Assura.

If the device is not supported per the LVS Table then the RCX rules for that device will not apply until the LVS table is updated to support the device.

Section is subject to change with the ultimate goal of making all the discrepancies to zero.

All Flows

Description	Model Structure	Modeled RX		Actual CAD RX		RX Discrepancy	Modeled CX	Actual CX	CX Discrepancy
		Contacts	Sheets	Contacts	Sheets		Sheet	Sheet	
All Periphery FETs	mXXXX d g s b w l m ad as pd ps nrd nrs	none	diff(min)	licon/mcon/vias	diff(ext)/poly/li/m1/m2-m3	none	poly/licon/li	li/m1/m2-m3	li-negligible
20V NDEFETs NONISO	xXXXX d g s b w l m ad as pd ps nrd nrs	none	diff(min)/licon	mcon/vias	diff(ext)/poly/li/m1/m2-m3	none	poly/licon/li	li/m1/m2-m3	li-negligible
20V NDEFETs ISO	xXXXX d g s b w l m ad as pd ps nrd nrs	none	diff(min)/licon	mcon/vias	diff(ext)/poly/li/m1/m2-m3	none	poly/licon/li/dnwdiode_psub	li/m1/m2-m3	li-negligible
20V PDEFETs	xXXXX d g s b w l m ad as pd ps nrd nrs	none	diff(min)/licon	mcon/vias	diff(ext)/poly/li/m1/m2-m3	none	poly/licon/li	li/m1/m2-m3/dnwhvdiode_psub	li-negligible
Cell FETs	NOT EXTRACTED FROM LAYOUT	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
All Diodes	dXXXX n1 n2 area pj	licon	diff	licon/mcon/vias	poly/li/m1/m2-m3	licon-negligible	Junction	li/m1/m2-m3	none
RF ESD Diodes	xesd_XXXX n1 n2 area pj	licon/mcon/ via	li/m1/m2	via2	m3	none	li/m1/m2	m3	none
Parasitic PNP	qXXXX nc nb ne ns pnppar m	licon/mcon	diff/li	mcon/vias	li/m1/m2-m3	li/mcon-neglible	na	li/m1/m2-m3	none
Parasitic PNP (5X)	qXXXX nc nb ne ns pnppar5x m	licon/mcon	diff/li	mcon/vias	li/m1/m2-m3	li/mcon-neglible	na	li/m1/m2-m3	none
Parasitic NPN	qXXXX nc nb ne ns npnpar m	licon/mcon	diff/li	mcon/vias	li/m1/m2-m3	li/mcon-neglible	na	li/m1/m2-m3	none
Non-precision Resistors	rXXXX a b l w m	none	sheet layer	licon/mcon/vias	none (no sheet resistance where sheet layer & res id layer intersect)	none	none	junction/li/m1/m2-m3	parasitic capacitance to substrate (tool limitation)
Precision poly resistor	xXXXXX hrpoly_X_X r0 r1 b l w m	licon/mcon	poly/li	via	m1/m2-m3	none	poly-sub	m1/m2-m3	li-negligible
MIM Capacitor (2-terminal)	xXXXX xcmimc2 c0 c1 w l m	via2	m3	N/A	poly/li/m1/m2	m2 (of the device) -negligible	capm-m2	li/m1/m2-m3	routing layers underneath device
MIM Capacitor (3-terminal)	xXXXX xcmimc c0 c1 b w l m	via2	m3	N/A	poly/li/m1/m2	m2 (of the device) -negligible	m2-sub/capm-m2	(1) Carea of M2-sub (non-overlap CAPM w/ 0.14um upsize) (2) M3 Cap by 1 snap grid width	none
Isolated Pwell Resistor	xXXXX pwres r0 r1 b l w m	licon/mcon	pwell/li	vias	m1/m2-m3	none	none	junction/m1/m2-m3	li-negligible
Vertical Parallel Plate Cap	xXXXX xcmvpp c0 c1 b m (note: no special RCX implementation for VPP required since black-box LVS will be used)	mcon/via	li/m1/m2	none (black box LVS)	none (black box LVS)	none	li/mcon//m1/via/m2	none (black box LVS)	Parasitic capacitance to routing above
Vertical Parallel Plate Cap over MOSCAP	xXXXX xcmvpp2_* c0 c1 b m	mcon/via	li/m1/m2	none (black box LVS)	none (black box LVS)	none	li/mcon//m1/via/m2	none (black box LVS)	Parasitic capacitance to routing above
4-terminal Vertical Parallel Plate Cap (M3 Shielded)	xXXXXX xcmvpp*x*_m3shield c0 c1 b term4 m=	licon/mcon/ via	poly/li/m1/m2	via3/via4	m3/m4/m5	none	poly/licon/li/mcon/m1/via/m2/m3	m3-substrate (not m3-m2), neighboring metal to VPP metal	none
4-terminal Vertical Parallel Plate Cap (M5 Shielded)	xXXXXX xcmvpp*_*m5shield c0 c1 b term4 m=	licon/mcon/ via/via2/via3	poly/li/m1/m2/m3/ m4	via4	m5	none	poly/licon/li/mcon/m1/via/m2/m3/m4/m5	neighboring metal to VPP metal	none
3-terminal Vertical Parallel Plate Cap	xXXXXX xcmvpp*x* c0 c1 b m= xXXXXX xcmvpp*x*_lishield c0 c1 b m=	mcon/ via	li/m1/m2	via2/via3/via4	m3/m4/m5	none	li/mcon/m1/via/m2	neighboring metal to VPP metal	Parasitic capacitance to routing above
3-terminal Vertical Parallel Plate Cap (for S8Q/S8P only)	xXXXXX xcmvpp*x*_m3_lishield c0 c1 b m=	mcon/ via/via2	li/m1/m2/m3	via3/via4	m4/m5	none	li/mcon/m1/via/m2/via2/m3	neighboring metal to VPP metal	none
Varactor	xXXXXX xcnwvc c0 c1 b l w m	licon/mcon/via	diff/poly/li/m1/m2	via2	m3	none	poly/li/m1/m2	nwdiodemodel/m3*	none
Inductor	xXXXXX xindXXXX t1 t2 body (note: no special RCX implementation for inductor required since black-box LVS will be used)	via	m2/Cu	Nothing extracted within inductor.dg layer		none	m2/via/Cu	Nothing extracted within inductor.dg layer	none

*NOTE: The models includes M1/M2 capacitance. As a result of RCX extraction limitation M1/M2 routing over the varactor will have no capacitance extraction.
Routing and placement of devices over or under Precision resistors (xhrpoly_X_X) should be avoided.
The parasitic capacitance between 3-terminal MIMC and any routing/devices is not included in layout RCX, except M3 by 1 snap grid width
No artificial fringing capacitance is extracted for MIMC M2/M3 due to CAD algorithm after CAPM sizing
The parasitic capacitance between Precision resistors (xhrpoly_X_X) and any routing/devices is not included in layout RCX.
S8Q-5R is not supported for RF ESD diode RCX blocking.

The areaid:substratecut will be extracted as a 0.123 ohm two terms resistor.

Layout Extraction Rules for all flows

Resistance Rules

General (RES.-)

.X	Parasitic resistance is not extrated under a sheet layer with it's corresponding res.id layer.

Sheet Resistance(SR.-)

.X	Calibre now extracts deltaW by bucketing the sheet rho based on different widths. The accuracy of each bucket must be within 2% of the Sheet Rho Calculation using deltaW.
.met3	Parasitic resistance is calculated for all metal3 (if metal3 exists for the specific technology)
.met2	Parasitic resistance is calculated for all metal2.
.met1	Parasitic resistance is calculated for all metal1 with the exception of varactor which follow rule SR.xcnwvc.1
.li1	Parasitic resistance is calculated for all li1.
.poly.1	Parasitic resistance on gates is calculated to the center of the gate.
.poly.2	Parasitic resistance for poly is not extracted beyond the device terminal. The device terminal for all devices but MOS is at the edge of the poly.
.diff.1	Parasitic resistance is not extracted for any diffusion regions.
.diff.2	Extract NRD/NRS for MOSFETs (except extendedDrain Fets) per the equations defined in USC-206.
.xnwvc.1	Inside the Varactor device boundry (see rule PASSIVES.cnwvc.1) all layers listed in the model (m1 and below) will not have resistance extracted.

contact-to-gate space(CT.-)

.via	All vias will have parasitic resistance extracted.
.mcon	All mcons will have parasitic resistance extracted.
.licon.1	All licons that are connected to Poly and not connected to the poly of the xhrpoly_X_X device should have resistance extracted.
.licon.2	All licons that are connnected to non-precision resistors will have resistance extracted.
.licon.3	All licons that are connected to FETs will be extracted by RCX.
.licon.4	All licons on diff of PNP/NPN will be considered part of the device model.
.licon.5	All licons on tap of PNP/NPN will be considered part of the device model.
.licon.6	All licons on non-PNP tap regions will have parasitic resistance extracted.
.hrpoly.1	All licons and mcons that are part of the hrpoly resistor will not have parasitic resitance extracted, these contacts are in the models.
.pwres.1	All licons and mcons that are part of the pwell resistor will not have parasitic resitance extracted, these contacts are in the models.

Capacitance Rules

General(CAP.-)

.X.1	No capacitance is extracted due to contacts. (This is a generic layout extraction tool limitation.)

MOS Devices (MOS.-)

.mos.1	area between poly and diff should not have capacitance extracted.
.mos.2	li within .055um will not have fringing capacitance (npcon.4 = 0.055 for S8)
.mos.3	there will be no fringing caps between gates (as capacitance is shielded by the LICON and MCON).
.mos.4	All 20V NMOS ISO DEFETs will have the parasitic diodes included in the models.
.mos.5	All 20V NMOS ISO DEFETs will have the parasitic diode included in the models. This model also includes the 5th terminal Drain-Psub diode (DeepNwell - Psub).
.mos.6	All 20V PMOS DEFETs will have the parasitic DeepNwell-Psub diode included in the CAD extraction.
.mos.7	The only 20V DEFET instance parameter that the model uses from CAD extraction is m-factor. The model will over-write all other instance parameters from CAD extraction.

Resistors (RES.-)

.res.1	short devices must not have capacitance calculated across the device.
.res.2	fuse devices must have capacitance extracted.
.res.3	poly resistors that are not the precision poly resistors (xhrpoly_X_X) must have capacitance extracted.
.res.4	metops that are merged must have capacitance extracted.
.res.5	parasitic resistors for diff/nwell must have the junction diode extracted.
.res.6	Poly precision resistors must not have the poly-psub parasitic capacitance extracted extracted (RCX should also exclude head/tail poly-psub capacitance).
.res.7	For Poly precision resistors xhrpoly_X_1, the device recognition layer is defined by growing the poly.rs AND rpm.dg layers 0.50 um(head/tail distance from ID layer) in all directions. Poly-Field/Diff/Well, and Poly-Poly, Poly-L not have capacitance extracted inside this device recognition layer.
.res.8	For Poly precision resistors xhrpoly_X_2, the device recognition layer is defined by growing the poly.rs AND rpm.dg layers 1.18 um(head/tail distance from ID layer) in all directions. Poly-Field/Diff/Well, and Poly-Poly, Poly-L not have capacitance extracted inside this device recognition layer.

Capacitors (PASSIVES.-)

.cnwvc.1	capacitance will not be extracted for field, diff, Poly, Li, M1, M2 up to 0.465u away from device recognition layer (as defined in LVS table of TDR)
.cmimc.1	capacitance will not be extracted between M2 to field, diff, Poly, Li, M1 up to 0.14u away from device recognition layer (intersection of M2 overlapping CAPM that's connected to VIA2, 3-terminal MiM only)
.crfesd.1	capacitance will not be extracted for field, diff, Poly, Li, M1, M2 up to 0.4u away from device recognition layers (diode terminals)

.xcmvpp.1	For Cap extraction of xcmvpp11p5x11p7_polym50p4shield, the metal5 exposed for metal5-metal5 capacitance extraction is an extra 0.4um from the extraction of xcmvpp11p5x11p7_polym5shield. For any layer below metal5, the capcitance extraction of xcmvpp11p5x11p7_polym50p4shield is the same as xcmvpp11p5x11p7_polym5shield (Note: for the unit cel xcmvpp11p5x11p7_polym50p4shield metal5 is pulled in 0.4um from the cell edge)

Bipolar Devices

none

Discussion of RCX discrepencies

1) Non-precision poly resistors	These resistors do not extract capacitance to substrate. This needs to be accounted for manually by using ICPS_0150_0210 (cap per perimeter) and ICPS_2000_4000 (cap per area). See ETD.
2) Un-shielded VPP's	Any routing above an un-shielded VPP will not be extracted. The impact of this on total capacitance and parasitic capacitance is already comprehended in the model corners. However, cross-talk is not modeled. Also, parasitic cap is routed to ground and this may not be ideal for the scenario. The parasitic cap can be estimated using RescapWeb.

Table L. Created Layer Algorithms (all flows except *HHNEC)

Mask Layer [c]	Drawn Layer [dg]	Created Layer [cl] Algorithm (Does not include CAD bias compensating process bias)	Add waffle pattern?	Run Add / Drop?	TSAC ?	Layer Use / Flow #
Field Oxide	diff,tap	**FOM** = diff:dg OR tap:dg OR size *pnp_emitter* by 0.050 [for S8P-10R-GSMC and S8P12-10R-GSMC] **FOM** = diff:dg OR tap:dg OR size pnp_emitter by 0.030 [rest of the flows]	yes	add, drop	yes	See Table C1
Deep N-Well	dnwell	**DNM** = dnwell:dg sized by cdnm.3	no	none	no	See Table C1, all exc. 29
		DNM = dnwell:dg sized by cdnm.3	no	none	no	29
P-Well Block Mask	nwell	**PWBM** = pwbm:dg	no	none	no	See Table C1
P-Well Drain Extended	pwdem	**PWDEM** =pwdem:dg	no	none	no	See Table C1
N-Well	nwell, hvi, thkox, pwell.rs, areaid.en	**NWM** = (nwell:dg not (hvi or thkox)) sized by cnwm.3a OR ((nwell_all AND (hvi or thkox)) sized by cnwm.3b) *Exempt (nwell hole overlapping pwell.rs)) from cnwm.3a sizing *Exempt ((nwell hole inside (hvi or thkox)) overlapping (areaid.en or pwell.rs)) from cnwm.3b sizing (i.e. The nwell holes for DEPMOS and isolated pwell resistor inside (hvi or thkox) should not be sized by cnwm.3b) Keepout: ((empty nwell hole inside (hvi or thkox)) overlapping areaid.en) sized by cnwm.3c	no	none	no	See Table C1
High Vt PCh	hvtp, nwell, lvtn	**HVTPM** is drawn for the High Vt LV PMOS, high Vt varactor and PMOS core FET; Created over [(LV nwell not overlapping varactors) NOT lvtn] OR [hvtp AND (LV nwell overlapping varactor)]; NOTE: Do not "OR" hvtp to this expression	no	add, drop	no	See Table C1
HLow VT PCh Radio	hvtr	**HVTRM** = hvtr:dg	no	none	no	See Table C1
Low Vt Nch	nwell, hvtp, areaid.ce, lvtn	**LVTNM** is drawn as well as created; **LVTNM** is created in [nwell AND (hvtp OR areaid.ce)] OR (LV nwell overlapping varactor); Merge if space is less than minimum	no	add, drop	no	See Table C1
N-Core Implant	ncm, nwell, areaid:ce, diff, hvtp	**NCM** is drawn in the core. It is created in periphery over [(LV nwell not overlapping varactors) NOT lvtn] OR [hvtp AND (LV nwell overlapping varactor)]. Merge only ncm inside areaid.ce to created layer before checking for violations Note: The algorithm performs a merging procedure to remove space violations	no	none	no	See Table C1
Tunnel Mask	tunm	**TUNM** = tunm:dg SIZED BY ctunm.3	no	none	no	All exc. 15, 40, 41
		TUNM = tunm:dg SIZED BY ctunm.3 OR natfet.dg	no	none	no	15, 50, 41
ONO Mask	tunm	**ONOM** = tunm:dg SIZED by clonom.3	no	none	no	See Table C1
Low Voltage Oxide	hvi, thkox	**LVOM** = (hvi or thkox):dg OR tunm:dg	no	none	no	See Table C1, all exc. 29
		LVOM = (hvi or thkox):dg OR tunm:dg sized by clvom.3	no	none	no	29
Resistor Protect	rpm, urpm	**RPM** = rpm.dg	no	none	no	See Table C1
		RPM = (rpm.dg OR urpm.dg) sized by crpm.3 / curpm.5	no	none	no	45
Reverse resistor	poly, rpm	**RRPM** = (poly.dg AND rpm.dg) sized by crrpm.3	no	none	no	45
Ultra poly resistor	poly, urpm	**URPM** = (poly.dg AND urpm.dg) sized by curpm.3	no	none	no	45
Poly 1	poly	**P1M** = poly:dg	yes	add, drop	yes	See Table C1
N-tip Implant	nwell, hvi, thkox, ldntm	**NTM** is created over nwell, ldntm and ((hvi or thkox) outside areaid.ce). Merge minimum space features.	no	add, drop	no	1,2,3,7,8, 10,11,13, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26
	nwell, hvi, thkox, ldntm	**NTM** is created over nwell, ldntm and ((hvi or thkox) outside areaid.ce) or clntm:dg. Merge minimum space features. Size by cntm.3	no	add, drop	no	29, 30
	nwell, hvi, thkox, ldntm, rpm	**NTM** is created over nwell, ldntm, rpm and ((hvi or thkox) outside areaid.ce). Merge minimum space features.	no	add, drop	no	4,12,15, 27, 28, 32
	nwell, hvi, thkox, ldntm	**NTM** is created over nwell, ldntm and (hvi or thkox). Merge minimum space features.	no	add, drop	no	5, 14
High Volt. N-tip	hvi, thkox, diff, tap, nwell, areaid.ce; hvntm	**HVNTM** is created over (ndiff AND (hvi or thkox)) outside areaid.ce, esd_nwell_tap. The source of the 16V PFET is excluded; The created layer would be merged with drawn layer; Merge minimum space features. For the S8P12-10R-UMC flow, apply a global sizing of 0.1 um/side after final cleanup. Keepout checks are done before the final global sizing of 0.1um/side is applied. Keepouts include: ndiff not in hv sized by 0.185 pdiff sized by 0.185 ptap not diff butting edge sized by 0.185 ptap butting edge diff prec_resistor sized by 0.185 (in flows that support precision resistors)	no	none	no	See Table C1

Mask Layer [c]	Drawn Layer [dg]	Created Layer [cl] Algorithm (Does not include CAD bias compensating process bias)	Add waffle pattern?	Run Add / Drop?	TSAC ?	Layer Use / Flow #
Lightly Doped N-tip	ldntm	**LDNTM** = ldntm:dg [for all flows except S8P12-10R-UMC] **LDNTM** = ldntm.dg SIZED BY 0.1 um/side [for S8P12-10R-UMC flow]	no	none	no	See Table C1, all exc. 29
		LDNTM = ldntm:dg sized -0.05				29
Nitride Poly Cut	npc	**NPCM** = npc:dg	no	none	no	See Table C1
		NPCM = clnpcm:dg (defined by CLDRC rules for this case)	no	none	no	45
P+ Implant	psdm	**PSDM** = psdm:dg OR size (psdm overlapping *pnp_emitter*) by 0.050 [for S8P-10R-GSMC and S8P12-10R-GSMC] PSDM = psdm:dg OR size (psdm overlapping pnp_emitter) by 0.030 [rest of the flows]	no	add, drop	no	See Table C1
N+ Implant	nsdm	**NSDM** = nsdm:dg	no	add, drop	no	See Table C1
Local Intr Cont.1	licon1	**LICM1 =** liconA => (licon or single slotted_licon on prec_resistor terminal) liconB => (multi slotted_licon on prec_resistor terminal) (liconA spaced <= 0.330 sized by clicm1.3a) (liconA spaced > 0.330 sized by clicm1.3b) (liconB edge length == 0.19 expand by clicm1.3a when spaced <=0.330) (liconB edge length == 0.19 expand by clicm1.3b when spaced> 0.33)	no	add, drop	no	See Table C1
Local Intrcnct 1	li1	**LI1M** = li1:dg	no	add, drop	yes	See Table C1
Metal Contact	mcon	**CTM1** = mcon1:dg	no	none	no	See Table C1
Metal 1	met1	**MM1** = met1:dg	yes	none	yes	See Table C1
Metal 1 - Cu	met1	**MM1 = met1:dg; slotting algorithm Table M2**	yes	none	no	See Table C1
Via	via	**VIM** = via:dg	no	none	no	See Table C1
Capacitor MiM on M2	capm	**CAPM** = capm:dg	no	none	no	See Table C1
Metal 2	met2	**MM2** = met2:dg	yes	none	yes	See Table C1
Metal 2 - Cu	met2	**MM2 = met2:dg; slotting algorithm Table M2**	yes	none	no	See Table C1
Via 2-TNV	via2	**VIM2** = via2:dg	no	none	no	See Table C1
Via 2-S8TM	via2	**VIM2** = via2:dg	no	none	no	See Table C1
Via 2-PLM	via2	**VIM2** = via2:dg	no	none	no	See Table C1
Metal 3-TLM	met3	**MM3** = met3:dg	no	none	no	See Table C1
Metal 3-S8TM	met3	**MM3** = met3:dg	no	none	yes	See Table C1
Capacitor MiM on M3	capm	**CAPMM3** = capm:dg	no	none	no	See Table C1
Metal 3-PLM	met3	**MM3** = met3:dg	no	none	yes	See Table C1
Metal 3-Cu	met3	**MM3 = met3:dg; slotting algorithm Table M2**	no	none	yes	See Table C1
Pad Via	via3	**VIM3** = via3:dg	no	none	no	See Table C1
Via3-PLM	via3	**VIM3** = via3:dg	no	none	no	See Table C1
Inductor-TLM	met3	**INDM** = met3:dg	no	none	yes	See Table C1
Capacitor MiM on M4	cap2m	**CAPMM4** = cap2m:dg	no	none	no	See Table C1
Metal 4	met4	**MM4** = met4:dg	no	none	yes	See Table C1
Metal 4-Cu	met4	**MM4 = met4:dg; slotting algorithm Table M2**	no	none	yes	See Table C1
Via4	via4	**VIM4** = via4.dg	no	none	no	See Table C1
Metal 5	met5	**MM5** = met5.dg	no	none	yes	See Table C1
Nitride Seal Mask	nsm	**NSM** = nsm	no	none	no	See Table C1
Pad (scribe protect)	pad	**PDM =** pad:dg OR (cpmm:dg AND areaid.mt) SIZED by cpdm.4, cpdm:mk removed inside the die for *-fab25 flows	yes	none	no	See Table C1
ViaTop	vimt	**VIMT** = viatop.dg	no	none	no	by request only
Mtest	mmtest	**MMTEST** = cmtest.dg	no	none	no	by request only
Polyimide and Polyimide_ExtFab	pad, fuse.dg, areaid.dt, areaid.ft	PMM is created by removing polyimide from the bondpad, the fuses, photo diode arrays, and if in the die, the laser targets; In addition to this, When "NoScribePMM" switch is used: a) PMM is generated over the scribe When "NoScribePMM" switch is not used: The following keepouts are generated for PMM pmm_keepout1 = Diecut_pmm sized by 7.5um pmm_keepout2 = Edge on areaid.ft obtained by projecting areaid.dt is sized up by 7.5um pmm_keepout3 = EXPAND/EXTEND corner of areaid.ft by 7.5um pmm_keepout = pmm_keepout1 OR pmm_keepout2 OR pmm_keepout3 Now, Final_PMM = Scribe NOT pmm_keepout For bondpads within die, the actual enclosure is cpmm.4 + a snap grid. This is done to preserve the 45 degrees at the bondpad corners ; pmm.dg can be drawn to fix cldrc errors; pmm.dg can be drawn to fix cldrc errors	no	drop	no	See Table C1

Mask Layer [c]	Drawn Layer [dg]	Created Layer [cl] Algorithm (Does not include CAD bias compensating process bias)	Add waffle pattern?	Run Add / Drop?	TSAC ?	Layer Use / Flow #
Polyimide_ExtFab	pad, areaid.dt, areaid.ft	Die: PDMM is to remove passivation and polyimide from the bondpads: pad:dg sized by cpdmm.5. **PDMM=pad:dg SIZED by cpdmm.5** Scribe: 50 um wide PDMM stripe centered in Scribe for any Scribe width with exceptions of s8*_md* For modules (instances) inside areaid:mt, it follows die rule, i.e., pad:dg sized by cpdmm.5. For bondpads within die, the actual enclosure is cpdmm.5 + a snap grid, to preserve the 45 degrees at the bondpad corners	no	drop	no	See Table C1
DECA PBO [3]	pi1	**PI1** = pi1.dg	no	no	no	See Table C1
Cu Inductor/Redist. [3]	rdl	**RDL** = rdl.dg	no	no	no	See Table C1
Polyimide 2 [3]	pi2	**PI2** = pi2.dg	no	no	no	See Table C1
Under Bump Metal [3]	ubm	**UBM** = ubm.dg; No data for the frame	no	no	no	See Table C1
Bumps [3]	bump	**BUMP** = bump.dg; No data for the frame	no	no	no	Internal DECA WLCSP layer

* No verification is supported for the S8SS-5R flow
See Layer Information Table for CAD flow numbers

Notes:

1. Notches and extrusions on created layers due to the sizing would be removed if not complying with s.1 and s.2 rules.

2. TSAC = Total Surface Area Check - run by CAD (thicknesses specified in the EDR). In addition to drawn layers + waffles, the drawn mask data (in fab blocks) inside (areaid.ft NOT areaid.dt) must be accounted in the TSAC calculation
3. These layers are required for WLCSP packaging

Table L1A. Additional drawn layers streamed out with tapeout gds data

Name	Drawn layer	Algorithm	HHNEC GDS	Flow
Amkor				
areaid.cuPillar	areaid.cr	See CLDRC and the diagram, AMS-771 Pillar ID indicates the location of Cu pillars for Amkor. The layer is a rectangular mark for pads > 39x39, offset to lower left wrt pad outline. For 39x39 pads, it reproduces pad shape.	81.55	s8pfhd-10r

Section M1: Waffles

Waffling is done using the SmartFill technology aka intelligent waffling introduced by mentor (CWR-75). For FOM/P1M, waffling is done in two stages with the second stage being more aggressive. The waffles for metal are done in a single pass with the flexibility to use different sizes/spaces. Waffles do not get OPC.

The waffles are created throughout the die/frame by default. Waffles will not be created over the waffleDrop layers.

Designers cannot draw waffle layers

For re-tapeouts, sometimes only one of the dependent masks is taped out e.g. FOM while using POLY mask from previous tapeout or vice-versa. This gives rise to problem the waffles created for new FOM (P1M) which can overlap old P1M (FOM) waffles (undesirable due to added capacitance). See SIDK-47 for design productivity win to avoid waffle overlap.

	Boundary	Switch	Waffle drop	Pad:dg keepout
Die	*die*	waffle_chip	active	yes
Frame	*frame*	waffle_frame	active	no

Table M1. Created Layer dummy structures ("waffles")

CD (not subject to CAD bias/OPC):

Name	Flow # (See Layer Info Table for flow names)	pass	Length and width	Space to itself	Space to keepout	Space to *inductor_metal*	Space to FOM	Space to *capacitor:dg*	space to *poly_waffles*	space to *fom_waffles*	space to *NSM_keepout*	space to areaid.dt	Space to fuse:dg	Space to previous shape	Space to scribe edge when die pad ring exists
FOM [5]	all	1	4.08	1.32	0.50	0.00	N/A	N/A	0.30	1.32	1.00	3.00	mf.10	N/A	13.00
	all	2	2.50	1.32	0.50	0.00	N/A	N/A	0.30	1.32	1.00	3.00	mf.10	1.32	13.00
	all	3	1.50	1.32	0.50	0.00	N/A	N/A	0.30	1.32	1.00	3.00	mf.10	1.32	13.00
	all	3	0.50	0.40	0.50	0.00	N/A	N/A	0.30	0.40	1.00	3.00	mf.10	1.32	13.00
P1M	all	1	0.72	0.36	1.00	0.00	0.30	N/A	0.36	0.30	1.00	3.00	mf.10	N/A	13.00
	all	2	0.72	0.36	1.00	0.00	0.30	N/A	0.36	0.30	1.00	3.00	mf.10	0.36	13.00
	all	2	0.54	0.36	1.00	0.00	0.30	N/A	0.36	0.30	1.00	3.00	mf.10	0.36	13.00
	all	2	0.48	0.36	1.00	0.00	0.30	N/A	0.36	0.30	1.00	3.00	mf.10	0.36	13.00
MM1 (4,7)	all	1	2.00	0.20	3.00	0.00	N/A	3.00	N/A	N/A	1.00	3.00	mf.6	N/A	13.00
	all	1	1.00	0.20	3.00	0.00	N/A	3.00	N/A	N/A	1.00	3.00	mf.6	0.20	13.00
	all	1	0.58	0.20	0.50	0.00	N/A	3.00	N/A	N/A	1.00	3.00	mf.6	0.20	13.00
	04,06,07,11,15,16,17,18,24,25,28,45	1	0.30	0.30	0.30	0.00	N/A	3.00	N/A	N/A	1.00	3.00	mf.6	0.20	13.00
MM2 (4,7)	all	1	2.00	0.20	3.00	0.00	N/A	3.00	N/A	N/A	1.00	3.00	mf.6	N/A	13.00
	all	1	1.00	0.20	3.00	0.00	N/A	3.00	N/A	N/A	1.00	3.00	mf.6	0.20	13.00
	all	1	0.58	0.20	0.50	0.00	N/A	3.00	N/A	N/A	1.00	3.00	mf.6	0.20	13.00
	04,06,07,11,15,16,17,18,24,25,28,45	1	0.30	0.30	0.30	0.00	N/A	3.00	N/A	N/A	1.00	3.00	mf.6	0.20	13.00
MM3 (4,7)	04,06,07,11,15,16,17,18,24, 25,28,29,34, 45	1	2.00	0.30	3.00	0.00	N/A	3.00	N/A	N/A	1.00	3.00	mf.6	N/A	13.00
		1	1.00	0.30	3.00	0.00	N/A	3.00	N/A	N/A	1.00	3.00	mf.6	0.30	13.00
		1	0.58	0.30	0.50	0.00	N/A	3.00	N/A	N/A	1.00	3.00	mf.6	0.30	13.00
	04,06,07,11,15,16,17,18,24,25,28,45	1	0.40	0.30	0.30	0.00	N/A	3.00	N/A	N/A	1.00	3.00	mf.6	0.30	13.00
MM4 (4,7)	04,06,07,11,15,16,17,18,24, 25,28,29,34,45	1	2.00	0.30	3.00	0.00	N/A	N/A	N/A	N/A	1.00	3.00	mf.6	N/A	13.00
		1	1.00	0.30	3.00	0.00	N/A	N/A	N/A	N/A	1.00	3.00	mf.6	0.30	13.00
		1	0.58	0.30	0.50	0.00	N/A	N/A	N/A	N/A	1.00	3.00	mf.6	0.30	13.00
	04,06,07,11,15,16,17,18,24,25,28,45	1	0.40	0.30	0.30	0.00	N/A	N/A	N/A	N/A	1.00	3.00	mf.6	0.30	13.00
MM5	04,06,07,11,15,16,17,18,24,25,28,45	1	3.00	1.60	3.00	0.00	N/A	N/A	N/A	N/A	1.00	3.00	mf.6	N/A	N/A

Keepout layers:

Name	Flow #	pass	diff:dg	tap:dg	fom.dy [6]	poly:dg	pwbm.dg	pwell resistor	licon:dg	li:dg	fsm:dg	mcon:dg	met1:dg	via:dg	via2:dg	met3.dg	via3.dg	met4.dg	via4.dg	met5:dg	pad:dg (2)	fuse:dg	*NSM_keepout*	cfom:dg (3)	cfom:mk NOT	*fom_waffles*	cp1m:mk NOT	*poly_waffles*	*cfsm:mk*	*cmm1:mk*	*cmm2:mk*	*cmm3.mk*	*cmm4.mk*	*cmm5.mk*	*cfom:wp*	*cp1m:wp*	*cmm1:wp*	*cmm2:wp*	*cmm3.wp*	*cmm4.wp*	*cmm5.wp*	*cfsm:wp*	*mm1_slots*	*mm2_slots*	*mm3_slots*	*mm4_slots*	capacitor.dg	*inductor_metal*	areaid.dt	*photoArray*
FOM [5]	all	1	X	X	X	X	X	X														X	X	X	X	X	X	X							X													X	X	X
	all	2	X	X	X	X	X	X														X	X	X	X	X	X	X							X													X	X	X
	all	3	X	X	X	X	X	X														X	X	X	X	X	X	X							X													X	X	X
	all	3	X	X	X	X	X	X														X	X	X	X	X	X	X							X													X	X	X
P1M	all	1	X	X	X	X		X														X	X	X	X	X	X	X								X												X	X	X
	all	2	X	X	X	X		X														X	X	X	X	X	X	X								X												X	X	X
	all	2	X	X	X	X		X														X	X	X	X	X	X	X								X												X	X	X
	all	2	X	X	X	X		X														X	X	X	X	X	X	X								X												X	X	X
MM1 (4,7)	all	1										X	X	X							X	X	X							X							X						X				X	X	X	X
	all	1										X	X	X							X	X	X							X							X						X				X	X	X	X
	all	1										X	X	X							X	X	X							X							X						X				X	X	X	X
	04,06,07,11,15,16,17,18,24,25,28,45	1										X	X	X							X	X	X							X							X						X				X	X	X	X
MM2 (4,7)	all	1												X	X						X	X	X								X							X						X			X	X	X	X
	all	1												X	X						X	X	X								X							X						X			X	X	X	X
	all	1												X	X						X	X	X								X							X						X			X	X	X	X
	04,06,07,11,15,16,17,18,24,25,28,45	1												X	X						X	X	X								X							X						X			X	X	X	X
MM3 (4,7)	04,06,07,11,15,16,17,18,24, 25,28,29,34, 45	1													X	X	X				X	X	X									X							X						X		X	X	X	X
		1													X	X	X				X	X	X									X							X						X		X	X	X	X
		1													X	X	X				X	X	X									X							X						X		X	X	X	X
	04,06,07,11,15,16,17,18,24,25,28,45	1													X	X	X				X	X	X									X							X						X		X	X	X	X
MM4 (4,7)	04,06,07,11,15,16,17,18,24, 25,28,29,34,45	1															X	X	X		X	X	X										X							X						X		X	X	X
		1															X	X	X		X	X	X										X							X						X		X	X	X
		1															X	X	X		X	X	X										X							X						X		X	X	X
	04,06,07,11,15,16,17,18,24,25,28,45	1															X	X	X		X	X	X										X							X						X		X	X	X
MM5	04,06,07,11,15,16,17,18,24,25,28,45	1																	X	X	X	X	X											X							X							X	X	X

Additional rules for FOM waffles wrt Nwell

Min spacing between FOM waffles to nwell	difftap.9
Min enclosure of FOM waffles by nwell	difftap.8
Min spacing between FOM waffles to HV nwell	difftap.18
Min enclosure of FOM waffles by HV nwell	difftap.17

Additional rules for waffles wrt areaid.ft

Min enclosure of FOM, P1M and Metal waffles by areaid.ft		nsm.3a

Note : Areas indicated by text.dg with text "falseScribe" are removed from density calc in frame

Waffles will be streamed out on a different layer for S8PF-10R-HHNEC

Notes:

1. Metal* waffling are independent from FOM and P1M waffling.
2. Pad:dg is a keepout for waffles only in the *die* (bond-pad and seal ring opening). If inside areaid:mt (etest pads) it is not considered keepouts
3. Cfom:dg specified only for clarification: cfom:dg is included in cfom:mk
4. Multiple size waffles generated for metal in single pass; For FOM/P1M: The waffle is generated in two stages; the second one is more aggressive
5. FOM waffles uses the 2000um and 200um window size and step respectively in the DFM FILL algorithm
6. fom.dy to be "OR"ed with the generated FOM waffles; fom.dy layer will not receive OPC
7. 0.4um X 0.4um met3/met4 waffles not supported in S8P-10R-GSMC, S8P12-10R-GSMC, S8P12-10R-UMC, S8PF-10R-UMC. No waffles <0.500 um permitted for *F25 flows.
8. Waffling would not receive mbOPC but is subject to the process bias sizing in the same way as drawn features

Section M2: FOM Dummy Rules

FOM waffles can be drawn using the fom dummy layer to improve pattern density in the chip;
These do not overlap nsdm/psdm and are highly resistive; fom.dy to be "OR"ed with the
generated FOM waffles and will not get OPC; Final FOM PD check should also account for fom.dy

(fomdmy.-)	FOM Dummy Rules	Value (um)
1	Min width of fom.dy	0.50
1a	Max width of fom.dy	25.00
2	Min spacing between fom.dy	0.40
3	Min spacing, no overlap, between fom.dy and fomDmy_keepout_1	0.50
4	Min spacing, no overlap, between fom.dy and areaid.sl	1.00
6	Min spacing, no overlap, between fom.dy to fuse	3.25
7	Min spacing, no overlap, between fom.dy and nsdm/psdm	0.13
8	Min enclosure of fom.dy by nwell	0.18
9	Min spacing of fom.dy to nwell	0.34
10	Min enclosure of fom.dy by HV_nwell	0.43
11	Min spacing of fom.dy to HV_nwell	0.33
12	Min enclosure of fom.dy by areaid.ft	0.50
13	Min spacing between fom.dy to areaid.dt	0.50